

Abstract

Process complexity has emerged as a promising lens for capturing and analysing the dynamics of organizational processes. While a variety of complexity metrics exist, their relationships, temporal behaviour, and sensitivity to process drift remain poorly understood, as there is a lack of comprehensive comparative analyses focussing on process complexity metrics. However, this understanding is essential for using process complexity as a meaningful tool to study process dynamics.

This thesis presents a systematic empirical analysis of multiple complexity metrics across different event log datasets. It investigates how different metrics relate to one another, how they evolve over time, and how they respond to incremental process changes. It thus answers three research questions:

RQ1: How are different process complexity metrics related to each other?

RQ2: How can process complexity metrics be used to capture dynamics over time?

RQ3: How do process complexity metrics react to process drift?

To answer these questions, three sub-analyses are performed, consisting of a correlation analysis, a temporal dynamics analysis, and a process drift simulation. The findings show that complexity metrics capture distinct facets of process behaviour, showing varying degrees of correlation, temporal patterns, and responsiveness to drift. Some metrics exhibit nonlinear bursts at certain tipping points, others follow steady trends, and their sensitivity to drift varies substantially, with some metrics exhibiting near drift-invariant behaviours. Additionally, the findings indicate that there are clusters of process complexity metrics with shared dependencies and strong positive relationships, while others are invariant, and some exhibit opposing change behaviour, representing trade-offs. These findings advance the understanding of process complexity as a multidimensional construct and offer practical guidance for selecting and applying complexity metrics to study process dynamics. Furthermore, future research opportunities are identified in comprehensive evaluations, more robust process drift simulation approaches, and the potential of process complexity to detect process drift and change patterns.

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Abbreviations

BPM	Business Process Management
EPA	Extended Prefix Automata
EEPA	Enriched Extended Prefix Automata
IEEE	Institute of Electrical and Electronic Engineers, Inc.
PM4PY	Process Mining for Python
TQM	Total Quality Management

1 Introduction

Processes are the core of how organizations function, adapt, and create value (March & Simon, 1958; Nelson & Winter, 1982). While traditional views often assume processes to be relatively stable structures, recent research has emphasized that processes are inherently dynamic, changing over time due to both endogenous adaptations and exogenous influences (Feldman & Pentland, 2003; Grisold, Janiesch, et al., 2024). As such, it has become increasingly important to find ways to systematically observe, measure, and understand these dynamics (Feldman et al., 2016; vom Brocke et al., 2024). One promising approach for capturing such process dynamics lies in analyzing the structural characteristics of processes using quantitative metric, particularly those that measure process complexity. Broadly speaking, process complexity refers to features such as variability, intricacy, or unpredictability in how processes unfold (Vidgof et al., 2023). Process complexity has been employed in various ways: as a proxy for the dynamics of digitized processes (Franzoi et al., 2024; Goh & Pentland, 2019; B. T. Pentland et al., 2012; B. T. Pentland & Goh, 2021; B. T. Pentland, Liu, et al., 2020; B. T. Pentland et al., 2021; B. T. Pentland, Mahringer, et al., 2020), an indicator of process model quality (Augusto et al., 2022; Mendling, 2008), or an evaluative tool for simulation models (Chapela-Campa et al., 2023), among other things. More recently, scholars have begun to compute process complexity directly from event log data and to explore its relationship with process performance (Vidgof et al., 2023). Such event log data and other forms of digital trace data are particularly well suited to capturing the dynamics of socio-technical processes (Feldman et al., 2016; Franzoi et al., 2023; vom Brocke et al., 2024), constituting longitudinal data that represents how underlying processes unfold over time, thus enabling analyses of processual dynamics over time (Langley et al., 2013; B. Pentland et al., 2021) and of emergent structures and patterns (Langley et al., 2013). Importantly, process complexity itself is a multidimensional construct, reflecting different aspects of processes (Lauterbach et al., 2020). Consequently, a multitude of complexity metrics are in use across academic and practical contexts, each emphasizing different aspects of process structure, behavior, or variability (Augusto et al., 2022; Vidgof et al., 2023).

Despite the valuable insights of this body of work, three important problems remain: First, there is still lack a clear understanding of how different complexity metrics relate to one another. Second, little is known about how these metrics behave over time when applied to event log data. And third, there is a lack of understanding how different complexity metrics respond to process drift in the form of incremental or structural changes in processes over time. These are crucial questions if process complexity is to be used as a meaningful lens for studying process dynamics. To address these challenges, this study

seeks to advance the understanding of process complexity in dynamic contexts by posing the following research questions:

RQ1: How are different process complexity metrics related to each other?

RQ2: How can process complexity metrics be used to capture dynamics over time?

RQ3: How do process complexity metrics react to process drift?

This thesis answers these questions by conducting a systematic, empirical analysis of a broad range of process complexity metrics across a diverse set of openly available event log data sets. This approach involves measuring interrelationships among complexity metrics, tracing their evolution over time, and evaluating their sensitivity to process drift. The findings highlight that process complexity metrics exhibit complex relationships, with different dynamics over time and in reaction to process drift. Some complexity metrics exhibit non-linear bursts at tipping points, while others exhibit linear or steady trends over time. While some complexity metrics form highly correlated clusters and reflect similar patterns, others diverge substantially, suggesting that they represent trade-offs that capture different facets of process behavior. Moreover, the findings show that certain metrics are more sensitive to incremental process drift, making them both more prone to be affected by process drift, and potentially more suitable to detect process drift.

In doing so, this thesis makes several contributions. First, it provides a comparative understanding of commonly used process complexity metrics and their empirical relationships, offering guidance on metric selection in different contexts. Second, it demonstrates the temporal behavior of complexity metrics and showcases potential tipping points, trade-offs, and constant trends. Third, it shows how sensitivity to process drift varies across metrics, which is important to consider when selecting appropriate measures to study dynamics.

The remainder of this thesis is structured as follows. Section 2 summarizes the foundations of Business Process Management (BPM), process mining, and prior research on process complexity. Section 3 presents the approach for measuring and analyzing the selected process complexity metrics, including technical details of the implementation. Then, Section 4 outlines the findings of the thesis, explaining the results of the correlation analysis, the temporal dynamics, and the results of the process drift simulation. Section 5 discusses the key findings of the analyses, contributions of this thesis for Business Process Management and routine dynamics research, practical implications, as well as the limitations of this thesis and future research opportunities in the context of process complexity. Finally, Chapter 6 concludes the thesis.

2 Background

2.1 Business Process Management

Process complexity in business processes, the key underlying concept that forms the basis for this thesis, is deeply linked to business processes and their role in organizations, as well as the wider domain of process research. As such, this thesis is most basally grounded in the academic discipline of Business Process Management, in addition to other research streams and influences. Business Process Management provides an interdisciplinary approach for the management and optimization of business processes, combining research, heuristics and methods from information management and management science (W. M. P. van der Aalst, 2013). Due to the formation of the domain of Business Process Management as a synthesis of multiple previous approaches and traditions in the context of process management, there are multiple perspectives on which approaches constitute the key antecedents of domain (Hammer, 2015; Harmon, 2015; Rosemann & vom Brocke, 2015; W. M. P. van der Aalst, 2013). However, there are two key intellectual antecedents commonly noted in the literature on Business Process Management, which are rooted in the management and optimization of processes and process outcomes (Hammer, 2015; Harmon, 2015). These prior intellectual antecedents are combined to provide a comprehensive organizational approach (DeToro & McCabe, 1997), which “presents a more comprehensive array of improvement options” (DeToro & McCabe, 1997, p. 55) for business operations. Business Process Management thus is an integrated management system for end-to-end business processes (Hammer, 2015).

On one hand, the discipline of Business Process Management is rooted in the work of Taylor (1911), Shewhart (1986), and Demming (1953) on statistical process control, which led to the discipline of quality management and approaches such as Lean or Six Sigma (Harmon, 2015), and can be understood to be the method through which quality management techniques and programmes such as Total Quality Management (TQM) are performed (Demming, 1953; Hammer, 2015; Shewhart, 1986; Taylor, 1911). From this perspective, Business Process Management is a “structured approach to analyze, improve, control, and manage processes with the aim of improving the quality of products and services” (Elzinga et al., 1995, p. 119). Similar to the aforementioned quality management approaches, it is characterized by continuous measurement of outcomes and continuous improvement, which are seen as key to continued business performance (Elzinga et al., 1995; Hammer, 2015). However, this quality-driven perspective is limited by the granular definition of processes as sequences of work activities, reducing the applicability to strategic, overarching processes, and by the common goal of reducing variation within

processes for consistent process performance – which does not necessarily translate into optimal process performance (Hammer, 2015).

The other contributing intellectual antecedent stems from the work of Hammer (1990), Hammer and Champy (1993), and Davenport (1993), which first developed and introduced the discipline of Business Process Reengineering, which primarily contributed to the study of business processes and process management through its focus on process design instead of the optimization of the process execution, and through its perspective on processes as end-to-end activities within organizations that create value for customers (Hammer, 2015). In this sense, reengineering is the “fundamental rethinking and radical redesign” (Hammer & Champy, 1993, p. 35) of processes, with explicit goal of improving process outcomes through this redesign (Davenport, 1993; Hammer, 1990; Hammer & Champy, 1993). While this approach had the caveat of overcoming fragmented process management responsibilities, instead enabling the design and management of large-scale, end-to-end and high-leverage processes, and thus yielding a greater impact on the overall process outcomes (Hammer, 1990, 2015; Hammer & Champy, 1993), Business Process Redesign still lacked the perspective of being a continuous effort as part of an ongoing process management cycle, instead being positioned as a somewhat episodic effort triggered by the need for process redesign (Hammer, 2015). Business Process management thus emerged as the synthesis of these approaches, yielding an integrated system for the management of end-to-end business processes through a process management cycle.

Viewed through the lens of knowledge domains, Business Process Management combines knowledge from information technology and computer science with knowledge from the domain of management science, and applies this aggregated knowledge to business processes (W. M. P. van der Aalst, 2004, 2013). Due to its holistic approach, this application involves a wide range of capabilities and building blocks. Among others, there are six core elements that represent critical success factors for the adoption and integration of Business Process Management in organizations. This six elements include the *strategic alignment* of the BPM efforts with the overall strategy of the organizations, appropriate *BPM governance*, *BPM methods* as the tools and techniques through which BPM activities are performed, supporting *information technology* in the form of BPM-related IT solutions and information systems, as well as the *people* that are involved in BPM activities, and the *culture* within the organization, which can likewise have an impact on the success of BPM efforts and initiatives (Rosemann & vom Brocke, 2015).

The goal of BPM in practice is the monitoring and management of processes, the be able to optimize and (re-)design them effectively, to improve and maintain process outcomes

(Hammer, 2015). In that vein, organizations and academics strive to facilitate value creation from data in the context of BPM. To this end, there has been a drive in recent years towards automation and automated process (re-)design, with the goal of leveraging such potentials to improve outcomes (Beerepoot et al., 2023). Part of this drive is reflected in the growth of the discipline of process mining, which aims to determine the characteristics and structure of processes based off the respective event trace data that is recorded and stored in information systems as event logs, whereby event traces in such logs consist of a sequence of events in the underlying process, as recorded in the system (Reijers et al., 2015).

2.2 Process Mining

Process mining links the actual processes and process data to their respective process models and insights generated from that data. As such, process mining aims to “discover, monitor and improve real processes” (W. van der Aalst, 2016), bridging the gap between machine learning and data analytics on one hand and process analytics on the other hand (W. van der Aalst, 2016). Process mining thus sits at the intersection of data science and process science, and encompasses techniques and activities commonly found in machine learning and data analysis projects (W. M. P. van der Aalst, 2022). These include the extraction of data from information systems that are used to support processes in organizations, and which record and store information pertaining to process activities and events as event data in database tables and merging such datasets. The extracted and merged data can then be explored, selected, filtered and cleaned in a data manipulation phase, applying data manipulation techniques commonly found in the domains of data science or data mining or data mining, or visualizing data to gain additional insights into the properties of the data. Additionally, this data manipulation phase is also concerned with scoping the dataset, selecting appropriate subsets of the data for the underlying process that is to be investigated, and transforming the data to fit the required format (W. M. P. van der Aalst, 2022). During this, issues such as incomplete records, inconsistencies, duplicates, or other data quality issues that can result from the initial data recording, such as errors during a manual data entry, and which could negatively impact later process mining activities and the application of process mining algorithms, can and should be addressed (Suriadi et al., 2017).

Process mining extends across three types of techniques and applications, which aim to achieve different results with the underlying event data. Firstly, *process discovery* aims to automatically construct process models from event trace data, conforming to the underlying ground truth model as best as possible, without information about that ground truth model. Connected to that is *process conformance* checking, whereby models,

generated or ground-truth, are compared against the available event trace data, enabling monitoring of processes, as well as evaluation of the appropriateness and correctness of the examined models. Finally, process mining also entails techniques for *process enhancement*, which involves utilizing of event trace data to improve the ground-truth processes, by providing insights into their effectiveness and optimization potential (W. van der Aalst, 2016).

In the realm of process discovery, there has been significant research into the development of reliable and well-performing methods and algorithms for the application on real-world event trace data. The associated process discovery algorithms produce a process model from the relations between the events in the event log on which they are employed, thereby producing an objective picture of possible control flows that are present in the event log (van Dongen et al., 2009). However, producing models that are neither overly complex, nor over- or underfitted to the event trace data used to generate the model, still poses a difficult problem in academia (Augusto et al., 2017; van Dongen et al., 2009). There is thus a wide range of models currently in use in practice and academia, trading off different quality metrics such as precision, fitness, or simplicity against each other (Augusto et al., 2022).

As outlined before, automatically discovered and generated models can deviate from the observed behaviour or desired control flow due to outliers in the data, which reflect exceptional event sequences in cases. Most processes follow a pareto-distribution, whereby roughly “80% of [...] cases can be explained by only 20% of the process variants” (W. M. P. van der Aalst, 2022). However, the remaining cases that require the majority of the process variants in turn are poised to cause a most of the process performance and compliance issues with the process (W. M. P. van der Aalst, 2022). As such, there is a need to maintain a view on the variability of the process, and the distribution between cases and required process variants.

Understanding how processes are performed, how they can be improved, and how they change over time is thus one of the essential goals of process mining (Hand et al., 2001). In this vein, analyzing and understanding the dynamic changes of processes over time and in reaction to both intentional and non-intentional changes in the process design and execution form a distinct field of study in process mining research, investigating the effects of concept drift on processes.

2.3 Concept Drift, Process Drift, and Process Dynamics

Due to the application of methods from data mining and associated domains, process mining is affected by similar problems and challenges also found in those domains. In

particular, process mining research also commonly faces the problems concept drift. Where concept drift in data mining relates to changes in the relation between input data and the target variable during the runtime of data analyses or machine learning algorithms (Sato et al., 2022), it can likewise affect process mining efforts and analyses, where it relates to changes to the underlying process during the analysis in question (Vidgof, 2025). Processes are changing and evolving in practice, with the process at the start of a given event log likely being different from the process at the end of the event log (Bose et al., 2011). As such, process mining techniques such as process discovery algorithms that require a steady state process that does not change or evolve over time, particularly in the timeframe for which the analysis is performed, can be negatively affected by and potentially inapplicable for event logs exhibiting such concept drift. In this context, this concept drift, whereby the underlying process of an event log changes over the timeframe of the event log, is also known as process drift (Bose et al., 2011). However, this phenomenon is multifaceted, with multiple mechanisms that can cause such drift, and different types of drift that exhibit specific characteristics (Gama et al., 2014). There are four types of process drift commonly found in academia: sudden drift, gradual drift, recurring drift, and incremental drift (Bose et al., 2011; Gama et al., 2014). In sudden drift, a process is substituted with another process instantaneously, with all new traces belonging to the new process. In gradual drift, both processes can coexist for a limited time, with traces in the period of overlap belonging to either of these processes. In recurring drift, this change between two processes is occurring intermittently, with processes switching on recurring periodic or non-periodic basis. Finally, incremental drift refers to a substitution of processes via smaller, incremental changes taking place over a longer time period.

Additionally, process drift can be momentary, for example if caused by technical issues or non-permanent changes or tests, which translates into outliers in the event log, which can be filtered out utilizing the data manipulation techniques established in process mining (Vidgof, 2025). On the other hand, permanent process drift, which represents lasting and significant changes to the underlying process, is of particular interest for further analyses and the focal point of most process drift detection techniques.

The concept of process drift is of particular interest for researchers in the field of process mining and process science. Key problems have been the detection of change points, also known as drift points, or drift intervals, the localization and characterisation of change and drift in processes, and the unravelling and sensemaking of the underlying process evolution (Bose et al., 2011). While initial research into the detection of process drift and drift points or intervals focussed on statistical analyses of event log data (Maaradji et al., 2015, 2017), there have been recent contributions to the field making use of machine

learning and computer vision techniques to detect such drift points (Kraus & van der Aa, 2024; Yeshchenko et al., 2022)., as well as contributions that connect process drift to process complexity metrics, outlining the potential for drift detection techniques utilizing process complexity metrics (Vidgof, 2025).

In the context of understanding and making sense of process dynamics and process evolution, there has also been growing interest in process dynamics from the domain of social sciences, using the lens of routine dynamics to make sense of process drift and process dynamics. The routine dynamics perspective holds that patterning is a mechanism for dynamics and stability in processual routines (Feldman et al., 2016), whereby patterning refers to the process of changing the configuration of paths in a narrative network which describes the organizational routines (B. T. Pentland & Goh, 2021). There are multiple levels of analysis that can be chosen to analyse organizational phenomena, using routine dynamics to understand the dynamics of changing routines (B. T. Pentland & Feldman, 2005). In this sense, the variation and selective retention of patterns of action is both necessary and sufficient to explain the dynamics of organizational routines (B. T. Pentland et al., 2012), and can generate a wide range of unique patterns and sequences through exogenous and endogenous change (B. T. Pentland et al., 2011). Event trace data in the form of event logs is particularly useful for process and routine dynamics research (Feldman et al., 2016). This perspective further shapes the discussion about process complexity, with the routine dynamics perspective being the basis for several process complexity metrics.

2.4 Business Process Complexity and Complexity Metrics

2.4.1 Foundations of Business Process Complexity

Process Complexity refers to a range of metrics that capture the variability, intricacy, or unpredictability of processes (Vidgof et al., 2023), and has a multi-faceted influence on the execution and performance on processes (Vidgof, 2025). Understanding and measuring this variability in the process, and other metrics related to the complexity of the process, enables researchers and practitioners to address these issues, for example through standardization (Münstermann et al., 2010; B. T. Pentland, Mahringer, et al., 2020; Schreiber & Abbad-Andaloussi, 2024). The reduction of complexity through standardization can influence process performance, even though this effect is dependent of the industry that the organization operates in (Münstermann et al., 2010). Additionally, recent work by (Vidgof et al., 2023) has shown that process complexity explains throughput times of processes, which are linked to the process performance. Process complexity has been employed in various ways: as a proxy for the dynamics of digitized

processes (Franzoi et al., 2024; Goh & Pentland, 2019; B. T. Pentland, Liu, et al., 2020; B. T. Pentland, Mahringer, et al., 2020), an indicator of process model quality (Augusto et al., 2022; Mendling, 2008), or an evaluative tool for simulation models (Chapela-Campa et al., 2023).

In that sense, process complexity both motivates and challenges process standardization and process mining efforts (Vidgof et al., 2023). The ability to correctly measure and interpret process complexity is particularly relevant due to the fact that process complexity can change dynamically over time, decreasing or increasing due to conscious or emergent changes to the underlying processes or organizational changes (Grisold, van der Aa, et al., 2024). Understanding and making sense of what causes changes thus requires additional context (Grisold, van der Aa, et al., 2024).

Importantly, process complexity itself is a multidimensional construct, akin to the multifaceted influence of process complexity on the process performance, reflecting different aspects of processes (Lauterbach et al., 2020). Consequently, a multitude of complexity metrics are in use across academic and practical contexts, each emphasizing different aspects of process structure, behavior, or variability (Augusto et al., 2022; Vidgof et al., 2023). While most complexity metrics are based on the control-flow perspective of processes, relatively few metrics also integrate more context, such as available properties of the underlying data in the event log that represent details and context from the process execution (Vidgof, 2025; Vidgof & Mendling, 2023).

2.4.2 Process Complexity Measurement and Event Log Data

The measurement and quantification of process complexity, to both enable further data-driven process analysis and provide further context for other process mining and business process management approaches, as well as understanding process dynamics, has received increased attention in recent years, with a wide variety of process complexity metrics proposed in prior research studies (Augusto et al., 2022; Grisold et al., 2022; Hansson et al., 2023; Schreiber & Abbad-Andaloussi, 2024; Vidgof et al., 2023; Zhou et al., 2023). Each of these metrics captures a unique perspective of process complexity, and provides a unique lens for understanding and making sense of it, as process complexity is a multi-faceted and multi-dimensional concept (Lauterbach et al., 2020). However, while there is a wide range of potential process complexity metrics found in academia, not all of them are applicable to event trace data (Zhou et al., 2023). Metrics must be state of the art, and appropriate to calculate the respective complexity metrics based on the event log itself, instead of primarily being applicable to discovered or ground-truth models as would be the case for control-flow complexity based metric, among others (Augusto et al., 2022). The initial selection of complexity metrics is thus based on the

initial literature focussing on process complexity metrics by Augusto et al. (2022) and Vidgof et al. (2023), and further complemented by other key contributions from the fields of information systems research (Grisold et al., 2022), business process management and process science (B. T. Pentland, Liu, et al., 2020), process mining (Chapela-Campa et al., 2023), computer science (Schreiber & Abbad-Andaloussi, 2024), and sociology and organizational research (Hansson et al., 2023; Lindberg et al., 2016).

Category	Process Complexity Metric	Label	Source
Entropy	Variant Entropy	var-e	(Augusto et al. 2022)
	Sequence Entropy	seq-e	(Augusto et al., 2022)
	Normalized Variant Entropy	nvar-e	(Augusto et al., 2022)
	Normalized Sequence Entropy	nseq-e	(Augusto et al., 2022)
Enriched Entropy	Enriched Variant Entropy	en-var-e	(Vidgof & Mendling, 2023)
	Enriched Sequence Entropy	en-seq-e	(Vidgof & Mendling, 2023)
	Enriched Normalized Variant Entropy	en-nvar-e	(Vidgof & Mendling, 2023)
	Enriched Normalized Sequence Entropy	en-nseq-e	(Vidgof & Mendling, 2023)
Size	Number of Events	#-e	(Günther, 2009)
	Number of Event Types	#-et	(Günther, 2009)
	Number of Sequences	#-seq	(Günther, 2009)
	Minimum Sequence Length	min-seq-len	(W. van der Aalst, 2016)
	Average Sequence Length	avg-seq-len	(W. van der Aalst, 2016)
	Maximum Sequence Length	max-seq-len	(W. van der Aalst, 2016)
	Average Time Difference between Consecutive Events	avg-td-e	(Günther, 2009)
Variation	Number of Acyclic Paths in Transition Matrix	#-acyclic-paths	(Hærem et al., 2015)
	Number of Ties in Transition Matrix	#-ties	(Hærem et al., 2015)
	Lempel–Ziv Complexity	lempel-ziv	(B. T. Pentland, 2003)
	Number of Unique Sequences	#-unique-seq	(W. van der Aalst, 2016)
	Percentage of Unique Sequences	perc-unique-seq	(W. van der Aalst, 2016)
	Average Number of Distinct Events per Sequence	avg-distinct-e	(Günther, 2009)
	Order Variation	order var	(Lindberg et al., 2016)
	Activity Variation	activity-var	(Lindberg et al., 2016)
	Repertoire	repertoire	(Hansson et al., 2023)
	Routinization	routinization	(Hansson et al., 2023)
	Probability of Action Pairs	prob-act-pairs	(Grisold et al., 2022)
Distance	Average Affinity	affinity	(Günther, 2009)
	Structure	structure	(Günther, 2009)
	Deviation from Random	dev-random	(B. T. Pentland, 2003)
	Average Edit Distance	avg-edit-distance	(B. T. Pentland, 2003)
	Structural Process Variety	structural-var	(Schreiber & Abbad-Andaloussi, 2024)
	Control-Flow Log Distance	cf-log-dist	(Chapela-Campa et al., 2023)
	n-Gram Distance	ngram-dist	(Chapela-Campa et al., 2023)

Table 2.1 List of process complexity metrics for event log data in academia

There is a multitude of process complexity measures proposed and used in academia (Augusto et al., 2022; Vidgof et al., 2023; Zhou et al., 2023), which investigate complexity from different perspectives, be it from the process model perspective or more recently from the event log perspective (Vidgof et al., 2023). Accordingly, these metrics rely on different types of data, on which they are to be applied. These metrics each focus on different aspects of complexity, by measuring and quantifying certain characteristics of the process, be it via the model or the event log. Characteristics of models used for complexity metrics include their size, their graph properties, and their respective control flow as the order of activities within the process, among others (Polančič & Cegnar, 2017; Zhou et al., 2023).

In the case of event logs, such characteristics can include the size of an event log, variation within the event log, distance between sequences in the event log, or most recently the graph entropy of the event log (Augusto et al., 2022; Vidgof et al., 2023). While these measures have been validated empirically (Augusto et al., 2022; Polančič & Cegnar, 2017; Vidgof et al., 2023; Zhou et al., 2023), there are currently no academic publications that focus on the interplay of such metrics. Similarly, there is research potential investigating the development of process complexity over time (Vidgof et al., 2023), as well as in investigating the connection between process complexity and other characteristics of event log data (Augusto et al., 2022).

2.4.3 Size-Based Complexity Metrics

Basic metrics that calculate the complexity of event logs via the size of the event log were introduced early, focussing on the number of individual events, event types and sequences (Günther, 2009), on the length of event sequences (W. van der Aalst, 2016), or on the time difference between sequential events (Günther, 2009). Such metrics are relatively simple to compute, but do not necessarily provide an appropriate estimate of the event log complexity, as they are focussed on the structure of the event log (Augusto et al., 2022). The minimum, average, and maximum sequence length, in part introduced by W. van der Aalst (2016), represent such basic measures of event log complexity derived from the event log structure, and are calculated intuitively by measuring the number of events across all traces in an event log. Similarly, the number of sequences simply counts the total number of traces in an event log (W. van der Aalst, 2016).

Günther (2009) introduced both basic structural notions for describing and handling event logs, and multiple structural process complexity metrics, which are referred to as size-based complexity metrics in this thesis, and which have been used to conceptualize and measure process complexity in multiple recent contributions (Augusto et al., 2022; Vidgof & Mendling, 2023). The number of events, introduced by Günther (2009) as the

magnitude of an event log, is calculated by summing up the count of events across all traces in an event log. The number of event types, introduced as the variety, is calculated by counting the number of unique event types across all events and all traces in a given event log. Similarly, the number of sequences, introduced as the support of a given event log, is calculated by counting the number of traces in that event log. The average time difference between consecutive events was also introduced by Günther (2009) as a performance characteristic of processes and their respective event logs, building on the structural complexity metric introduced as the time granularity of a given trace, which is defined as the smallest difference between the timestamps of two consecutive events in that trace, and which is used to calculate the time granularity of a given event log as the mean of the time granularities of all traces within that event log. As such, the average time difference between consecutive events applies the basic calculation of the time granularity to capture the mean timestamp difference instead of the minimal timestamp difference. While the latter had first been introduced as a structural process complexity metric to yield additional context for timestamp-sensitive process mining techniques, the former has been predominantly used in recent contributions (Augusto et al., 2022; Vidgof & Mendling, 2023).

2.4.4 Variation-Based Complexity Metrics

Another set of metrics introduced in academic literature aims to quantify and measure the variation within data, in the case process complexity and event log complexity by measuring the variability of processes through the variability of events in the event logs. This is done by counting the number of acyclic paths in the transition matrix of the event log (Hærem et al., 2015), the number of ties in that transition matrix (Hærem et al., 2015), by counting the number and computing the share of unique sequences in the event log (W. van der Aalst, 2016), counting the average number of unique events per sequence (Günther, 2009) or by computing the Lempel-Ziv complexity of the event log (B. T. Pentland, 2003). For the number of acyclic paths the number of distinct event types, called the number of vertices (“v”), and the number of distinct direct-follows transitions between event types, called the number of edges (“e”), are counted before applying an exponential formula to approximate the number of acyclic paths (Hærem et al., 2015).

$$\text{Number of Acyclic Paths} = 10^{0.08 * (1 + e - v)}$$

The number of ties metric is likewise based on the network representation of the process, summing up the overall number of ties between nodes in the network. It is operationalised using a graph representation of the event log, and sums up the number of paths from the root of the graph for each end node of the graph, thereby across all trace variants (Hærem et al., 2015). The number of unique sequences and the percentage of unique sequences

are again intuitive metrics for the variation across traces, and computed by counting the total number of unique traces across an event log for the former and dividing it by the total number of traces for the latter (W. van der Aalst, 2016). The Lempel-Ziv complexity on the other hand counts the number of sub-sequences that are necessary to describe a sequence, such as a trace, without repetition (B. T. Pentland, 2003). It essentially describes how repetitive or random sequences are and can thus be applied to traces and aggregated across entire event logs. The average number of distinct events is another metric introduced by Günther (2009), which is computed by counting the number of distinct events per trace, and calculating the average of that value across all traces in an event log.

Other metrics such as the order variation focus on the variation in the sequencing of event types in individual traces by counting the total number of transitions between event types and dividing it by the total number of events in the event log (Lindberg et al., 2016). Similarly, the activity variation is based on counting the occurrences of all event types in an event log and measuring their relative shares, followed by calculating the Shannon entropy over those shares (Lindberg et al., 2016), and is thereby also linked to the concept of entropy introduced by Shannon (1948). Two further metrics examine patterns in trace data. While the Repertoire measures the absolute number of unique, recognizable patterns across traces (Hansson et al., 2023), the Routinization measures the degree to which traces follow such recognizable patterns (Hansson et al., 2023). Finally, the Probability of Action Pairs measures the probability of recurrent sequences of event types in traces, by calculating the transition matrix for all event types in a log or sub-log, and the resulting transition probability matrix for all event types (Grisold et al., 2022), which is derived by dividing the number of direct-follows transitions between event types by the total number of transitions from the source event type, across the entire event log.

2.4.5 Distance-Based Complexity Metrics

Another group of metrics focusses on the similarity of event traces within event logs, with the rationale that homogenous events logs translate to less complex processes, while more heterogenous event logs signal complexity. These metrics include the average affinity as a measure for the similarity of event traces (Günther, 2009), which is computed by calculating the similarity of the direct-follows transitions between different unique traces in the event log for all unique traces, and then normalizing by the number of unique traces. The structure, also introduced by Günther (2009), is computed counting the number direct-follows transitions between event types and dividing it by the maximum possible number of direct-follows transitions between event types. The structure metrics is the inverse proportion of this value. Another metric is the average edit distance, which is

computed by first calculating the pairwise Levenshtein distances (Levenshtein, 1965) between all traces in the event log, and then calculating the average of these (B. T. Pentland, 2003). The deviation from random (B. T. Pentland, 2003) measures how different the observed transitions are from a fully random transition matrix, and is computed by first computing the overall transition matrix between event types across all traces, then calculating the squared deviations from the expected baseline for equally distributed transitions, and then aggregating these deviations and normalizing them. Another distance-based metric is the structural process variety, also utilizing the pairwise Levenshtein distances (Levenshtein, 1965) to calculate a distance matrix all traces, followed by the application of agglomerative hierarchical clustering, based on which the unique structural variety value is measured as the sum of the distances between the clusters yielded by the agglomerative hierarchical clustering (Schreiber & Abbad-Andaloussi, 2024). For relative complexity metrics, the control-flow log distance measures distances between logs by calculating the pairwise Damerau-Levenshtein distance (Damerau, 1964) between all traces in two event logs, performing an optimal matching between traces to minimize the sum of distances, and then calculating the average across all normalized distances (Chapela-Campa et al., 2023). Finally, the n-gram log distance, another relative metric, is computed by decomposing two logs into n-grams consisting of n events each, aggregating the absolute differences between the frequencies of the n-grams, and normalizing this distance by the total sum of frequencies of all n-grams across both event logs (Chapela-Campa et al., 2023).

2.4.6 Entropy-Based and Enriched-Entropy-Based Complexity Metrics

Another distinct group of process complexity metrics is based on the application of graph entropy to processes in general and event logs in particular. According to (Dehmer & Mowshowitz, 2011), graph entropy can be interpreted as the structural information content of a graph, and thus be used as a measure of the graphs complexity, building on the basic properties of entropy (Shannon, 1948). Entropy-based metrics fulfil the requirement of monotonicity (Polyvyanyy, Solti, et al., 2020), which enables their application for analyzing process models and event logs on the language level (Polyvyanyy, Alkhamash, et al., 2020; Polyvyanyy, Solti, et al., 2020). However, the application of graph entropy measures for event logs requires an adequate representation of the event log in a graph structure, without introducing additional biases and abstractions. Without representation inherent to the entropy-based complexity metric, complexity measurement would be dependent on the process discovery algorithm chosen to produce the process model (Augusto et al., 2017), as different graphs of the same process can yield different graph entropy results (Dehmer & Mowshowitz, 2011). To this end, (Augusto et al., 2022) use the Extended Prefix Automaton (EPA) as introduced by

(Muñoz-Gama & Carmona, 2010) to partition the event data. For the variant entropy metric, the graph entropy is calculated over the size of the partitions generated by the EPA, based on the graph entropy measures introduced by (Dehmer & Mowshowitz, 2011). The normalized variant entropy is a normalization of the variant entropy metric, returning results in the range of $[0,1]$ (Augusto et al., 2022).

Variant Entropy:

$$E_v = |S| * \log(|S|) - \sum_{i=1}^{\max(C)} |\{s \in S | C(s) = i\}| * \log(|\{s \in S | C(s) = i\}|)$$

Sequence Entropy:

$$E_s = |seq(S)| * \log(|seq(S)|) - \sum_{i=1}^{\max(C)} |seq_i(S)| * \log(|seq_i(S)|)$$

Normalized Variant Entropy:

$$\bar{E}_v = \frac{E_v}{|S| * \log(|S|)}$$

Normalized Sequence Entropy:

$$\bar{E}_s = \frac{E_s}{|seq(S)| * \log(|seq(S)|)}$$

While the variant entropy is based on the number of partitions generated by the EPA, the sequence entropy also accounts for the frequency of events per state, by calculating the graph entropy over the cardinality of events across states per partition, instead of over the cardinality of states per partition (Augusto et al., 2022). The normalized metrics are calculated analogously to those for the previously introduced entropy-based metrics. (Vidgof et al., 2023) further extend these entropy-based metrics, integrating the data variety in the measurement of process complexity, deriving enriched entropy-based metrics. These metrics are calculated analogously to the entropy-based metrics put forward by (Augusto et al., 2022), while accounting for different data across events in otherwise identical sequences (Vidgof et al., 2023). The enriched entropy metrics are computed analogously to the normal entropy-based metrics, while accounting for differences in the event attribute data to differentiate between distinct variants and sequences. To this end, the EPA is extended to this event attribute data, yielding an Enriched Extended Prefix Automaton (EEPA). This EEPA is then used to apply the same calculations highlighted before, resulting in the enriched entropy metrics.

3 Method

3.1 Research Design

To comprehensively answer the research questions guiding this thesis, an explorative research design is employed, conducting an empirical evaluation of relevant process complexity metrics on a set of event log datasets. These metrics are identified based on previous seminal contributions in the fields of process mining, information systems research, and computer science, providing different perspectives on the issue of process complexity measurability. The empirical evaluation of process complexity metrics builds on the methodological foundations established by Augusto et al. (2022) and Vidgof et al. (2023). By answering all three research questions, this thesis aims to deepen the understanding of process complexity metrics in academia. To this end, three analyses are performed. Firstly, a correlation analysis elucidates the relationships between metrics. Secondly, a temporal dynamics analysis examines dynamics over time exhibited by the process complexity metrics in real-world event logs. Thirdly, a process drift simulation investigates the dynamics of process complexity metrics under incremental process drift. Understanding the relationships of complexity measures and their characteristics regarding their application on event log data sets yields further insights into their applicability in practice. This should widen the current scope of research on process complexity metrics and provide new perspectives on the topic of process complexity, in part inspired by the related field of data science. The empirical results should further underline and expand on the findings of previous analyses of process complexity metrics (Augusto et al., 2022; B. T. Pentland, Liu, et al., 2020; Vidgof et al., 2023). In particular, the thesis expands on previous simulation approaches for process complexity in the context of process drift, by applying these approaches to a wider range of complexity metrics and to real-world event logs (Miller et al., 2012; B. T. Pentland, Liu, et al., 2020; B. T. Pentland et al., 2021; Vidgof, 2025).

3.2 Data Collection, Selection, Ingestion, and Transformation

To ensure comparability, the selection of datasets is aligned with established approaches in the BPM and process mining fields, selecting publicly available event log datasets, primarily from the Business Process Intelligence Challenges (BPIC), in line with previous studies on process complexity metrics using event log data (Augusto et al., 2022; Vidgof et al., 2023), accessing the data via the IEEE Task Force on Process Mining website (IEEE Task Force on Process Mining, 2024b). These datasets stem from different real-world organizations and backgrounds, representing a diverse set of processes, IT systems, and organizational contexts.

Event Log Name	Year	Log Type	Source
Purchase order handling process (BPI Challenge 2019)	2019	Real-world (Purchase orders)	(Boudewijn van Dongen, 2019)
Payment process of Common Agricultural Policy (BPI Challenge 2018)	2018	Real-world (Payments)	(Boudewijn van Dongen & F. (Florian) Borchert, 2018)
Loan application process of a Dutch financial institute (BPI Challenge 2017)	2017	Real-world (Financial institute)	(Boudewijn van Dongen, 2017)
Municipality log 1 (BPI Challenge 2015)	2015	Real-world (Municipalities)	(Boudewijn van Dongen, 2015a)
Municipality log 2 (BPI Challenge 2015)	2015	Real-world (Municipalities)	(Boudewijn van Dongen, 2015b)
Municipality log 3 (BPI Challenge 2015)	2015	Real-world (Municipalities)	(Boudewijn van Dongen, 2015c)
Municipality log 4 (BPI Challenge 2015)	2015	Real-world (Municipalities)	(Boudewijn van Dongen, 2015d)
Municipality log 5 (BPI Challenge 2015)	2015	Real-world (Municipalities)	(Boudewijn van Dongen, 2015e)
Change log (BPI Challenge 2014)	2014	Real-world (Management)	(Boudewijn van Dongen, 2014b)
Incident activity log (BPI Challenge 2014)	2014	Real-world (Management)	(Boudewijn van Dongen, 2014a)
Incident management log (BPI Challenge 2013)	2013	Real-world (Management)	(Ward Steeman, 2013b)
Problem management log, open problems (BPI Challenge 2013)	2013	Real-world (Management)	(Ward Steeman, 2013c)
Problem management log, closed problems (BPI Challenge 2013)	2013	Real-world (Management)	(Ward Steeman, 2013a)
Event log of a loan application process (BPI Challenge 2012)	2012	Real-world (Loan application)	(Boudewijn van Dongen, 2012)
Anonymized event log of a Dutch Academic Hospital (BPI Challenge 2011)	2011	Real-world (Hospital)	(Boudewijn van Dongen, 2011)
Hospital Billing - Event Log (Mannhardt)	2017	Real-world (Hospital, Billing)	(Mannhardt, 2017)

Table 3.1 Datasets in scope of thesis

The list of datasets in scope of this thesis is available in table 3.1. The datasets are available in the XES format for event log data in process mining (IEEE Task Force on Process Mining, 2024a). The code that is used to extract and transform these event log datasets, and the implementations of the process complexity metrics as well as the implementations for the correlation analysis, temporal dynamics analysis, and the process drift simulation is available in a GitHub repository¹.

¹ https://github.com/MrWaitomo/WWU_processcomplexity

Event Log Name	Number of Traces	Number of Unique Traces	Maximum Trace Length	Average Events per Trace	Number of Event Types	Median Trace Duration	Overall Log Duration
Purchase order handling process (BPI Challenge 2019)	251734	11973	990	6,34	42	64 days 01:03:30	26371 days 23:00:00
Payment process of Common Agricultural Policy (BPI Challenge 2018)	43809	28457	2973	57,39	41	266 days 14:38:10	1356 days 13:03:02
Loan application process of a Dutch financial institute (BPI Challenge 2017)	31509	15930	180	38,16	26	19 days 02:06:20	397 days 04:19:48
Municipality log 1 (BPI Challenge 2015)	1199	1170	101	43,55	398	61 days 08:57:51	1761 days 00:00:00
Municipality log 2 (BPI Challenge 2015)	832	828	132	53,31	410	108 days 12:00:12	1709 days 00:00:00
Municipality log 3 (BPI Challenge 2015)	1409	1349	124	42,36	383	38 days 11:00:25	1889 days 00:00:00
Municipality log 4 (BPI Challenge 2015)	1053	1049	116	44,91	356	96 days 09:51:19	1933 days 00:00:00
Municipality log 5 (BPI Challenge 2015)	1156	1153	154	51,11	389	77 days 04:09:48	1926 days 15:31:43
Change log (BPI Challenge 2014)	18000	872	1007	1,68	240	0 days 00:00:00	181 days 17:06:00
Incident activity log (BPI Challenge 2014)	46616	31725	178	10,01	39	0 days 18:21:00	450 days 11:51:00
Incident management log (BPI Challenge 2013)	7554	1511	123	8,68	4	7 days 13:10:44	783 days 08:22:43
Problem management log, open problems (BPI Challenge 2013)	819	108	22	2,87	3	1 days 17:32:18	2047 days 02:19:20
Problem management log, closed problems (BPI Challenge 2013)	1487	183	35	4,48	4	81 days 23:31:58	2332 days 07:59:24
Event log of a loan application process (BPI Challenge 2012)	13087	4366	175	20,04	24	0 days 19:24:17	165 days 15:26:10
Anonymized event log of a Dutch Academic Hospital (BPI Challenge 2011)	1143	981	1814	131,49	624	332 days 23:00:00	1172 days 00:00:00
Hospital Billing - Event Log (Mannhardt)	100000	1020	217	4,51	18	102 days 08:41:21	1131 days 22:45:38

Table 3.2 Dataset details and characteristics

These datasets cover both synthetic event logs and real-world event logs from a wide range of industry settings, including public sector governance (Boudewijn van Dongen & F. (Florian) Borchert, 2018), municipal (Boudewijn van Dongen, 2015a, 2015b, 2015c, 2015d, 2015e), manufacturing IT (Ward Steeman, 2013a, 2013b, 2013c), banking (Boudewijn van Dongen, 2017), or hospital billing processes (Boudewijn van Dongen, 2011; Mannhardt, 2017). Additionally, the datasets cover a wide range of different sizes, degrees of uniqueness, and overall variation across events, event types, sequences, and other properties. While some datasets are comprised of only a small share of unique traces (Boudewijn van Dongen, 2019), others consist overwhelmingly of unique traces (Boudewijn van Dongen, 2015a, 2015b, 2015c, 2015d, 2015e). These key characteristics of the individual datasets are available in table 3.2.

For the import, handling, manipulation, and analysis of the datasets, the PM4PY library in Python is used, as it provides comprehensive capabilities and pre-built functions to do so. PM4PY was developed with the goal of supporting and enabling process mining in Python by the Fraunhofer Institute for Applied Information Technology (Berti et al., 2023). The basic functionalities of the PM4PY library, and the functions implemented by Vidgof (2024), yield first insights into the characteristics of each dataset. This includes the maximum length of traces, the number of unique events and traces, as well as the maximum, minimum, and median duration of traces in each log.

3.3 Complexity Metrics Implementation and Measurement

Building on prior work (Augusto et al., 2019; Augusto et al., 2022; Vidgof & Mendling, 2023; Vidgof et al., 2023), the thesis extends existing approaches to process complexity measurement by incorporating additional metrics and systematically analysing them across a diverse set of event logs. All complexity metrics are implemented, measured, and evaluated in Python, a high-level, object-oriented programming language commonly used for data mining, data science, or process mining. To ascertain the validity and computational correctness of the complexity metric implementation, established baseline functionalities provided by Vidgof (2024) are adapted and extended, leveraging the established process mining processing standards and functions of the PM4PY library for data ingestion and transformation (Berti et al., 2023). For the purpose of this paper, the focus lies on a detailed analysis on a suitable selection of metrics that capture key dimensions of process complexity and enable quantitative comparisons.

Category	Process Complexity Metric	Label	Source
Entropy	Variant Entropy	var-e	(Augusto et al. 2022)
	Sequence Entropy	seq-e	(Augusto et al., 2022)
	Normalized Variant Entropy	nvar-e	(Augusto et al., 2022)
	Normalized Sequence Entropy	nseq-e	(Augusto et al., 2022)
Enriched Entropy	Enriched Variant Entropy	en-var-e	(Vidgof & Mendling, 2023)
	Enriched Sequence Entropy	en-seq-e	(Vidgof & Mendling, 2023)
	Enriched Normalized Variant Entropy	en-nvar-e	(Vidgof & Mendling, 2023)
	Enriched Normalized Sequence Entropy	en-nseq-e	(Vidgof & Mendling, 2023)
Size	Number of Events	#-e	(Günther, 2009)
	Number of Event Types	#-et	(Günther, 2009)
	Number of Sequences	#-seq	(Günther, 2009)
	Minimum Sequence Length	min-seq-len	(W. van der Aalst, 2016)
	Average Sequence Length	avg-seq-len	(W. van der Aalst, 2016)
	Maximum Sequence Length	max-seq-len	(W. van der Aalst, 2016)
	Average Time Difference between Consecutive Events	avg-td-e	(Günther, 2009)
Variation	Number of Acyclic Paths in Transition Matrix	#-acyclic-paths	(Hærem et al., 2015)
	Number of Ties in Transition Matrix	#-ties	(Hærem et al., 2015)
	Lempel–Ziv Complexity	lempel-ziv	(B. T. Pentland, 2003)
	Number of Unique Sequences	#-unique-seq	(W. van der Aalst, 2016)
	Percentage of Unique Sequences	perc-unique-seq	(W. van der Aalst, 2016)
	Average Number of Distinct Events per Sequence	avg-distinct-e	(Günther, 2009)
	Order Variation	order var	(Lindberg et al., 2016)
	Activity Variation	activity-var	(Lindberg et al., 2016)
	Probability of Action Pairs	prob-act-pairs	(Grisold et al., 2022)
Distance	Average Affinity	affinity	(Günther, 2009)
	Structure	structure	(Günther, 2009)
	Deviation from Random	dev-random	(B. T. Pentland, 2003)
	Average Edit Distance	avg-edit-distance	(B. T. Pentland, 2003)
	Structural Process Variety	structural-var	(Schreiber & Abbad-Andaloussi, 2024)

Table 3.3 Process complexity metrics in scope of thesis

These metrics are presented in table 3.3. Metrics are calculated log-wise and stored in a pandas data-frame in the same order in which the logs are imported. For metrics based on Levenshtein distance matrices, the event logs have been downsampled to 50,000 traces, to accommodate hardware restrictions. Some process complexity metrics from recent academic contributions are not in scope of the thesis due to technical limitations or methodological considerations. The repertoire and routinization metrics introduced by Lindberg et al. (2016) both require pattern mining using the VMSP algorithm, which does not produce functional results for all datasets in scope of the thesis, thus limiting the comparability and transferability of the results across datasets and evaluation methods. The control-flow log distance and the n-gram distance, introduced by Chapela-Campa et al. (2023), on the other hand only provide relative measures of complexity, requiring at

least two event logs or sub-logs of event logs to be applicable. Since this does not yield singular results per event log, these two metrics similarly cannot be compared to other, singular measurement results, thereby limiting their applicability, particularly for the correlation analysis. Finally, the probability of actions pairs, proposed by Grisold et al. (2022), provides transition matrices for all event types in a given event log, which are then compared to derive a relative complexity measurement. In addition to the relative nature of the score, the transition matrices themselves are well-suited for qualitative analyses of changes in the event transitions, but ill-suited for automated, quantitative evaluation, once again limiting the applicability of this metric for the evaluation in scope of this thesis. Despite these issues, all five metrics are promising and intriguing candidates for research into process complexity and process dynamics. This selection of process complexity metrics, and their measurement results, form the basis for the correlation analysis.

3.4 Correlation Analysis Implementation

To investigate the relationships between different process complexity metrics, the Pearson correlation (Pearson, 1896) and the Spearman rank correlation coefficient (Spearman, 1904) are calculated on the timeseries complexity measurement results across all datasets. This timeseries-based approach generates significantly more data points than cross-log correlations alone, enabling a more fine-grained and robust analysis of metric interrelationships, thus increasing statistical power of the correlation analysis and narrowing the confidence interval, thereby increasing the validity of the correlation analysis through increasing the probability that correlation values are statistically significant (Cohen, 1969). The systematic calculation of correlation coefficients at a sufficiently low significance level, termed the p-value, will quantify the relationships between the complexity metrics.

To address potential issues caused by outliers and non-normal distributed event log dataset measurement results, the Pearson correlation that is commonly used in research, for example in process mining (Augusto et al., 2022), and requires normally distributed data and linear relationships between data (Pearson, 1896) is complemented with the Spearman correlation. The latter does not require normally distributed data and is less sensitive to outliers, while also being applicable to non-linear relationships between data (Spearman, 1904; Winter et al., 2016).

To interpret the results of the correlation analysis, the commonly used thresholds for the probability value, termed the p-value, are used. These thresholds for the p-value at 0.01 and 0.05 are commonly used in process science (Augusto et al., 2022; Vidgof et al., 2023) and in statistics (Cohen, 1969). Additionally, the resulting statistically significant

correlations can be grouped, following the convention set out by Cohen (1963). Strong correlations are marked by a correlation coefficient whose absolute value is above 0.5, moderate correlations by a correlation coefficient whose absolute value is between 0.3 and 0.5, weak correlations by a correlation coefficient whose absolute value is between 0.1 and 0.3, and neutral correlations possess a correlation coefficient with an absolute value below 0.1.

3.5 Temporal Dynamics Analysis

To analyse process dynamics over time, a sliding window approach is applied to partition each event log into a sequence of temporally ordered sublogs, employing the strategy of temporal bracketing to decompose the timeframe into successive and distinct periods to ensure accurate analysis results (Langley, 1999). By computing the process complexity metrics for each sublog in the series, a timeseries of process complexity measurements is generated, which, in turn, provide insights into the dynamics of the process complexity metrics over time (Beerepoot et al., 2023), representing longitudinal data that enables analyses of process dynamics over time and of the dynamic emergence of structures and patterns (Langley et al., 2013). These dynamics enable an empirical analysis of previous theories about process drift in processes (B. T. Pentland, Liu, et al., 2020; B. T. Pentland et al., 2021). To this end, the time series measurements of the process complexity measurements are evaluated systematically. To identify patterns and structures in these dynamics, the time measurements are plotted against the earliest trace start date of the respective sublog, thereby producing a curve of the process complexity value of the metric over time. However, these curves must be plotted separately for each dataset, as each dataset has a distinct trace distribution and observation period. Additionally, plotting curves for all datasets in parallel yields highly complex and confusing figures that impede the analysis and clear identification of patterns.

Accordingly, the temporal dynamics analysis requires an initial pre-selection of metrics and datasets for a more in-depth analysis, which then serve as narrative examples that enable an exhaustive analysis of all findings. For this pre-selection, multiple aggregate statistics and tests are performed, to determine underlying variation characteristics and trends across all datasets and metrics. Firstly, the spectral entropy is calculated, which measures the frequency distribution complexity of the time series, thereby providing information about the fuzziness of the complexity curves (Powell & Percival, 1979). Secondly, the permutation entropy is calculated, which measures the local temporal order complexity of the time series, thereby providing information about the predictability of the pattern of complexity curve (Bandt et al., 2002; Bandt & Pompe, 2002). Thirdly, the volatility index is calculated, which measures the scale-free change magnitude of the time

series, thereby providing information about the volatility of the of the complexity metrics (Mandelbrot, 1963). Finally, the Mann-Kendall test is performed for each combination of dataset and complexity metric, which detects monotonic trends and is a non-parametric test, meaning that it can be applied to all distributions (Mann, 1945). This test yields a trend strength and a significance value, termed p-value, which is used to ascertain that only highly reliable trend estimates are considered. Based on these four methods, two datasets are chosen, for which a subset of metrics is analysed in detail in the form of narrative examples, highlighting patterns that are present across all metrics and datasets in scope.

3.6 Process Drift Simulation Implementation

For the process drift simulation, prior work by Pentland et al. (2020) is extended to real-world event log data, simulating process drift through variation across process iterations. The lexicon of event types in the respective event log dataset is extracted and used for all iterations of the process simulation. To derive the initial transition matrix and identify start and end event types (including their probabilities of serving as start or end events), the last window of event traces in the log is used. The window size is a simulation parameter and is consistently applied during the simulation. In each iteration, the selected window is used to calculate the transition probability matrix. During each iteration, the start event is sampled from the start events in the event log, with the chance weighted by the relative share of occurrences of the respective start event in the event log. The subsequent events are chosen randomly from the transition matrix from the previous iteration step, based on the transition probabilities. To simulate process drift, there is a chance (determined by a configurable variation factor) that the next event is chosen entirely at random from all event types in the log. Analogously to the choice of start events, each end event initially derived from the starting window has a weighted chance to act as the end of the simulated event trace, when it is the latest simulated event in the simulated event trace. If no end event triggers the end of the event trace, the maximum trace length of the underlying event log dataset acts as a length guard, with each simulation iteration being stopped when reaching that maximum trace length. The simulation was run for 10,000 iterations, with a window size of 200 traces, and a variation factor of 0.01. Based on the four aggregate statistics and trend tests established for the temporal dynamics analysis, two datasets are chosen, for which a subset of metrics is analysed in detail in the form of narrative examples, highlighting patterns that are present across all metrics and datasets in scope.

4 Findings

The findings are structured along the three research questions of the thesis. Section 4.1 presents the results of the correlation analysis, providing insights into the interrelationships among the selected process complexity metrics. Section 4.2 focuses on the temporal dynamics of these metrics, examining how they evolve over time. Finally, Section 4.3 explores how the metrics respond to process drift, analysing their sensitivity to incremental process change.

4.1 Complexity Correlation Analysis

As a first step, based on the principles set out in the methodology of this thesis and utilizing the implementation of the process complexity metrics, the individual complexity metrics are calculated for all datasets selected for this thesis. The results of the correlation analysis are presented across all datasets. The results of the correlation analysis, for both the Pearson and the Spearman correlation, represent aggregated statistics, yielding a single correlation value for each dataset and metric combination. The results are thus displayed in tables 4.1, 4.2, 4.3, 4.4, 4.5, and 4.6. Correlations are calculated for p-values of 0.05 and 0.01. Importantly, for all tables containing correlation coefficients as part of the correlation analysis, correlations with a p-value above or equal to the threshold of 0.05 are highlighted in bold and those with a p-value above or equal to 0.01 but below 0.05 underscored, thereby ensuring highly significant results. As such, the plain correlation coefficients without bold highlighting or underscoring represent the most statistically significant results, with a p-value below the 0.01 threshold.

Both correlations yield similar results given the total number of correlation observations ($n=378$), identifying 54 (Pearson) and 16 (Spearman) non-significant pairs of correlation measures with a p-value equal to or above the 0.01 threshold. Additionally, the correlations identify 46 (Pearson) and 15 (Spearman) non-significant pairs of correlation measures with a p-value equal to or above the 0.05 threshold. More Pearson than Spearman correlations are not statistically significant, highlighting that the Pearson correlation is looser than the Spearman correlation. Given the total number of correlation observations and pairs, the majority of correlations are statistically significant. The correlations can be grouped, following the convention set out by Cohen (1963). Strong correlations are marked by a correlation coefficient whose absolute value is above 0.5, moderate correlations by a correlation coefficient whose absolute value is between 0.3 and 0.5, weak correlations by a correlation coefficient whose absolute value is between 0.1 and 0.3, and neutral correlations possess a correlation coefficient with an absolute value below 0.1, thereby representing uncorrelated variables.

The following results will solely focus on the Pearson correlation coefficients with a p-value below 0.01, which leads to a minor reduction in overall observations, but further improves the statistical power and validity of the findings.

	var-e	nvar-e	seq-e	nseq-e	en-var-e	en-nvar-e	en-seq-e	en-nseq-e	#-ties
nvar-e	0.22								
seq-e	0.98	0.26							
nseq-e	0.23	0.70	0.23						
en-var-e	0.91	0.30	0.97	0.22					
en-nvar-e	0.18	0.95	0.24	0.56	0.30				
en-seq-e	0.91	0.31	0.97	0.22	1.00	0.30			
en-nseq-e	0.19	0.95	0.26	0.61	0.31	0.99	0.32		
#-ties	0.98	0.27	0.99	0.26	0.94	0.23	0.93	0.24	
lempel-ziv	0.87	0.40	0.93	0.31	0.95	0.37	0.95	0.39	0.93
#-unique-seq	0.73	0.40	0.78	0.31	0.81	0.37	0.81	0.39	0.78
perc-unique-seq	-0.08	-0.64	-0.16	0.00	-0.24	-0.80	-0.25	-0.77	-0.12
#-e	0.91	0.33	0.98	0.23	1.00	0.31	1.00	0.33	0.95
#-et	-0.07	-0.04	-0.09	0.35	-0.10	-0.15	-0.11	-0.12	-0.08
#-seq	0.31	0.37	0.47	0.11	0.65	0.44	0.65	0.45	0.37
min-seq-len	0.00	-0.29	-0.01	-0.18	<u>-0.03</u>	-0.31	<u>-0.04</u>	-0.31	-0.01
max-seq-len	0.32	0.00	0.35	0.08	0.36	-0.01	0.36	0.01	0.34
avg-seq-len	0.01	-0.32	-0.02	-0.09	-0.05	-0.37	-0.05	-0.37	0.00
avg-id-e	<u>-0.03</u>	-0.08	<u>-0.03</u>	-0.15	-0.04	-0.04	-0.04	-0.05	-0.04
dev_random	-0.01	-0.17	-0.05	0.26	-0.09	-0.29	-0.09	-0.27	-0.03
avg-edit-distance	-0.04	-0.24	-0.05	-0.20	-0.06	-0.24	-0.06	-0.24	-0.05
struct-var	0.97	0.29	0.96	0.29	0.91	0.25	0.91	0.26	0.98
order_var	-0.02	-0.19	-0.06	0.09	-0.10	-0.30	-0.10	-0.29	-0.05
activity_var	-0.07	-0.15	-0.10	0.27	-0.13	-0.27	-0.14	-0.25	-0.08
#-acyclic-paths	0.00	0.01	0.00	0.02	0.00	0.01	0.00	0.01	0.00
avg-distinct-e	-0.11	-0.33	-0.15	0.10	-0.19	-0.45	-0.19	-0.44	-0.13
structure	-0.06	-0.12	-0.06	0.14	-0.06	-0.20	-0.06	-0.18	-0.06
affinity	0.07	0.33	0.08	0.01	0.07	0.44	0.08	0.40	0.08

Table 4.1 Pearsson correlations of complexity metrics across all datasets – part 1.

Starting with the total numbers of statistically significant correlations, there are 71 correlation pairs with strong correlations, 67 pairs with moderate correlations, 108 pairs with weak correlations, and 78 total pairs with neutral or insignificant correlations. While the strongest positive correlations reach near perfect correlation coefficients at 1.00 or 0.99, for example between the number of events (“#-e”) and the enriched variant entropy (“en-var-e”) or the enriched sequence entropy (“en-seq-e”), the strongest negative correlations reach strong correlation coefficients at -0.80, between the percentage of unique sequences (“perc-unique-seq”) and the enriched normalized variant entropy (“en-nvar-e”).

There are multiple strong positive correlations between the entropy-based correlation metrics, with a distinct split between the normalized and the non-normalized metrics. The variant entropy (“var-e”) is strongly correlated with the sequence entropy (“seq-e”), with a correlation coefficient of 0.98, and with the enriched variant entropy (“en-var-e”) and the enriched sequence entropy (“en-seq-e”), both with the correlation coefficient 0.91. On

the other hand, there are only weak or neutral correlations between the non-normalized entropy metrics and the normalized entropy metrics, such as seen between the variant entropy (“var-e”) and the normalized variant entropy (“nvar-e”) with a correlation coefficient of 0.22, or the normalized sequence entropy (“nseq-e”) with a correlation coefficient of 0.23, with lower correlation coefficients for the enriched normalized metrics. The normalized metrics themselves are strongly correlated with each other. For example, the normalized variant entropy (“nvar-e”) and the normalized sequence entropy (“nseq-e”) exhibit a correlation coefficient of 0.71, while the correlation coefficient between the normalized variant entropy (“nvar-e”) and both enriched normalized entropy metrics (“en-nvar-e”, “en-nseq-e”) is 0.95, showing a near-perfectly correlated relationship between these metrics.

Other notable strong positive correlations can be seen between the variant entropy (“var-e”) and the number of ties (“#-ties”) with a correlation coefficient of 0.98, the lempel-ziv complexity (“lempel-ziv”) with a correlation coefficient of 0.87, the number of unique sequences (“#-unique-seq”) with a correlation coefficient of 0.73, the number of events (“#-e”) with a correlation coefficient of 0.91, and the structural variety (“struct-var”) with a correlation coefficient of 0.97. The same pattern of relationships can be seen for the sequence entropy (“seq-e”), in line with the strong correlation between both non-normalized entropy metrics. The enriched entropy metrics exhibit similar correlation relationships to the variant entropy and the sequence entropy, with the addition of a strong positive correlation between the enriched variant entropy (“en-var-e”) and the number of sequences (“#-seq”) with a correlation coefficient of 0.65, and between the enriched sequence entropy (“en-seq-e”) and the number of sequences (“#-seq”) with a correlation coefficient of 0.65.

Apart from the strong positive correlations, there are also strong negative correlations between some of entropy metrics and other complexity metrics. The normalized variant entropy (“nvar-e”) is strongly negatively correlated with the percentage of unique sequences (“perc-unique-seq”) with a correlation coefficient of -0.64, and the same is true for the enriched normalized variant entropy (“en-nvar-e”) with a correlation coefficient of -0.80.

In addition to the previously mentioned correlations with the entropy metrics, the number of ties (“#-ties”) exhibits strong positive correlations with the lempel-ziv complexity (“lempel-ziv”) with a correlation coefficient of 0.93, with the number of unique sequences (“#-unique-seq”) with a correlation coefficient of 0.78, with the number of events (“#-e”) with a correlation coefficient of 0.95, and with the structural variety (“struct-var”) with a correlation coefficient of 0.98. Similarly, the lempel-ziv complexity (“lempel-ziv”) exhibits strong positive correlations with the number of ties (“#-ties”) with a correlation coefficient of 0.93, with the number of unique sequences (“#-unique-seq”) with a correlation coefficient of 0.78, with the number of events (“#-e”) with a correlation coefficient of 0.95, and with the structural variety (“struct-var”) with a correlation coefficient of 0.98.

exhibits strong positive correlations with the number of unique sequences (“#-unique-seq”) with a correlation coefficient of 0.90, the number of events (“#-e”) with a correlation coefficient of 0.96, the number of sequences (“#-seq”) with a correlation coefficient of 0.60, and the structural variety (“struct-var”) with a correlation coefficient of 0.92.

	lempel- ziv	#-unique- seq	perc- unique-seq	#-e	#-et	#-seq	min-seq- len	max-seq- len	avg-seq- len
#-unique-seq	0.90								
perc-unique-seq	-0.26	-0.25							
#-e	0.96	0.82	-0.25						
#-et	-0.09	-0.11	0.41	-0.10					
#-seq	0.60	0.64	-0.49	0.63	-0.12				
min-seq-len	-0.06	-0.07	0.24	-0.04	0.06	-0.11			
max-seq-len	0.37	0.26	0.00	0.38	0.32	0.23	0.07		
avg-seq-len	-0.04	-0.10	0.37	-0.05	0.42	-0.16	0.65	0.57	
avg-td-e	-0.06	-0.05	-0.04	-0.04	-0.22	-0.04	-0.08	-0.09	-0.14
dev_random	-0.08	-0.11	0.50	-0.09	0.68	-0.20	0.20	0.11	0.36
avg-edit-distance	-0.07	-0.07	0.14	-0.06	0.04	-0.07	0.08	0.23	0.41
struct-var	0.92	0.83	-0.12	0.92	-0.03	0.40	-0.03	0.39	0.02
order_var	-0.13	-0.18	0.44	-0.10	0.51	-0.22	0.23	0.02	0.31
activity_var	-0.13	-0.16	0.50	-0.14	0.82	-0.21	0.20	0.10	0.37
#-acyclic-paths	0.00	0.00	0.01	0.00	0.05	0.00	0.00	0.00	0.00
avg-distinct-e	-0.21	-0.24	0.62	-0.19	0.73	-0.29	0.31	0.04	0.42
structure	-0.06	-0.11	0.32	-0.06	0.68	-0.06	0.16	0.15	0.32
affinity	0.06	-0.01	-0.50	0.08	-0.46	0.01	-0.15	-0.13	-0.27

Table 4.2 Pearsson correlations of complexity metrics across all datasets – part 2.

The number of unique sequences (“#-unique-seq”) is strongly positively correlated with the number of events (“#-e”) with a correlation coefficient of 0.82, the number of sequences (“#-seq”) with a correlation coefficient of 0.64, and the structural variety (“struct-var”) with a correlation coefficient of 0.83.

The percentage of unique sequences (“perc-unique-seq”) on the other hand exhibits strong positive correlations with the deviation from random (“dev_random”) with a correlation coefficient of 0.50, the activity variation with a correlation coefficient of 0.50, and the average number of distinct events (“avg-distinct-e”) with a correlation coefficient of 0.62. It is noteworthy that the first two relationships are right at the threshold for the correlation coefficient to be considered strong correlations. Additionally, there is a strong negative correlation between the percentage of unique sequences (“perc-unique-seq”) and the affinity (“affinity”), with a correlation coefficient of -0.50, also right at the threshold for strong correlations.

In addition to the correlations highlighted before, the number of events (“#-e”) is strongly positively correlated with the number of sequences (“#-seq”) with a correlation coefficient of 0.62, and strongly positively correlated with the structural variety (“struct-var”) with a correlation coefficient of 0.92.

The number of event types in turn is strongly positively correlated with the deviation from random (“dev_random”) with a correlation coefficient of 0.68, the order variation (“order_var”) with a correlation coefficient of 0.51, the activity variation (activity_var”) with a correlation coefficient of 0.82, the average number of distinct events (“avg-distinct-e”) with a correlation coefficient of 0.73, and the structure (“structure”) with a correlation coefficient of 0.68.

Except for the previously mentioned correlations, the number of sequences (“#-seq”) does not exhibit any other strongly correlations with the remaining complexity metrics. The minimum sequence length (“min-seq-len”) is only strongly positively correlated with the average sequence length (“avg-seq-len”), with a correlation coefficient of 0.65. The maximum sequence length (“max-seq-len”) is likewise strongly positively correlated with the average sequence length (“avg-seq-len”), with a correlation coefficient of 0.57. The average sequence length (“avg-seq-len”), in turn, does not exhibit strong correlations with any other complexity metrics outside those previously mentioned.

	avg-td-e	dev_random	avg-edit-distance	struct-var	order_var	activity_var	#-acyclic-paths	avg-distinct-e	structure
dev_random	-0.08								
avg-edit-distance	-0.07	-0.07							
struct-var	0.92	0.83	-0.12						
order_var	-0.13	-0.18	0.44	-0.10					
activity_var	-0.13	-0.16	0.50	-0.14	0.82				
#-acyclic-paths	0.00	0.00	0.01	0.00	0.05	0.00			
avg-distinct-e	-0.21	-0.24	0.62	-0.19	0.73	-0.29	0.31		
structure	-0.06	-0.11	0.32	-0.06	0.68	-0.06	0.16	0.15	
affinity	0.06	-0.01	-0.50	0.08	-0.46	0.01	-0.15	-0.13	-0.27

Table 4.3 Pearsson correlations of complexity metrics across all datasets – part 3.

The average time difference between events (“avg-td-e”) is strongly positively correlated with the structural variety (“struct-var”), with a correlation coefficient of 0.92, while the deviation from random (“dev_random”) is likewise strongly positively correlated with the structural variety (“struct-var”), with a correlation coefficient of 0.83.

Additionally, the average edit distance (“avg-edit-distance”) is strongly positively correlated with the activity variation (“activity_var”) with a correlation coefficient of 0.50, right at the threshold, and the avg distinct events (“avg-distinct-e”), with a correlation coefficient of 0.62. The average edit distance (“avg-edit-distance”) is also strongly negatively correlated with the affinity (“affinity”), with a correlation coefficient of -0.50, again right at the threshold to be considered a strong correlation.

Finally, the order variation (“order_var”) exhibits strong positive correlations with the activity variation (“activity_var”) with a correlation coefficient of 0.82, the average number of distinct events (“avg-distinct-e”) with a correlation coefficient of 0.73, and the

structure (“structure”) with a correlation coefficient of 0.68. The remaining metrics exhibit no strong correlations, neither positive nor negative.

In addition to strong correlations, neutral or insignificant correlations are also of interest to understand and measure relationships between complexity metrics. The variant entropy (“var-e”) exhibits a neutral correlation with the percentage of unique sequences (“perc-unique-seq”) with a correlation coefficient of -0.08, the number of event types (“#-et”) with a correlation coefficient of -0.07, the average edit distance (“avg-edit-distance”) with a correlation coefficient of -0.04, the activity variation (“activity_var”) with a correlation coefficient of -0.07, the average number of distinct events (“avg-distinct-e”) with a correlation coefficient of -0.11, the structure (“structure”) with a correlation coefficient of -0.06, and the affinity (“affinity”) with a correlation coefficient of 0.07. As before, the sequence entropy (“seq-e”) exhibits similar patterns, with the addition of neutral correlations with the deviation from random (“dev_random”) with a correlation coefficient of -0.05, and the order variation (“order_var”) with a correlation coefficient of -0.06. The enriched entropy metrics exhibit similar neutral correlations to the sequence entropy, while the four normalized entropy metrics exhibit neutral or near-neutral correlations to the non-normalized entropy metrics, as stated before.

The number of ties (“#-ties”) exhibits neutral correlations with the number of event types (“#-et”) with a correlation coefficient of -0.08, the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.04, the average edit distance (“avg-edit-distance”) with a correlation coefficient of -0.05, the order variation (“order_var”) with a correlation coefficient of -0.05, the activity variation (“activity_var”) with a correlation coefficient of -0.08, the structure (“structure”) with a correlation coefficient of -0.06, and the affinity (“affinity”) with a correlation coefficient of 0.08.

Furthermore, the lempel-ziv complexity (“lempel-ziv”) is neutrally correlated with the number of event types (“#-et”) with a correlation coefficient of -0.09, the minimum sequence length (“min-seq-len”) with a correlation coefficient of -0.06, the average sequence length (“avg-seq-len”) with a correlation coefficient of -0.04, the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.06, the deviation from random (“dev_random”) with a correlation coefficient of -0.08, the average edit distance (“avg-edit-distance”) with a correlation coefficient of -0.07, the structure (“structure”) with a correlation coefficient of -0.06, and the affinity (“affinity”) with a correlation coefficient of 0.06.

The number of unique sequences (“#-unique-seq”) is likewise neutrally correlated with the minimal sequence length (“min-seq-len”) with a correlation coefficient of -0.07, and also neutrally correlated with the average time difference between events (“avg-td-e”)

with a correlation coefficient of -0.05, and the average edit distance (“avg-edit-distance”) with a correlation coefficient of -0.07. In contrast, the percentage of unique sequences (“perc-unique-seq”) is only neutrally correlated with the average time difference between events (“avg-td-e”), with a correlation coefficient of -0.04.

Moving further across the correlation table, the number of events (“#-e”) is neutrally correlated with the average sequence length (“avg-seq-len”) with a correlation coefficient of -0.05, the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.04, the deviation from random (“dev_random”) with a correlation coefficient of -0.09, the average edit distance (“avg-edit-distance”) with a correlation coefficient of -0.06, the structure (“structure”) with a correlation coefficient of -0.06, and the affinity (“affinity”) with a correlation coefficient of 0.08.

In turn, the number of event types, in addition to the correlations highlighted previously, is neutrally correlated with the minimum sequence length (“min-seq-len”) with a correlation coefficient of 0.06, the average edit distance (“avg-edit-distance”) with a correlation coefficient of 0.04, and the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.05. The number of sequences (“#-seq”) is neutrally correlated with the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.04, the average edit distance (“avg-edit-distance”) with a correlation coefficient of -0.07, and the structure (“structure”) with a correlation coefficient of -0.06.

In regard to sequence length metrics, the minimum sequence length (“min-seq-len”) is neutrally correlated to the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.07, and the average time difference between events (“avg-td-e”) with a correlation coefficient of 0.08. The maximum sequence length (“max-seq-len”) is neutrally correlated to the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.09, and the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.04. The average sequence length does not exhibit any statistically significant neutral correlations where the p-value is below the threshold of 0.01.

In addition to the previously mentioned correlations, the average time difference between events (“avg-td-e”) is neutrally correlated with deviation from random (“dev_random”) with a correlation coefficient of -0.08, the average edit distance (“avg-edit-distance”) with a correlation coefficient of -0.07, the structure (“structure”) with a correlation coefficient of -0.06, and the affinity (“affinity”) with a correlation coefficient of 0.06.

Outside the previously raised correlations, the deviation from random (“dev_random”) is only neutrally correlated with the average edit distance (“avg-edit-distance”) with a

statistically significant correlation coefficient of -0.07. The average edit distance (“avg-edit-distance”) itself is not neutrally correlated with any additional metrics when focussing on statistically significant results with a p-value threshold of 0.01.

The structural variety (“struct-var”) is additionally neutrally correlated with the structure (“structure”) with a correlation coefficient of -0.06, and the affinity (“affinity”) with a correlation coefficient of 0.08. The order variation (“order_var”) is neutrally correlated with the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.05, and the activity variation (“activity_var”) is neutrally correlated with the structure with a correlation coefficient of -0.06. The remaining complexity metrics do not exhibit neutral correlations amongst each other.

As stated before, the Spearman method yields both a closer correlation with more statistically significant results, as well as slightly different correlation results for the individual pairs of correlation metrics, which is why these results are also examined in the following. Starting with the total numbers of statistically significant Spearman correlations, there are 176 correlation pairs with strong correlations, 71 pairs with moderate correlations, 88 pairs with weak correlations, and 27 total pairs with neutral or insignificant correlations. Overall, the Spearman correlations are significantly stronger, with more than 46% of the total 378 correlation observations falling into the group of strong correlations with a correlation coefficient above 0.5, compared to more than 18% or 71 observations for the Pearson correlation. While the strongest positive correlations reach near perfect correlation coefficients at 1.00 or 0.99, for example between the number of events (“#-e”) and the sequence entropy (“seq-e”) at 0.99, or the enriched normalized sequence entropy (“en-nseq-e”) and the enriched normalized sequence entropy (“en-nseq-e”) at 1.00, the strongest negative correlations reach strong correlation coefficients at -0.81, between the average number of distinct events (“avg-distinct-e”) and the average time difference between events (“avg-td-e”).

	var-e	nvar-e	seq-e	nseq-e	en-var-e	en-nvar-e	en-seq-e	en-nseq-e	#-ties
nvar-e	0.61								
seq-e	0.95	0.75							
nseq-e	0.72	0.64	0.62						
en-var-e	0.93	0.78	1.00	0.61					
en-nvar-e	0.53	0.96	0.69	0.57	0.73				
en-seq-e	0.92	0.78	0.99	0.6	1.00	0.73			
en-nseq-e	0.56	0.96	0.70	0.59	0.74	1.00	0.74		
#-ties	0.98	0.55	0.93	0.64	0.91	0.46	0.90	0.49	
lempel-ziv	0.94	0.65	0.98	0.54	0.97	0.58	0.96	0.59	0.94
#-unique-seq	0.87	0.88	0.95	0.69	0.96	0.83	0.97	0.85	0.83
perc-unique-seq	-0.23	-0.68	-0.45	-0.08	-0.50	-0.77	-0.50	-0.75	-0.2
#-e	0.93	0.70	0.99	0.54	0.98	0.64	0.98	0.65	0.94
#-et	0.54	-0.05	0.39	0.38	0.36	-0.12	0.35	-0.11	0.55
#-seq	0.79	0.91	0.90	0.61	0.93	0.90	0.93	0.90	0.74
min-seq-len	-0.14	-0.52	-0.30	-0.06	-0.34	-0.60	-0.35	-0.59	-0.09
max-seq-len	0.74	0.09	0.61	0.38	0.57	0.02	0.56	0.05	0.80
avg-seq-len	0.26	-0.43	0.08	0.05	0.03	-0.52	0.02	-0.51	0.34
avg-id-e	-0.17	0.42	0.02	-0.10	0.06	0.52	0.07	0.51	-0.22
dev_random	0.32	-0.19	0.15	0.29	0.11	-0.28	0.11	-0.27	0.32
avg-edit-distance	0.01	-0.33	-0.12	0.03	-0.15	-0.36	-0.15	-0.36	0.01
struct-var	0.97	0.59	0.93	0.68	0.91	0.52	0.91	0.55	0.97
order_var	0.12	-0.29	-0.01	0.08	-0.04	-0.37	-0.05	-0.37	0.15
activity_var	0.36	-0.14	0.21	0.31	0.17	-0.22	0.16	-0.21	0.36
#-acyclic-paths	0.83	0.29	0.71	0.56	0.68	0.19	0.67	0.22	0.84
avg-distinct-e	0.20	-0.33	0.05	0.10	0.01	-0.42	0.00	-0.41	0.24
structure	0.23	-0.26	0.11	0.12	0.08	-0.31	0.07	-0.30	0.25
affinity	0.01	0.43	0.14	0.09	0.18	0.49	0.19	0.47	-0.02

Table 4.4 Spearman correlations of complexity metrics across all datasets - part 1.

There are multiple strong positive correlations between the entropy-based correlation metrics. However, the split between the normalized and the non-normalized metrics is less significant than present within the Pearson correlations, with all entropy metrics exhibiting strong positive correlations with each other. The variant entropy (“var-e”) is strongly correlated with the sequence entropy (“seq-e”), with a correlation coefficient of 0.95, and with the enriched variant entropy (“en-var-e”) and the enriched sequence entropy (“en-seq-e”), both with the correlation coefficient 0.92 and 0.91, respectively. On the other hand, there are also moderately strong positive correlations between the non-normalized entropy metrics and the normalized entropy metrics, such as seen between the variant entropy (“var-e”) and the normalized variant entropy (“nvar-e”) with a correlation coefficient of 0.61, or the normalized sequence entropy (“nseq-e”) with a correlation coefficient of 0.72, with lower correlation coefficients for the enriched normalized metrics. The normalized metrics themselves are also strongly correlated with each other. For example, the normalized variant entropy (“nvar-e”) and the normalized sequence entropy (“nseq-e”) exhibit a correlation coefficient of 0.64, while the correlation coefficient between the normalized variant entropy (“nvar-e”) and both enriched normalized entropy metrics (“en-nvar-e”, “en-nseq-e”) is 0.96, showing a near-perfectly correlated relationship between these metrics.

Other notable strong positive correlations can be seen between the variant entropy (“var-e”) and the number of ties (“#-ties”) with a near perfect positive correlation coefficient of 0.98, the lempel-ziv complexity (“lempel-ziv”) with a correlation coefficient of 0.94, the number of unique sequences (“#-unique-seq”) with a correlation coefficient of 0.87, the number of events (“#-e”) with a correlation coefficient of 0.93, the number of event types (“#-et”) with a correlation coefficient of 0.54, the number of sequences (“#-seq”) with a correlation coefficient of 0.79, the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.74, the structural variety (“struct-var”) with a correlation coefficient of 0.97, and the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.83. The same pattern of relationships can be seen for the sequence entropy (“seq-e”), in line with the strong correlation between both non-normalized entropy metrics, except for the only moderately strong positive correlation between the sequence entropy and the number of event types (“#-et”) with a correlation coefficient of 0.39. The enriched entropy metrics exhibit similar correlation relationships to the sequence entropy.

Apart from the strong positive correlations, there are also strong negative correlations between some of entropy metrics and other complexity metrics. The normalized variant entropy (“nvar-e”) is strongly negatively correlated with the percentage of unique sequences (“perc-unique-seq”) with a correlation coefficient of -0.64, and the minimum sequence length (“min-seq-len”) with a correlation coefficient of -0.52. The same is true for the enriched normalized variant entropy (“en-nvar-e”) with a correlation coefficient of -0.72 and -0.60, respectively. Furthermore, the enriched variant entropy (“en-var-e”) is likewise strongly negatively correlated with the percentage of unique sequences (“perc-unique-seq”) with a correlation coefficient of -0.50, which is also true for the enriched sequence entropy (“en-seq-e”). The enriched normalized sequence entropy (“en-nseq-e”) exhibits the same patterns seen with the enriched normalized variant entropy (“en-nvar-e”).

In addition to the previously mentioned correlations with the entropy metrics, the number of ties (“#-ties”) exhibits strong positive correlations with the lempel-ziv complexity (“lempel-ziv”) with a correlation coefficient of 0.94, with the number of unique sequences (“#-unique-seq”) with a correlation coefficient of 0.83, with the number of events (“#-e”) with a correlation coefficient of 0.94, the number of event types (“#-et”) with a correlation coefficient of 0.55, the number of sequences (“#-seq”) with a correlation coefficient of 0.74, the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.80, the structural variety (“struct-var”) with a correlation coefficient of 0.97, and with the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.84.

Similarly, the lempel-ziv complexity (“lempel-ziv”) exhibits strong positive correlations with the number of unique sequences (“#-unique-seq”) with a correlation coefficient of 0.89, the number of events (“#-e”) with a correlation coefficient of 0.99, the number of sequences (“#-seq”) with a correlation coefficient of 0.82, the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.68, the structural variety (“struct-var”) with a correlation coefficient of 0.92, and the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.75.

	lempel- ziv	#-unique- seq	perc- unique-seq	#-e	#-et	#-seq	min-seq- len	max-seq- len	avg-seq- len
#-unique-seq	0.89								
perc-unique-seq	-0.37	-0.57							
#-e	0.99	0.92	-0.44						
#-et	0.48	0.23	0.42	0.40					
#-seq	0.82	0.98	-0.69	0.87	0.13				
min-seq-len	-0.21	-0.46	0.69	-0.28	0.35	-0.54			
max-seq-len	0.68	0.45	0.09	0.65	0.63	0.35	0.16		
avg-seq-len	0.19	-0.14	0.60	0.13	0.70	-0.25	0.68	0.65	
avg-id-e	-0.08	0.20	-0.70	-0.01	-0.68	0.31	-0.75	-0.45	-0.80
dev_random	0.23	0.00	0.58	0.15	0.82	-0.10	0.57	0.43	0.70
avg-edit-distance	-0.10	-0.23	0.49	-0.12	0.43	-0.28	0.35	0.23	0.52
struct-var	0.92	0.86	-0.25	0.92	0.53	0.78	-0.22	0.77	0.25
order_var	0.09	-0.17	0.60	0.01	0.69	-0.24	0.72	0.29	0.72
activity_var	0.30	0.05	0.52	0.21	0.90	-0.04	0.56	0.48	0.70
#-acyclic-paths	0.75	0.57	0.14	0.71	0.74	0.48	0.13	0.76	0.57
avg-distinct-e	0.16	-0.15	0.63	0.07	0.75	-0.24	0.75	0.43	0.84
structure	0.22	-0.04	0.47	0.13	0.88	-0.12	0.45	0.44	0.63
affinity	0.04	0.23	-0.48	0.11	-0.41	0.32	-0.18	-0.27	-0.34

Table 4.5 Spearman correlations of complexity metrics across all datasets - part 2.

The number of unique sequences (“#-unique-seq”) is strongly positively correlated with the number of events (“#-e”) with a correlation coefficient of 0.92, the number of sequences (“#-seq”) with a correlation coefficient of 0.98, the structural variety (“struct-var”) with a correlation coefficient of 0.86, and the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.57. Additionally, the number of unique sequences (“#-unique-seq”) is strongly negatively correlated with the percentage of unique sequences (“perc-unique-seq”) with a correlation coefficient of -0.57.

The percentage of unique sequences (“perc-unique-seq”) on the other hand exhibits strong positive correlations with the minimum sequence length (“min-seq-len”) with a correlation coefficient of 0.69, the average sequence length (“avg-seq-len”) with a correlation coefficient of 0.60, the deviation from random (“dev_random”) with a correlation coefficient of 0.58, the order variation (“order_var”) with a correlation coefficient of 0.60, the activity variation with a correlation coefficient of 0.50, and the average number of distinct events (“avg-distinct-e”) with a correlation coefficient of 0.63. Additionally, there is a strong negative correlation between the percentage of unique

sequences (“perc-unique-seq”) and the number of sequences (“#-seq”) with a correlation coefficient of -0.69, as well as with the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.70.

In addition to the correlations highlighted before, the number of events (“#-e”) is strongly positively correlated with the number of sequences (“#-seq”) with a correlation coefficient of 0.87, the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.65, the structural variety (“struct-var”) with a correlation coefficient of 0.92, and with the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.71.

The number of event types in turn is strongly positively correlated with the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.63, the average sequence length (“avg-seq-len”) with a correlation coefficient of 0.70, the deviation from random (“dev_random”) with a correlation coefficient of 0.82, the deviation from random (“dev_random”) with a correlation coefficient of 0.68, the structural variety (“struct-var”) with a correlation coefficient of 0.53, the order variation (“order_var”) with a correlation coefficient of 0.69, the activity variation (activity_var”) with a correlation coefficient of 0.90, the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.74, the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.75, and the structure (“structure”) with a correlation coefficient of 0.88.

Outside the previously mentioned correlations, the number of sequences (“#-seq”) exhibits a strongly negative correlation with the minimum sequence length (“min-seq-len”) with a correlation coefficient of -0.54, and a strongly positive correlation with the structural variety (“struct-var”) with a correlation coefficient of 0.78. The minimum sequence length (“min-seq-len”) is strongly positively correlated with the average sequence length (“avg-seq-len”) with a correlation coefficient of 0.65, the deviation from random (“dev_random”) with a correlation coefficient of 0.57, the order variation (“order_var”) with a correlation coefficient of 0.72, the activity variation (“activity_var”) with a correlation coefficient of 0.56, and the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.75. The minimum sequence length (“min-seq-len”) is also strongly negatively correlated with the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.75. The maximum sequence length (“max-seq-len”) is likewise strongly positively correlated with the average sequence length (“avg-seq-len”) with a correlation coefficient of 0.57, and with the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.76. The average sequence length (“avg-seq-len”) exhibits strong positive correlations with the deviation from random (“dev_random”) with a correlation coefficient of 0.70, the average

edit distance (“avg-edit-distance”) with a correlation coefficient of 0.52, the order variation (“order_var”) with a correlation coefficient of 0.72, the activity variation (“activity_var”) with a correlation coefficient of 0.70, the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.57, the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.84, and the structure (“structure”) with a correlation coefficient of 0.63. The average sequence length (“avg-seq-len”) also exhibits a strong negative correlation with the average time difference between events (“avg-td-e”), with a correlation coefficient of -0.80.

	avg-td-e	dev_random	avg-edit-distance	struct-var	order_var	activity_var	#-acyclic-paths	avg-distinct-e	structure
dev_random	-0.73								
avg-edit-distance	-0.49	0.48							
struct-var	-0.13	0.28	0.05						
order_var	-0.78	0.85	0.45	0.06					
activity_var	-0.75	0.92	0.47	0.32	0.88				
#-acyclic-paths	-0.46	0.62	0.26	0.82	0.45	0.59			
avg-distinct-e	-0.81	0.86	0.50	0.14	0.95	0.89	0.52		
structure	-0.69	0.72	0.42	0.21	0.70	0.87	0.40	0.73	
affinity	0.37	-0.37	-0.10	-0.04	-0.25	-0.37	-0.20	-0.27	-0.43

Table 4.6 Spearman correlations of complexity metrics across all datasets - part 3.

The average time difference between events (“avg-td-e”) is additionally strongly negatively correlated with the deviation from random (“dev_random”) with a correlation coefficient of -0.73, the order variation (“order_var”) with a correlation coefficient of -0.78, the activity variation (“activity_var”) with a correlation coefficient of -0.75, the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of -0.81, and the structure (“structure”) with a correlation coefficient of -0.69.

The deviation from random (“dev_random”) is likewise strongly positively correlated with the order variation (“order_var”) with a correlation coefficient of 0.85, the activity variation (“activity_var”) with a correlation coefficient of 0.92, the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.62, the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.86, and with the structure (“structure”) with a correlation coefficient of 0.72.

Additionally, the average edit distance (“avg-edit-distance”) is strongly positively correlated with average number of distinct events per trace (“avg-distinct-e”), with a correlation coefficient of 0.50, right at the threshold for strong correlations. The structural variety is exhibiting a strong positive correlation with the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.82.

Furthermore, the order variation (“order_var”) exhibits strong positive correlations with the activity variation (“activity_var”) with a correlation coefficient of 0.88, the average

number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.95, and the structure (“structure”) with a correlation coefficient of 0.70. Similarly, the activity variation also exhibits strong correlations with the number of acyclic paths (“#-acyclic-paths”) with a correlation coefficient of 0.59, the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.89, and with the structure (“structure”) with a correlation coefficient of 0.87, in addition to the correlations with other metrics mentioned beforehand.

Finally, the number of acyclic paths (“#-acyclic-paths”) is also strongly positively correlated with the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.52, and the average number of distinct events per trace (“avg-distinct-e”) is strongly positively correlated with the structure (“structure”) with a correlation coefficient of 0.73.

In addition to strong correlations, neutral or insignificant correlations are also of interest to understand and measure relationships between complexity metrics. Starting off, the variant entropy (“var-e”) exhibits no statistically significant neutral correlations with any other complexity metric. The normalized variant entropy (“nvar-e”) exhibits neutral correlations with number of event types (“#-et”) with a correlation coefficient of -0.05, and with the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.09. As before, the sequence entropy (“seq-e”) exhibits neutral correlations with the average sequence length (“avg-seq-len”) with a correlation coefficient of 0.08, and average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.05. The normalized sequence entropy (“nseq-e”) exhibits similar patterns, exhibiting neutral correlations with the percentage of unique sequence (“perc-unique-seq”) with a correlation coefficient of -0.08, the minimum sequence length (“min-seq-len”) with a correlation coefficient of -0.06, the average sequence length (“avg-seq-len”) with a correlation coefficient of 0.05, the order variation (“order_var”) with a correlation coefficient of 0.08, and the affinity (“affinity”) with a correlation coefficient of 0.09.

The enriched variant entropy (“en-var-e”) exhibits neutral correlations with the average time difference between events (“avg-td-e”) with a correlation coefficient of 0.06, the order variation (“order_var”) with a correlation coefficient of -0.04, and the structure (“structure”) with a correlation coefficient of 0.08. The enriched normalized variant entropy (“en-nvar-e”) exhibits no neutral correlations, while the enriched sequence entropy (“en-seq-e”) exhibits the same neutral correlation pattern as seen with the enriched variant entropy (“en-var-e”). The enriched normalized sequence entropy (“en-nseq-e”) in turn only exhibits a neutral correlation with the maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.05. The number of ties (“#-ties”)

exhibits neutral correlations with the minimum sequence length (“min-seq-len”) with a correlation coefficient of -0.09.

Furthermore, the lempel-ziv complexity (“lempel-ziv”) is neutrally correlated with the average time difference between events (“avg-td-e”) with a correlation coefficient of -0.08, the order variation (“order_var”) with a correlation coefficient of 0.09, and the affinity (“affinity”) with a correlation coefficient of 0.04.

The number of unique sequences (“#-unique-seq”) is likewise neutrally correlated with the activity variation (“activity_var”) with a correlation coefficient of 0.05, and also with the structure (“structure”) with a correlation coefficient of -0.04. The percentage of unique sequences (“perc-unique-seq”) is neutrally correlated with maximum sequence length (“max-seq-len”) with a correlation coefficient of 0.09.

Moving further across the correlation table, the number of events (“#-e”) is statistically significantly neutrally correlated with the average number of distinct events per trace (“avg-distinct-e”) with a correlation coefficient of 0.07. The number of event types (“#-et”), in addition to the correlations highlighted previously, is not neutrally correlated with any other complexity metric. The number of sequences (“#-seq”) is additionally neutrally correlated with the activity variation (“activity_var”) with a correlation coefficient of -0.04.

In regard to sequence length metrics, the minimum sequence length (“min-seq-len”) is neutrally correlated to no additional complexity metrics. The same is true for the maximum sequence length (“max-seq-len”) and the average sequence length (“avg-seq-len”), with all three metrics not exhibiting any additional statistically significant neutral correlations where the p-value is below the threshold of 0.01.

In addition to the previously mentioned correlations, the average time difference between events (“avg-td-e”) is neutrally correlated with no other metrics. The same is true for the deviation from random (“dev_random”). The average edit distance (“avg-edit-distance”) is also neutrally correlated with the structural variety (“struct-var”) with a correlation coefficient of 0.05.

In turn, the structural variety (“struct-var”) is additionally neutrally correlated with the order variation (“order_var”) with a correlation coefficient of 0.06, and the affinity (“affinity”) with a correlation coefficient of -0.04. The order variation (“order_var”) is not neutrally correlated with additional complexity metrics, and the same is true for the activity variation (“activity_var”), as well as for the remaining complexity metrics.

4.2 Temporal Dynamics Analysis

4.2.1 Temporal Dynamics Analysis Across all Datasets

The temporal dynamics across all datasets and metrics require a statistical representation to reduce the dimensionality of the data and provide insights into the underlying dynamics. Due to the number of datasets and metrics in scope, plotting and comparing each combination of dataset and metric is not possible in the scope of this thesis. Instead, narrative examples of the key temporal dynamics are presented and analysed. To this end, the spectral entropy, the permutation entropy, and the volatility index, as calculated for each combination of dataset and complexity metric, for the timeseries created with the sliding window approach, are presented in figures 4.1, 4.2, 4.3 and 4.4. Additional annotated figures are available in the appendix. The results of this statistical analysis inform and motivate the selection of datasets and metrics for the narrative examples.

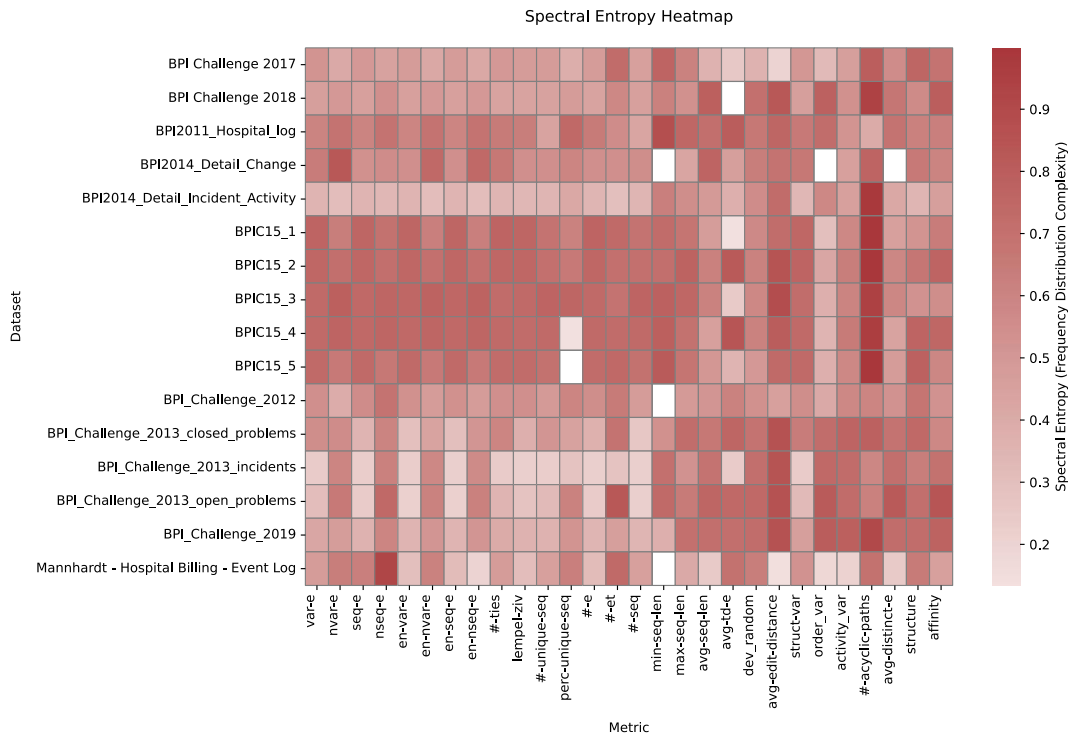


Figure 4.1 Spectral entropy heatmap for temporal dynamics

The spectral entropy heatmap in figure 4.1 provides an overview over the frequency distribution complexity of the data, in essence highlighting how fuzzy or smooth the time series is, with values close to zero representing smooth temporal dynamics curves, while values close to one represent fuzzy temporal dynamics curves with a complex underlying frequency distribution. The data highlights that all datasets exhibit distinct frequency distribution complexities for the complexity metrics, with spectral entropy measurements

varying significantly between different datasets for the same metrics. In particular, the Mannhardt hospital event log and the BPI 2011 hospital event log exhibit varied patterns for the spectral entropy across the complexity metrics, with entropy values between 0.1, as seen with the average edit distance (“avg-edit-distance”) for the Mannhardt hospital event log, and 0.9, as seen with the normalized sequence entropy (“nseq-e”) for the Mannhardt hospital event log or the minimum sequence length (“min-seq-len”) for the BPI 2011 hospital event log. Additionally, both hospital event logs exhibit distinct patterns, despite representing similar processes and organizational backgrounds, thereby providing comparable, yet distinct perspectives on the temporal dynamics. Other datasets exhibit similar patterns across the complexity metrics, yet with more non-significant or non-computable results, such as for both BPI 2014 datasets, or with fewer varied frequency distributions across the complexity metrics, such as seen with the BPI 2013 event logs. Similarly, the BPI 2015 event logs exhibit largely homogenous entropy measurements across all five datasets, thereby limiting their applicability to provide distinct perspectives on temporal dynamics. Focussing on metrics, the number of acyclic paths (“#-acyclic-paths”) exhibits higher spectral entropy measurements across multiple datasets, which is particularly visible for the BPI 2015 event logs.

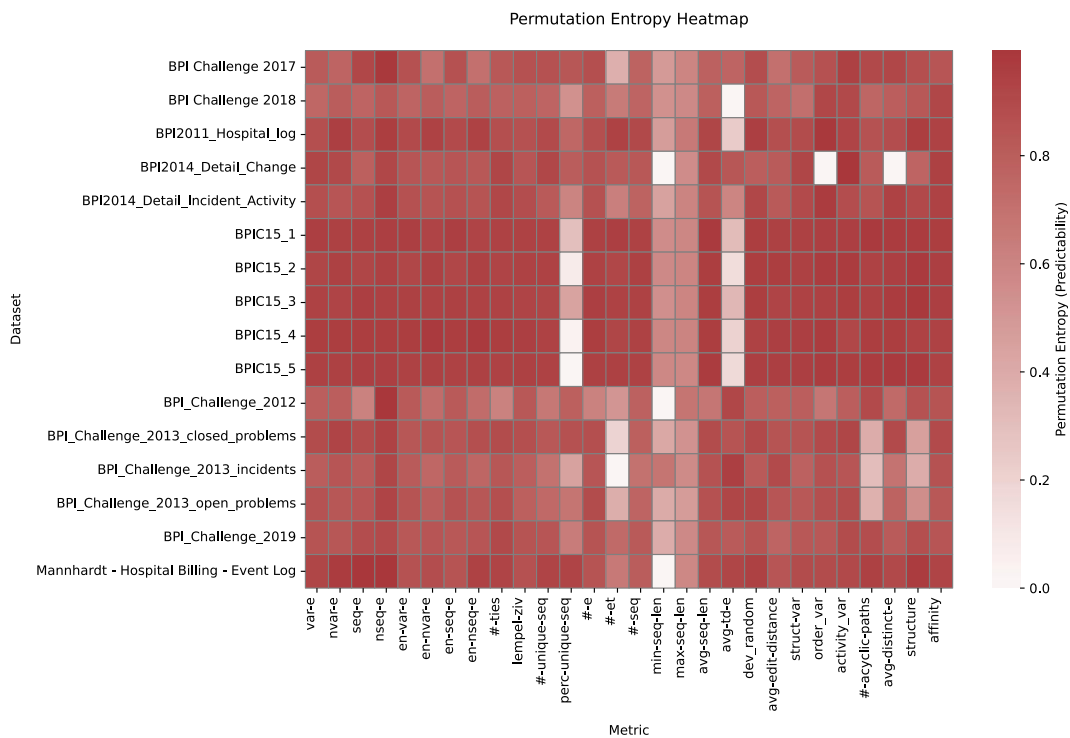


Figure 4.2 Permutation entropy heatmap for temporal dynamics

The permutation entropy heatmap in figure 4.2 provides an overview over the predictability of the data, in essence highlighting how complex the local temporal ordering of the time series is, with values close to zero representing predictable temporal

dynamics curves, while values close to one represent hard-to-predict temporal dynamics curves with a complex local ordering. The data highlights that all datasets exhibit distinct local ordering complexities for the complexity metrics, with permutation entropy measurements varying significantly between different datasets for the same metrics. In particular, the Mannhardt hospital event log and the BPI 2011 hospital event log exhibit varied patterns for the permutation entropy across the complexity metrics, with entropy values between 0.0, as seen with the minimum sequence length (“min-seq-len”) for the Mannhardt hospital event log, and 1.0, as seen with the normalized variant entropy (“nvar-e”), the sequence entropy (“seq-e”), the normalized sequence entropy (“nseq-e”), or the structure (“structure”) for the Mannhardt hospital event log or the order variation (“order_var”) or the deviation from random (“dev_random”) for the BPI 2011 hospital event log. In this case, these two datasets exhibit patterns that are representative of the remaining datasets, with only slight variations across datasets. Generally, the permutation entropy measurements are significantly higher than the spectral entropy measurements across all datasets and metrics, signifying low predictability for the temporal dynamics curves of the complexity metrics across all such combinations. Focussing on the individual metrics, the percentage of unique sequences (“perc-unique-seq”) exhibits a distinct pattern of relatively low permutation entropy measurements across all datasets, particularly visible with the BPI 2015 datasets. The same is true for the number of acyclic paths (“#-acyclic-paths”) and the structure (“structure”), highlighted by the BPI 2013 permutation entropy measurements.

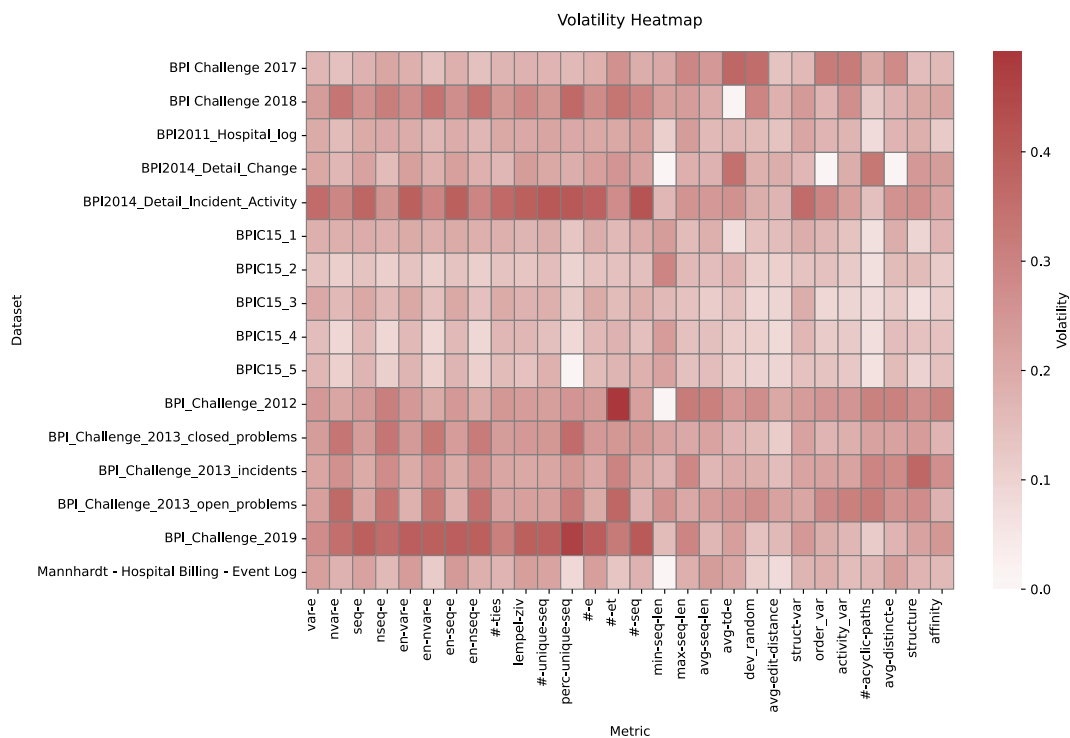


Figure 4.3 Volatility index heatmap for temporal dynamics

The volatility heatmap in figure 4.3 provides an overview over the volatility of the data, in essence highlighting how large the magnitude of change of the data is, representing the standard deviation of the data or time series. When applied to normalized time series, values close to zero represent near-constant curves temporal dynamics curves, while values close to one represent temporal dynamics curves with frequent swings across the entire range of the data. The data highlights that all datasets exhibit distinct volatility patterns for the complexity metrics. Some datasets, such as the BPI 2017, the BPI 2018, the BPI 2014 detail change, and the BPI 2012 event logs exhibit low volatilities across all complexity metrics. The remaining datasets exhibit varied volatilities across all complexity metrics. Low-volatility combinations of datasets and metrics represent near-constant curves, limiting their interpretability. In turn, datasets such as the Mannhardt hospital event log and the BPI 2011 hospital event log, with varied volatilities and select high-volatility metrics provide distinct perspectives on temporal dynamic across the time series. Overall, all complexity metrics exhibit varied and dataset-dependent volatilities. Where some complexity metrics, such as the percentage of unique sequences (“perc-unique-seq”) or the structural variety (“struct-var”) exhibit relatively high volatilities across all datasets compared to the remaining complexity metrics, others exhibit relatively low volatilities, such as those exhibited by the normalized sequence entropy (“nseq-e”), the deviation from random (“dev_random”), the average edit distance (“avg-edit-distance”), or the affinity (“affinity”).

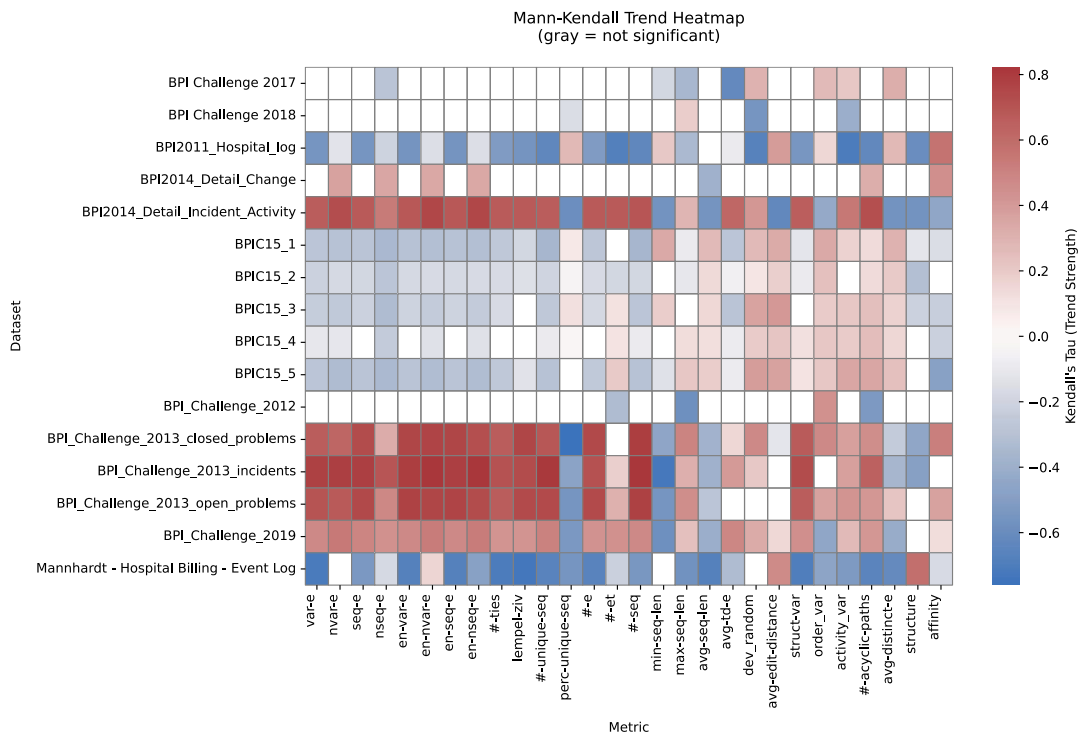


Figure 4.4 Mann-Kendall trend test heatmap for temporal dynamics

While the normalized entropy and the volatility index provide measures of the variability of the data, the presence of persistent dynamics in the timeseries is another key point of interest. The timeseries data of the process drift simulation is thus also tested for monotonic trends with the non-parametric Mann-Kendall test, with a p-value of 0.05. The resulting test statistics are displayed in figure 4.4. Non-significant results with a p-value equal to or above the threshold 0.05 are filtered out and thus displayed in white, while increasing trends are highlighted with a red gradient and decreasing trends with a blue gradient. Multiple datasets do not yield significant trends for the majority of complexity metrics in scope, such as the BPI 2017, BPI 2018, BPI 2014 Detail Change, and BPI 2012 event log datasets. The remaining datasets exhibit varying trends, either majority increasing trends such as seen with the BPI 2013 event logs, or majority decreasing trends such as seen with the Mannhardt hospital event log or the BPI 2011 hospital event log. At the same time, no dataset exhibits purely increasing or decreasing trends. Similarly, no single complexity metrics exhibits solely increasing or decreasing trends.

4.2.2 Temporal Dynamics Analysis Mannhardt Hospital Event Log

Based off the aggregate statistics and trend test, the Mannhardt hospital event log is chosen as a narrative example, highlighting select key metrics and examining their respective temporal dynamics through their complexity curves over time. These curves are plotted against the trace start data, providing insights into how each process complexity metric fluctuates throughout the observation period. Each single complexity measurement is displayed at the earliest trace start date in the respective sliding window. The complexity measurement results are displayed before normalization, in their original value range. Based on the aggregate statistics, only a limited selection of metrics is presented, which exhibit distinct temporal dynamics, and serve as narrative examples for these dynamics.

The narrative examples include the number of acyclic paths (“#-acyclic-paths”), the normalized sequence entropy (“nseq-e”), the percentage of unique sequences (“perc-unique-seq”), the activity variation (“activity_var”), the structural variety (“struct-var”), the affinity (“affinity”), the structure (“structure”), and the average edit distance (“avg-edit-distance”).

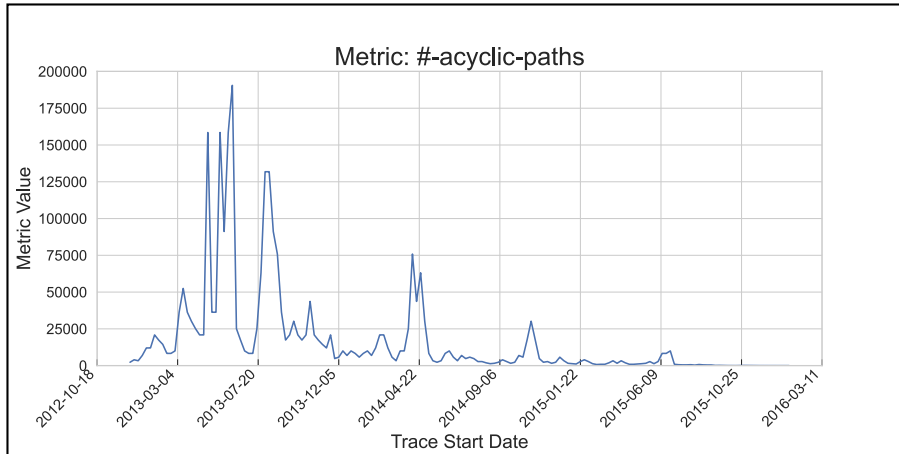


Figure 4.5 Complexity metric: Number of acyclic paths. Temporal dynamics for Mannhardt hospital event log.

Figure 4.5 illustrates the temporal evolution of the number of acyclic paths (“#-acyclic-paths”). The curve shows clear temporal fluctuations in the metric values, with pronounced phases of increase, multiple sharp peaks, and subsequent decline. The metric begins at low levels in late 2012, at the beginning of the time series, followed by a strong upward movement in the first half of 2013. During mid-2013, the curve exhibits several sharp spikes, reaching the highest recorded values in the observation period during the second quarter of 2013. After this volatile phase, the metric decreases notably towards the end of 2013. From early 2014 onward, the curve is characterized by lower overall levels compared to 2013, though short-lived peaks reappear around spring 2014 and again in early 2015. Beyond these episodes, the metric stabilizes at consistently lower values, with only minor fluctuations. Toward the end of the observation window (late 2015 to early 2016), values remain close to zero, and no further significant peaks are observed.

In summary, the curve highlights distinct phases: an initial rise and period of high complexity as measured by the number of acyclic paths in early- and mid-2013, followed by a transition to lower complexity values after 2014, with occasional isolated spikes. Overall, the curve of the time series measurements for the number of acyclic paths exhibits sudden and repeated spikes or bursts, followed by decreasing values with intermittent spikes of lower magnitude.

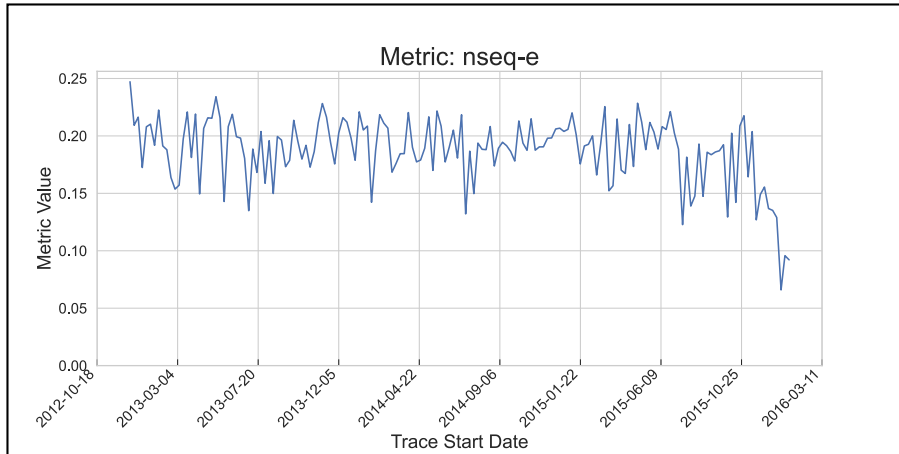


Figure 4.6 Complexity metric: Normalized sequence entropy. Temporal dynamics for Mannhardt hospital event log.

Figure 4.6 in turn illustrates the temporal evolution of the normalized sequence entropy (“nseq-e”). The curve displays relatively stable values over most of the observation period, with fluctuations occurring within a narrow band. From late 2012 until mid-2015, the metric consistently oscillates between approximately 0.15 and 0.23, with frequent short-term variations but no strong upward or downward long-term trend. This period is marked by small-scale volatility around a stable central level. A change in the curve is visible from late 2015 onward. The metric begins to decline more noticeably, with values falling below 0.15 in early 2016 and reaching the lowest point in the entire series by the end of the observation window.

In summary, the curve of the normalized sequence entropy time series is characterized by an extended phase of relative stability, followed by a late-phase decrease in 2015–2016. This pattern highlights a transition from consistent oscillations around a stable level toward a pronounced decline in the final months of the observation period.

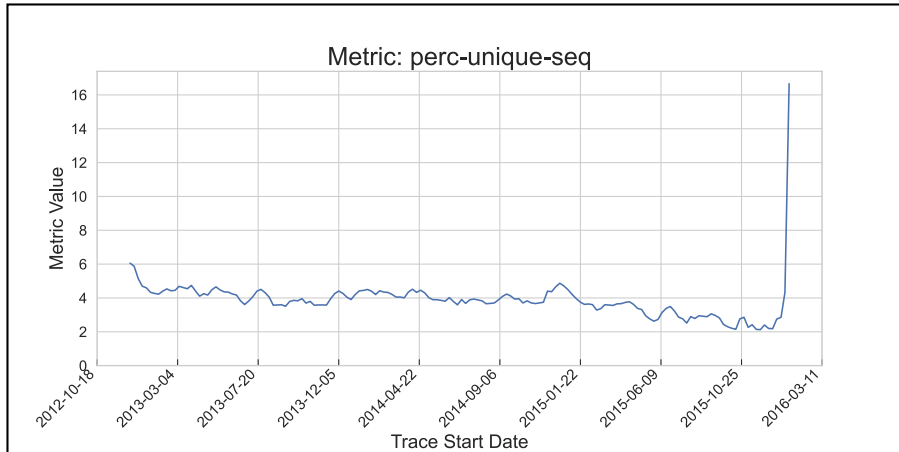


Figure 4.7 Complexity metric: Percentage unique sequences. Temporal dynamics for Mannhardt hospital event log.

Moving to the next complexity metric, figure 4.7 illustrates the temporal evolution of the percentage of unique sequences (“perc-unique-seq”). The curve shows a long phase of relative stability followed by a pronounced increase at the very end of the time window. From late 2012 until mid-2015, values fluctuate within a narrow range, mostly between 2% and 5%, with only minor oscillations and no sustained upward or downward movement. During this extended period, the curve remains relatively flat, showing small local variations but maintaining overall stability. A gradual decline can be observed from mid-2015 onward, as the metric decreases slightly below earlier levels, reaching its local minimum in late 2015. After this period of decline, the curve exhibits a sharp rise beginning in early 2016, culminating in an abrupt spike that reaches the highest value of the entire observation window.

In summary, the curve of the time series for the percentage of unique sequences is characterized by a prolonged phase of stability, a modest decline in late 2015, and a sudden, extreme increase in early 2016. This dynamic highlights the presence of a strong late-phase shift in the metric, contrasting with its otherwise stable temporal behavior.

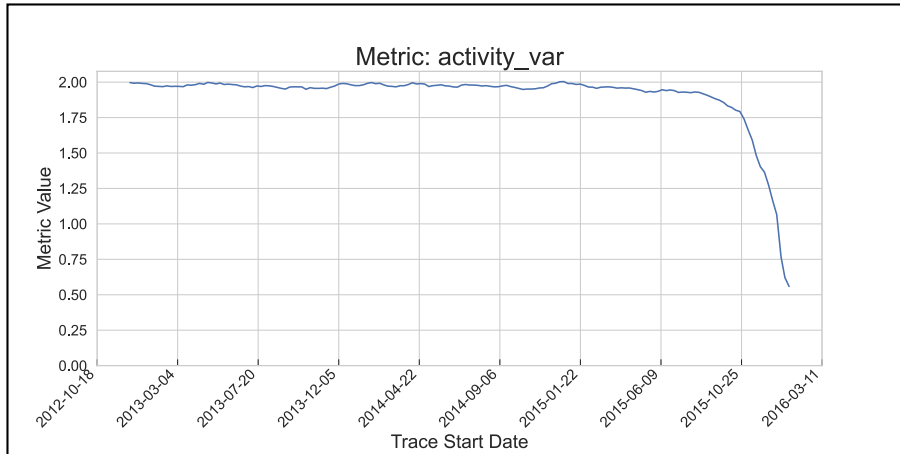


Figure 4.8 Complexity metric: Activity variation. Temporal dynamics for Mannhardt hospital event log.

Next, figure 4.8 illustrates the temporal evolution of the activity variation (“activity_var”). For most of the time span, the curve remains at a consistently high level with minimal variation, followed by a sharp decline in the final phase. From late 2012 until approximately mid-2015, the metric fluctuates only slightly around values close to 2.0, indicating a period of remarkable stability. Minor oscillations are present but do not alter the overall flat trend during these years. Beginning in late 2015, the curve exhibits a noticeable downward trajectory. This decline accelerates in early 2016, with values dropping steeply and reaching the lowest levels of the entire observation period by the end of the time window.

In summary, the curve of the time series for the activity variation is characterized by a prolonged phase of stability at high levels, followed by a sudden and substantial decrease in the closing months of the observation period. This marks a clear shift from long-term consistency to rapid decline.

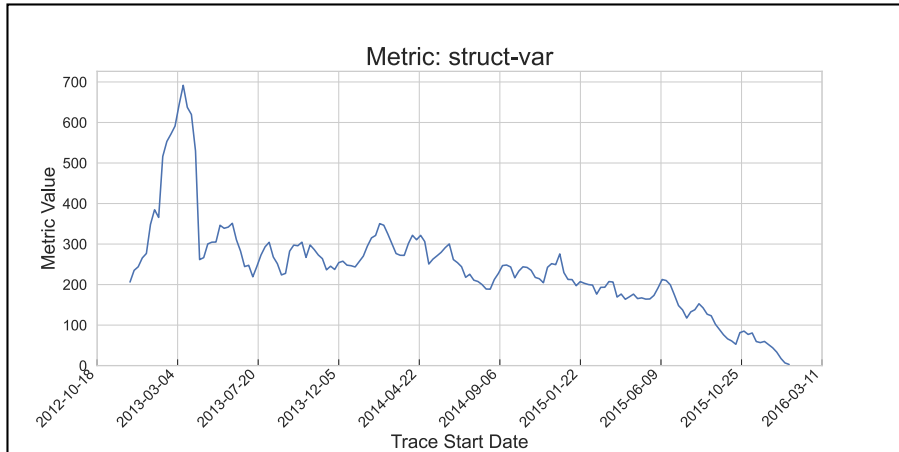


Figure 4.9 Complexity metric: Structural variety. Temporal dynamics for Mannhardt hospital event log.

Figure 4.9 illustrates the temporal evolution of the structural variety (“struct-var”). The curve displays an initial increase, followed by a peak phase, and then a gradual but steady decline over time. From late 2012 until the first half of 2013, the metric rises sharply, reaching its maximum value in early-2013, at the end of the first quarter of 2013. This marks the highest point across the entire time window. After this peak, the curve drops but stabilizes at moderately elevated levels, fluctuating between approximately values of 250 and 350 during late 2013 and 2014. Beginning in 2015, the overall trajectory shifts downward more clearly, with the metric showing a gradual decline interspersed with smaller oscillations. This downward trend continues through 2015 and into 2016, culminating in the lowest observed values by the end of the observation period.

In summary, the curve of the time series for the structural variety is characterized by three distinct phases: an initial sharp increase and peak, a period of moderate stability with fluctuations, and a prolonged downward trend toward minimal values.

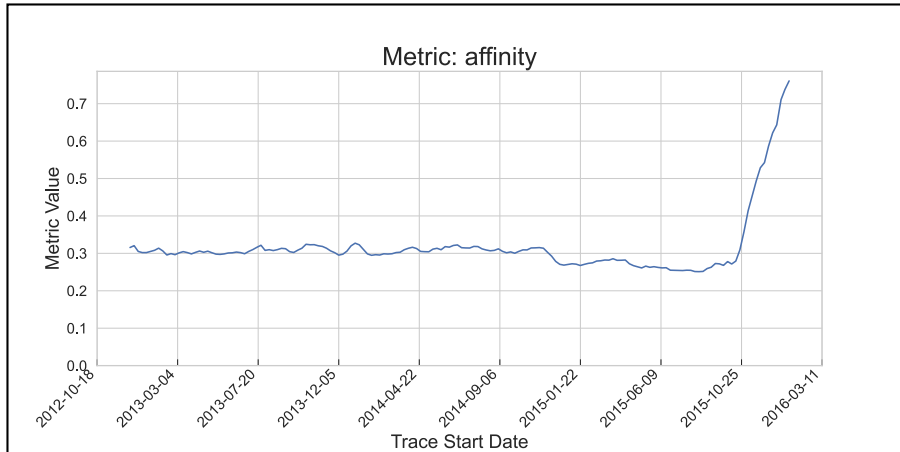


Figure 4.10 Complexity metric: Affinity. Temporal dynamics for Mannhardt hospital event log.

As another narrative example, figure 4.10 illustrates the temporal evolution of the affinity (“affinity”). The curve is characterized by a long phase of stability at moderate levels, followed by a sharp and sustained increase in the final months. From late 2012 until approximately mid-2015, the metric remains relatively stable, fluctuating only slightly around values close to 0.3. During this extended period, no substantial upward or downward movements are visible, and the curve retains a flat overall shape with only minor local variations. A slight downward trend is visible from the end of 2014 until late 2015, with values sinking below 0.3, and instead moving between 0.3 and 0.25. Starting in late 2015, the curve begins to rise. This upward movement intensifies markedly in early 2016, with the metric accelerating rapidly to reach its highest values of the entire observation window by the end of the time span.

In summary, the curve of the time series for the affinity exhibits a stable or near-stable long-term phase followed by a distinct late-phase surge. The transition from flat stability to steep growth highlights a pronounced shift in dynamics in the final segment of the observation period.

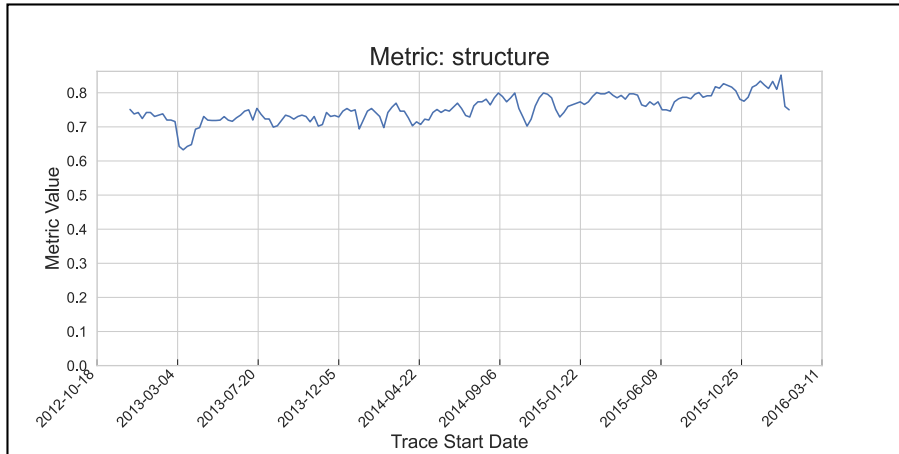


Figure 4.11 Complexity metric: Structure. Temporal dynamics for Mannhardt hospital event log.

As the next example figure 4.11 illustrates the temporal evolution of the structure (“structure”). The curve is characterized by moderate oscillations around a relatively high and stable level, with a slight upward tendency visible toward the later stages. From late 2012 through mid-2013, the metric fluctuates around values between 0.65 and 0.75. Notably, there is a brief dip in early 2013, with values decreasing from above 0.7 to below 0.65, followed by recovery. In the following years (2013–2014), the curve maintains this level while showing small short-term fluctuations. From 2014 onward, the metric gradually shifts upward, oscillating mostly between 0.7 and 0.8, while retaining its overall stable profile. The highest values are reached in late 2015 and early 2016, where the curve briefly approaches or slightly exceeds 0.8.

In summary, the curve of the time series for the structure metric exhibits a stable long-term trajectory with a brief dip in the beginning, and with small oscillations and a gradual upward drift over time, culminating in the highest values at the end of the observation window.

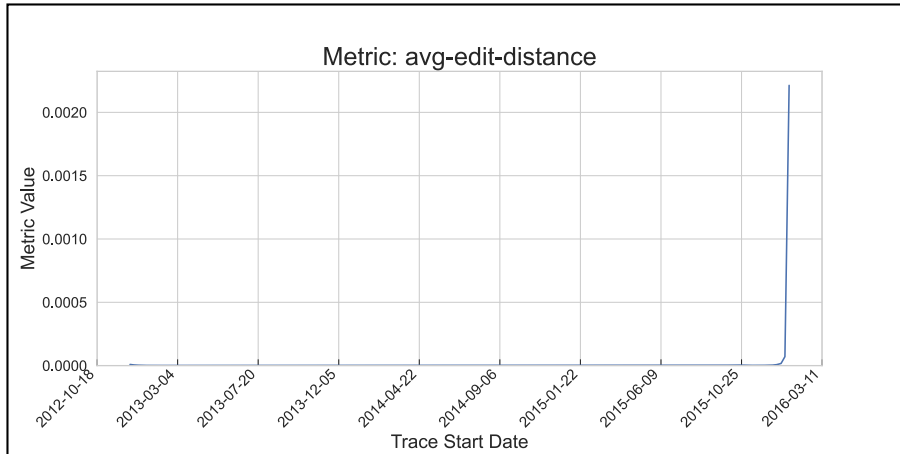


Figure 4.12 Complexity metric: Average edit distance. Temporal dynamics for Mannhardt hospital event log.

Finally, figure 4.12 illustrates the temporal evolution of the average edit distance (“avg-edit-distance”). For nearly the entire observation period, the curve remains at values close to zero, showing no visible variation from late 2012 through most of 2015. A marked change occurs at the very end of the time window. Beginning in early 2016, the curve rises sharply, leading to a steep upward spike. This increase culminates in the highest recorded values of the metric, reached at the close of the observation period. In summary, the curve of the time series for the average edit distance metric is characterized by a prolonged phase of near-zero stability, followed by a sudden and steep rise in the final months. This creates a distinctive flat–spike pattern across the observed time span.

4.2.3 Temporal Dynamics Analysis BPI 2011 Hospital Event Log

To provide further narrative examples of the temporal dynamics from a comparable background, with representative entropy patterns and insightful metric trends, curves are plotted for the time series of the same set of metrics, but for the BPI 2011 hospital event log. This event log contains traces from an observation window of similar length to the Mannhardt hospital event log.

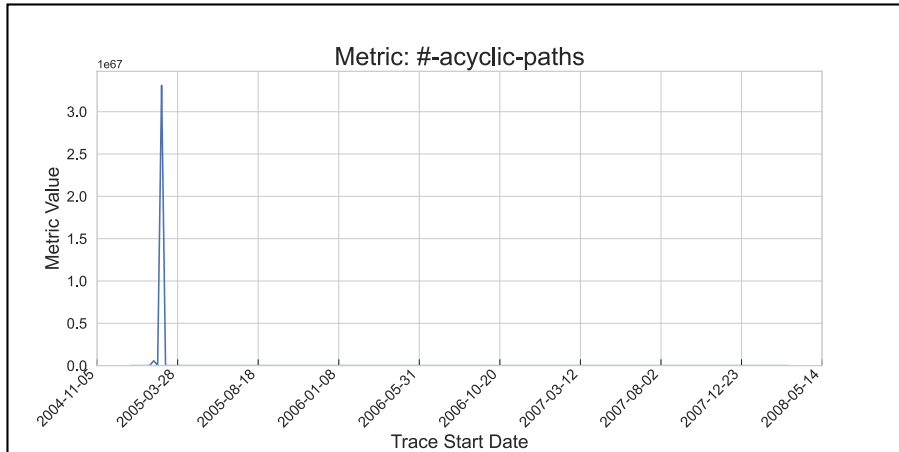


Figure 4.13 Complexity metric: Number of acyclic paths. Temporal dynamics for BPI 2011 hospital event log.

Starting off, figure 4.13 illustrates the temporal evolution of the number of acyclic paths (“#-acyclic-paths”). The curve is dominated by a single extreme peak occurring early in the observation period, after which the metric remains flat at zero for the remainder of the timeline. Specifically, from late 2004 until early 2005, the metric rises abruptly, culminating in an exceptionally high spike around March 2005. This peak represents the maximum value in the entire observation window by several orders of magnitude. Immediately after this point, the metric drops back to relative near-zero levels and remains at that level without further variation from mid-2005 through 2008. While the latter levels of the metric do not actually zero, the relative scale of the metric, with the maximum value exceeding $3 * 10^{67}$, diminishes any other measurements to relative near-zero levels.

In summary, the curve of the time series for the number of acyclic paths exhibits a highly skewed temporal profile: a short-lived but extreme early spike followed by a prolonged phase of near-zero values throughout the rest of the observation period.

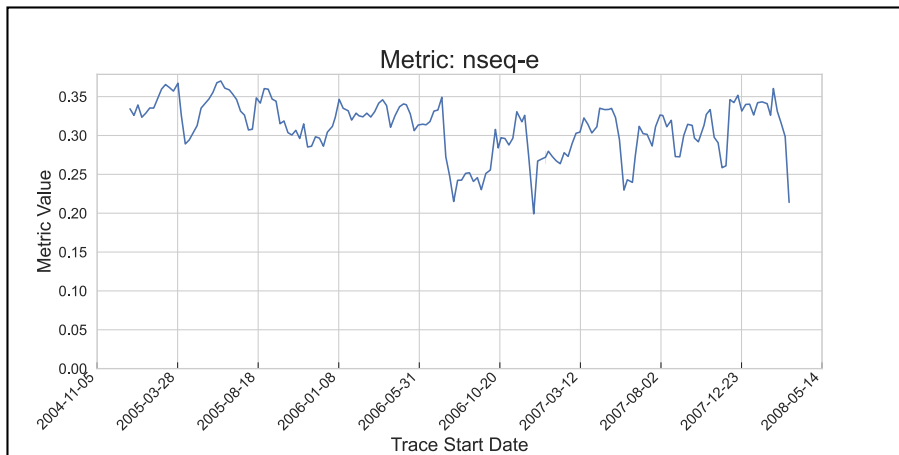


Figure 4.14 Complexity metric: Normalized sequence entropy. Temporal dynamics for BPI 2011 hospital event log.

Moving to the next narrative example, figure 4.14 illustrates the temporal evolution of the normalized sequences entropy (“nseq-e”). The curve shows consistent oscillations within a moderate range, with no long-term upward or downward trajectory. From late 2004 to 2008, the metric fluctuates primarily between values of 0.25 and 0.35, with short-term peaks occasionally exceeding 0.35 and troughs dropping below 0.25, reaching values around 0.2. Several of these sharper declines occur intermittently, for example around late 2006 and early 2008, but are followed by quick recoveries to previous levels. While there is a consistent downward trend with lower highs and lower lows between mid-2005 and the end of 2006, this trend is broken from then onwards, instead reverting to reach values previously only reached in the beginning of the observation period. In the last segment of the curve, starting in early 2008, there is a sharp decline in the curve, similar to previous troughs.

In summary, the curve of the time series for the normalized sequence entropy is characterized by stable long-term behaviour combined with frequent short-term oscillations. While brief dips and peaks punctuate the series, the metric consistently remains within a narrow band across the entire observation period.

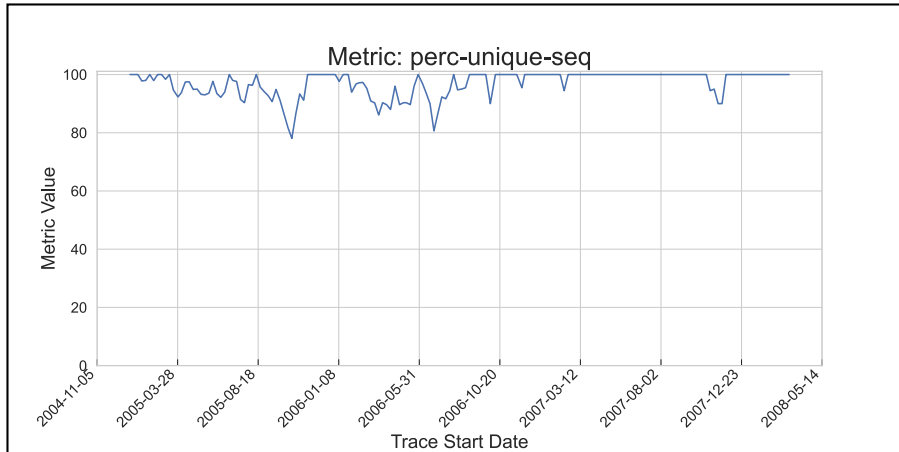


Figure 4.15 Complexity metric: Percentage unique sequences. Temporal dynamics for BPI 2011 hospital event log.

Next, figure 4.15 illustrates the temporal evolution of the percentage of unique sequences (“perc-unique-seq”). Overall, the curve remains at consistently high values, close to 100%, with only short-term fluctuations. From late 2004 through 2008, the metric generally oscillates within a narrow band between roughly 85% and 100%. Most values lie near the upper boundary, and extended stretches reach 100% without interruption. Occasional dips appear, most noticeably in 2006 and 2007, when the curve briefly declines to lower levels before quickly returning to near-maximum values.

In summary, the curve of the time series for the percentage of unique sequences is characterized by a long-term pattern of consistently high values, punctuated by intermittent dips, but always returning to levels close to 100% throughout the observation period.

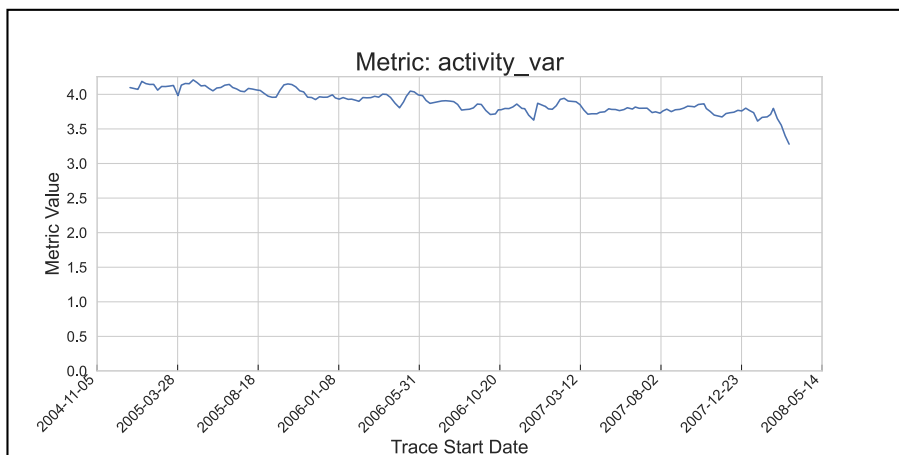


Figure 4.16 Complexity metric: Activity variation. Temporal dynamics for BPI 2011 hospital event log.

Figure 4.16 illustrates the temporal evolution of the activity variation (“activity_var”). The curve shows a generally stable profile with a slight gradual decline over time, ending with a more pronounced drop in the final phase. From late 2004 through roughly mid-2006, the metric fluctuates around values close to 4.0, with only minor short-term variation. After this period, a slow downward trend becomes visible, with values gradually decreasing toward the range of 3.7 to 3.8 during 2006–2007. In early 2008, the decline accelerates, and the curve falls more clearly, reaching its lowest point at the end of the observation window with a value of around 3.25, and with a near-linear downward slope.

In summary, the curve of the time series for the activity variation is characterized by an extended phase of stability near the level of the value 4.0, followed by a slow downward drift and a sharper late-phase decline.

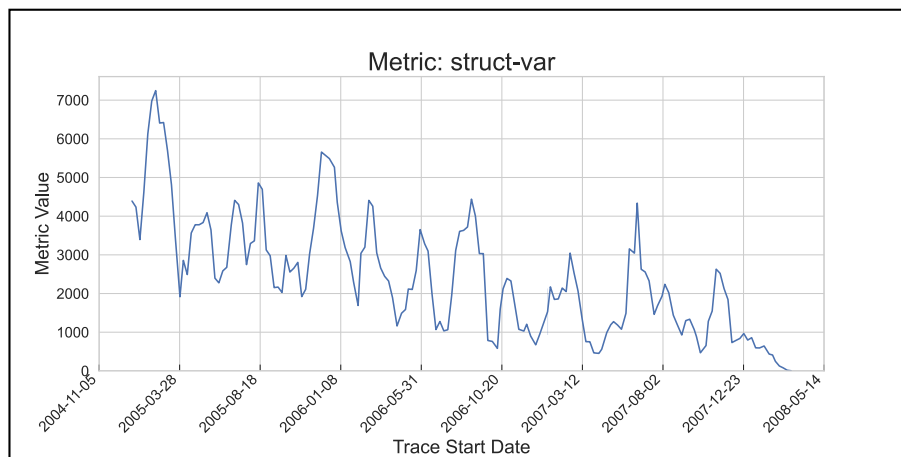


Figure 4.17 Complexity metric: Structural variety. Temporal dynamics for BPI 2011 hospital event log.

Figure 4.17 illustrates the temporal evolution of the structural variety (“struct-var”). The curve exhibits substantial fluctuations, with alternating peaks and troughs, complemented by a gradual long-term decline. From late 2004 through early 2005, the metric rises sharply, reaching one of its highest peaks at values above 7000. After this initial surge, the curve shows a recurring oscillatory pattern: frequent rises and falls between approximately 1000 and 5000, with peaks occurring at irregular intervals throughout 2005 and 2006. Beginning in 2007, the oscillations persist but with generally lower amplitude, as both peaks and troughs trend downward over time. By late 2007 and early 2008, the metric declines more markedly, with values falling below 1000 and steadily approaching zero at the close of the last segment of the observation period.

In summary, the curve of the time series for the structural variety is characterized by an initial sharp peak, an extended period of strong oscillations, and a long-term downward trend leading to minimal values at the end of the observation period.

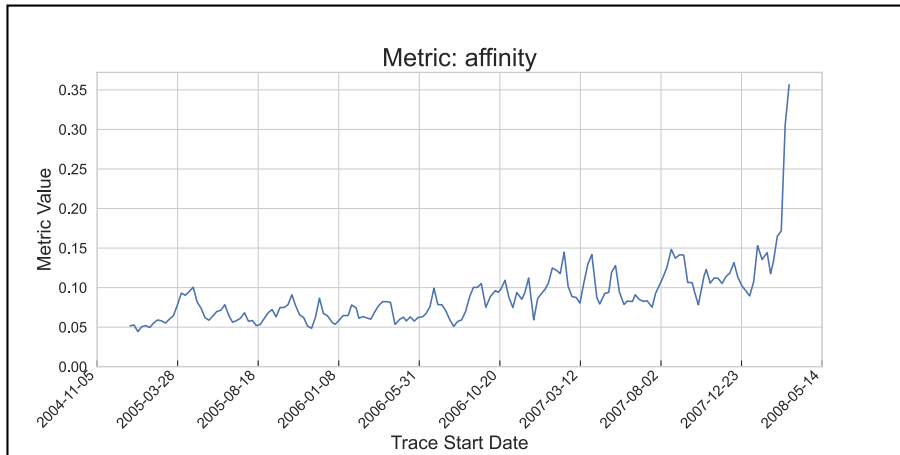


Figure 4.18 Complexity metric: Affinity. Temporal dynamics for BPI 2011 hospital event log.

Moving to the next narrative example, Figure 4.18 illustrates the temporal evolution of the affinity (“affinity”). The curve reveals a gradual long-term upward trend, combined with short-term fluctuations, culminating in a steep increase at the end of the timeline. From late 2004 to around 2006, the metric remains at relatively low levels, fluctuating between 0.05 and 0.10, with small oscillations. Over the following years (2006–2007), the curve shows a gradual rise, with values more frequently reaching the range of 0.10 to 0.15, while still exhibiting short-term volatility. Similarly, the amplitude of the oscillations increases during that timeframe. In 2008, the upward trend accelerates. The curve rises sharply in the final months of the observation window, ultimately reaching the highest recorded values of the entire series.

In summary, the curve of the time series for the affinity metric is characterized by a gradual increase over the long term, accompanied by short-term oscillations, and a pronounced late-phase surge that defines the end of the time span.

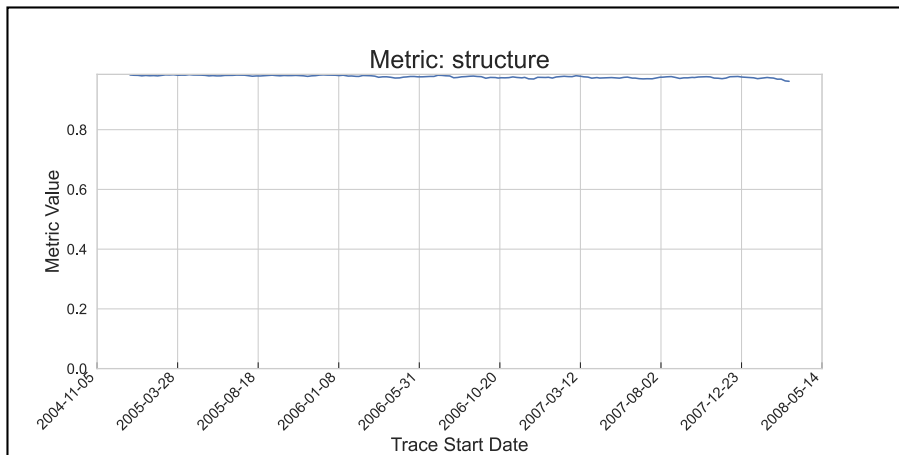


Figure 4.19 Complexity metric: Structure. Temporal dynamics for BPI 2011 hospital event log.

Next, figure 4.19 illustrates the temporal evolution of the structure (“structure”). The curve is marked by an extended phase of stability and consistency, with values remaining close to the upper bound throughout the entire timeline. From late 2004 through 2008, the metric fluctuates only slightly around values near 0.95 to 0.97. These small oscillations are subtle and do not interrupt the overall flat trajectory. No substantial upward or downward movements are visible across the entire time span, and the curve maintains a nearly horizontal shape. In summary, the curve of the time series for the structure metric is characterized by long-term stability at very high levels, with only minor local variation and no significant changes in direction or trend.

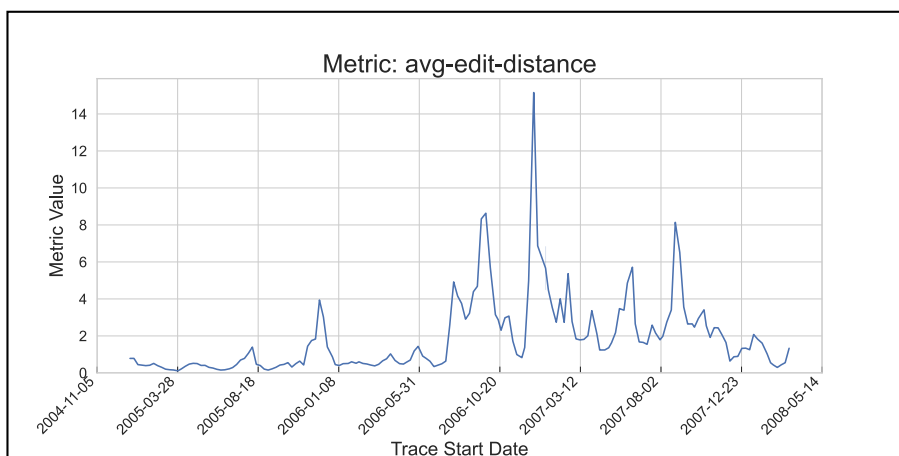


Figure 4.20 Complexity metric: Average edit distance. Temporal dynamics for BPI 2011 hospital event log.

Finally, figure 4.20 illustrates the temporal evolution of the average edit distance (“avg-edit-distance”). The curve shows an overall fluctuating pattern, marked by recurring peaks of varying intensity, including several sharp spikes, before tapering off toward

lower values. From late 2004 through 2005, the metric remains relatively low, with values mostly close to 0 and only minor oscillations. Starting in 2006, the curve begins to exhibit more pronounced variability, with a succession of peaks and troughs. This culminates in a major spike in early 2007, where the metric reaches its highest recorded value of the entire time span. Following this extreme peak, the curve continues to fluctuate throughout 2007, showing several additional elevated peaks, though none as high as the early 2007 maximum. In 2008, the oscillations persist but at generally reduced levels, with the curve trending back downward toward smaller values by the end of the period.

In summary, the curve of the time series for the average edit distance is characterized by a long low-level phase, a highly volatile mid-period with recurring peaks and a dominant spike, and a subsequent decline to lower levels in the final phase.

4.3 Simulation of Process Drift

4.3.1 Simulation Results Across all Datasets

The process drift simulation results across all datasets and metrics require a statistical representation to reduce the dimensionality of the data and provide insights into the underlying dynamics and simulation behaviour. Due to the number of datasets and metrics in scope, plotting and comparing each combination of dataset and metric is not possible in the scope of this thesis. Instead, narrative examples of the key process simulation results are presented and analysed, in the form of curves representing the process complexity over the course of the process drift simulation. To this end, the spectral entropy, the permutation entropy, and the volatility index, as calculated for each combination of dataset and complexity metric, for the timeseries of measurements created during the simulation, are presented in figures 4.21, 4.22, 4.23 and 4.24. Additional annotated figures are available in the appendix. The results of this statistical analysis inform and motivate the selection of datasets and metrics for the narrative examples.

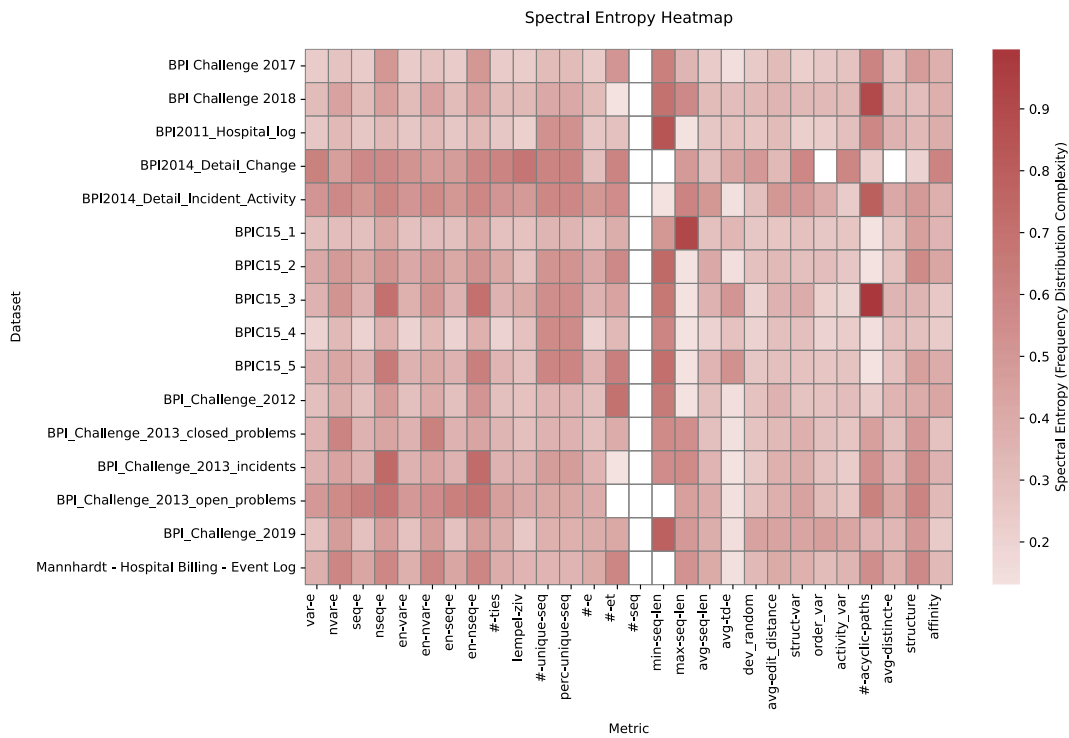


Figure 4.21 Spectral entropy for process drift simulation

The spectral entropy heatmap in figure 4.21 provides an overview over the frequency distribution complexity of the data, in essence highlighting how fuzzy or smooth the time series is, with values close to zero representing smooth temporal dynamics curves, while values close to one represent fuzzy temporal dynamics curves with a complex underlying frequency distribution. In contrast to the results of the temporal dynamics analysis, the data highlights that all datasets exhibit similar frequency distribution complexities for the complexity metrics, with limited variation across datasets. Focussing on the previously selected narrative examples, the Mannhardt hospital event log and the BPI 2011 hospital event log exhibit similar patterns for the spectral entropy across the complexity metrics, with slight variations exhibited by specific metrics. These include the minimum sequence length (“min-seq-len”), with a missing spectral entropy due to constant metric values for the Mannhardt hospital event log, and a spectral entropy of 0.8 for the BPI 2011 hospital event log. Otherwise, both event log datasets exhibit largely the same patterns as the other datasets. Focussing on metrics, the spectral entropy measurements vary significantly between different datasets for a limited selection of metrics, which include the number of acyclic paths (“#-acyclic-paths”), the minimum sequence length (“min-seq-len”), and the maximum sequence length (“max-seq-len”). The number of sequences (“#-seq”) is another notable outlier in the data, with no spectral entropy measurements visible in the heatmap, due to calculation errors caused by the constant nature of the number of sequences over the course of the simulation, as it is based

on one of the simulation parameters. Otherwise, most metrics exhibit largely the same distribution of spectral entropy measurements across datasets, with low to moderate entropy values.

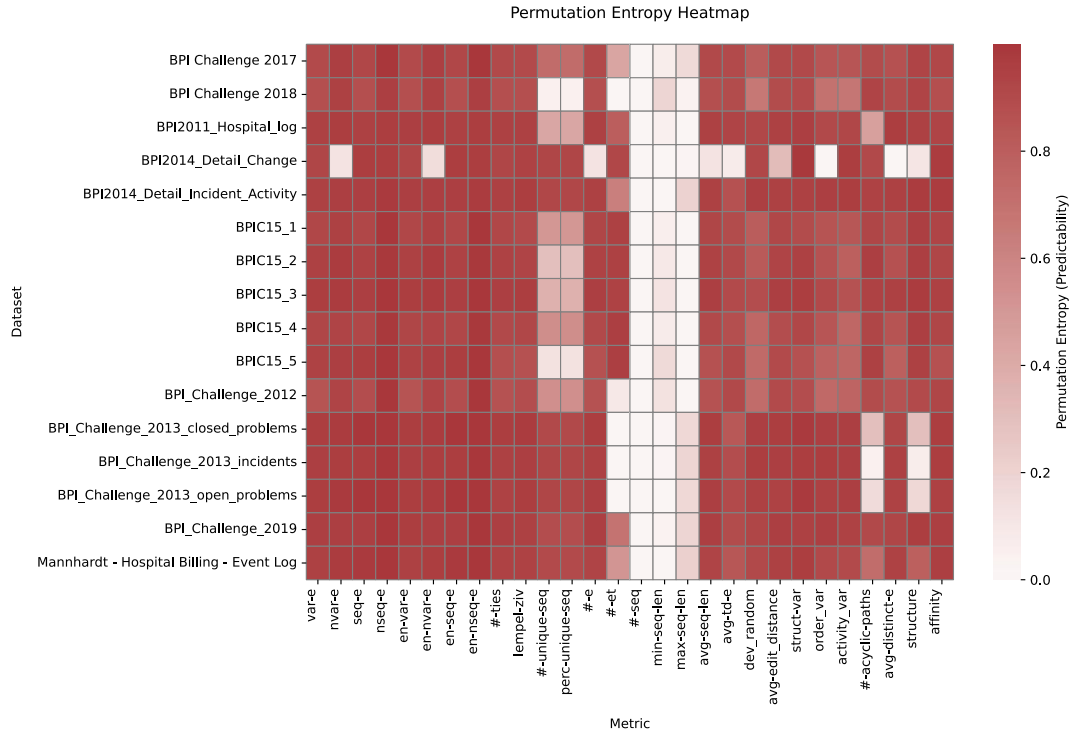


Figure 4.22 Permutation entropy heatmap for process drift simulation

The permutation entropy heatmap in figure 4.22 provides an overview over the predictability of the data, in essence highlighting how complex the local temporal ordering of the time series is, with values close to zero representing predictable temporal dynamics curves, while values close to one represent hard-to-predict temporal dynamics curves with a complex local ordering. The data highlights that all datasets exhibit homogenous permutation entropies, with limited variation across datasets, in line with the spectral entropy measurements for the process drift simulation. In particular, the Mannhardt hospital event log and the BPI 2011 hospital event log exhibit shared patterns for the permutation entropy across the complexity metrics. In this case, these two datasets exhibit patterns that are representative of the remaining datasets. Focussing on the outliers, the minimum sequence length (“min-seq-len”) and maximum sequence length (“max-seq-len”) both exhibit low permutation entropy measurements close to zero. The number of sequences (“#-seq”) is another notable outlier in the data, with no permutation entropy measurements visible in the heatmap, due to calculation errors caused by the constant nature of the number of sequences over the course of the simulation, as it is based on one of the simulation parameters. Apart from that, the number of acyclic paths (“#-acyclic-paths”) and the structure (“structure”) also exhibit distinct patterns, with notably

lower permutation entropy measurements for the BPI 2013 datasets. Otherwise, most metrics exhibit largely the same distribution of permutation entropy measurements across datasets, with permutation entropy measurements close to one for all datasets and almost all metrics.

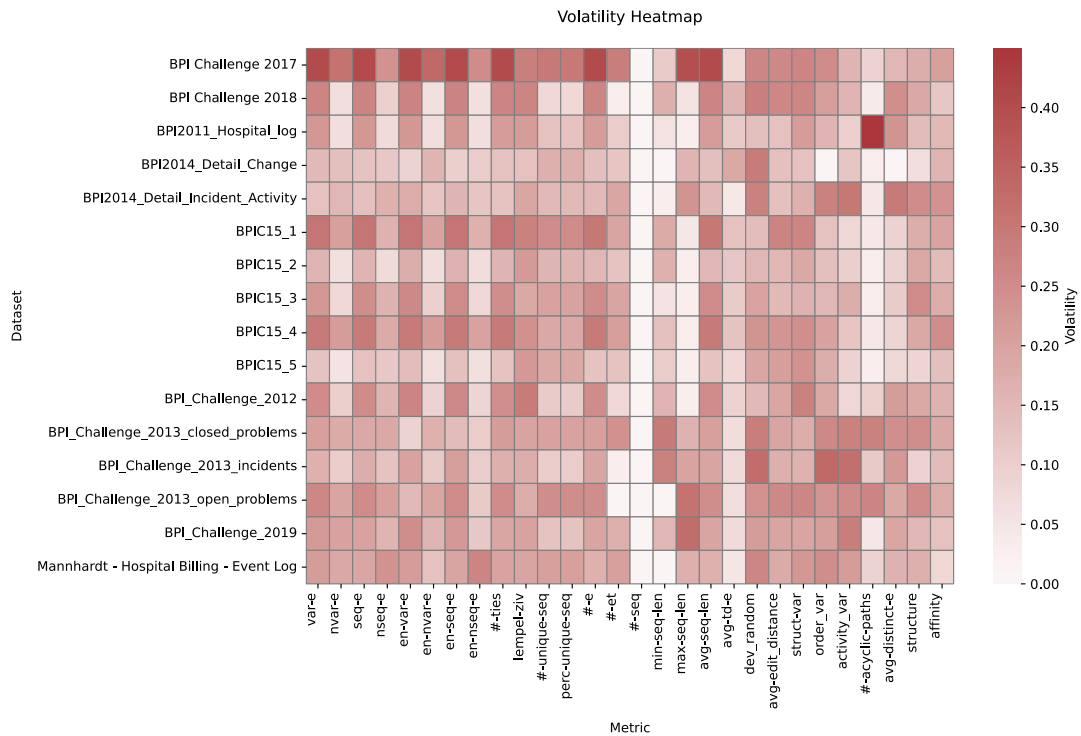


Figure 4.23 Volatility index heatmap for process drift simulation

The volatility heatmap in figure 4.23 provides an overview over the volatility of the data, in essence highlighting how large the magnitude of change of the data is, representing the standard deviation of the data or time series. When applied to normalized time series, values close to zero represent near-constant curves temporal dynamics curves, while values close to one represent temporal dynamics curves with frequent swings across the entire range of the data. The data highlights that all datasets exhibit mostly homogenous volatility patterns, with variation mainly across metrics. Datasets such as the Mannhardt hospital event log and the BPI 2011 hospital event log still exhibit varied volatilities and select high-volatility metrics, providing distinct perspectives process drift dynamics across the simulation time series. An example of this are the relatively high volatility of the number of acyclic paths (“#-acyclic-paths”) exhibited by the Mannhardt hospital event log measurement, with a volatility index of 0.4, and the same measurement for the BPI 2011 hospital event log, with a volatility index of 0.1 by comparison. In addition, the volatility index measurements of the BPI 2017 dataset exhibit stronger volatilities relative to the remaining datasets. Overall, all complexity metrics exhibit low to moderate, dataset-dependent volatility patterns. The number of sequences (“#-seq”) again is an

outlier in the data, with no permutation entropy measurements visible in the heatmap, due to the same reasons outlined before.

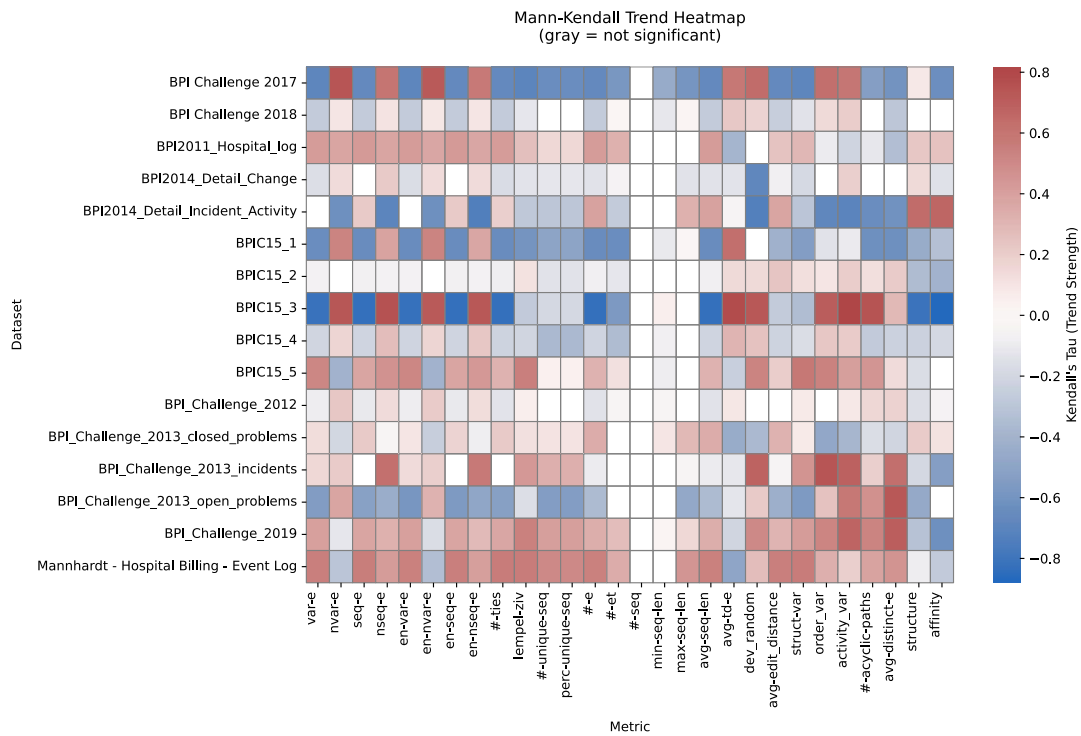


Figure 4.24 Mann-Kendall trend test heatmap for process drift simulation

While the normalized entropy and the volatility index provide measures of the variability of the data, the presence of persistent dynamics in the timeseries is another key point of interest. The timeseries data of the process drift simulation is thus also tested for monotonic trends with the non-parametric Mann-Kendall test, with a p-value of 0.05. The resulting test statistics are displayed in figure 4.24. Non-significant results with a p-value equal to or above the threshold 0.05 are filtered out and thus displayed in white, while increasing trends are highlighted with a red gradient and decreasing trends with a blue gradient. Multiple datasets do not yield significant trends for the majority of complexity metrics in scope, such as the BPI 2015-2 and the BPI 2012 event log datasets. The remaining datasets exhibit varying trends, either majority increasing trends such as seen with the Mannhardt hospital event log, or majority decreasing trends such as seen with the BPI 2017 event log. At the same time, no dataset exhibits purely increasing or decreasing trends. The Mannhardt hospital event log and the BPI 2011 hospital event log again entail increasing, decreasing, and ambiguous trends, providing distinct perspectives on process drift dynamics while also representing the wider patterns exhibited by all datasets. Similarly, no single complexity metrics exhibits solely increasing or decreasing trends. Overall, the patterns found with the Mann-Kendall trend

test are heterogenous across datasets and across metrics, in contrast to the spectral entropy, permutation entropy, and volatility measurements.

4.3.2 Simulation Results Mannhardt Hospital Event Log

Based off the aggregate statistics and trend test, the Mannhardt hospital event log is again chosen as a narrative example for the process drift simulation results, highlighting select key metrics and examining their respective process drift dynamics through their complexity curves over time. These curves are plotted against the number of simulation iterations, providing insights into how each process complexity metric fluctuates throughout the observation period. The complexity measurement results are displayed before normalization, in their original value range. Based on the aggregate statistics, only a limited selection of metrics is presented, which exhibit distinct temporal dynamics, and serve as narrative examples for these dynamics.

As before for the temporal dynamics, the narrative examples include the number of acyclic paths (“#-acyclic-paths”), the normalized sequence entropy (“nseq-e”), the percentage of unique sequences (“perc-unique-seq”), the activity variation (“activity_var”), the structural variety (“struct-var”), the affinity (“affinity”), the structure (“structure”), and the average edit distance (“avg-edit-distance”).

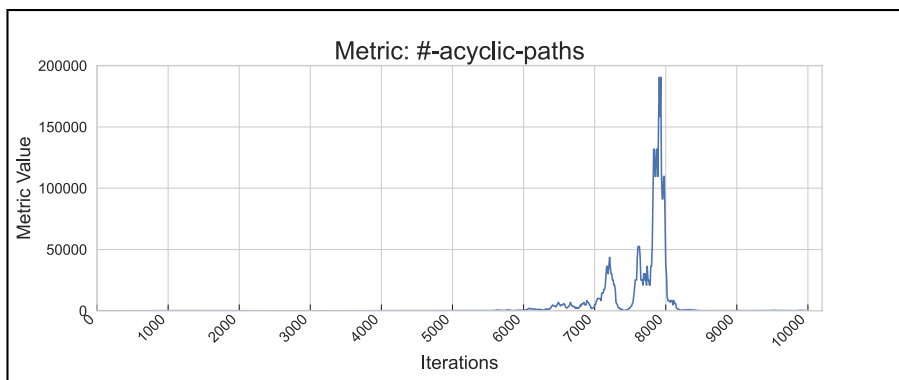


Figure 4.25 Complexity metric: Number of acyclic paths. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.25 illustrates the process drift simulation results for the number of acyclic paths (“#-acyclic-paths”). The curve is dominated by a pronounced late-phase peak, preceded by a long period of near-zero values and moderate intermediate fluctuations. From the beginning of the simulation up to approximately iteration 6000, the metric remains at or close to zero, indicating no notable changes during this extended phase. Starting around iteration 6500, the curve begins to rise gradually, showing intermittent increases and moderate peaks. A sharp escalation occurs between iterations 7500 and 8200, where the

metric spikes dramatically, reaching its highest value of nearly 200,000. This marks the dominant feature of the curve. Immediately after this peak, the metric declines steeply, returning to near-zero levels by iteration 8500 and remaining flat through the end of the simulation.

In summary, the curve of the time series for the number of acyclic paths is characterized by an extended low-level phase, followed by a late surge with sharp peaks or spikes, and a rapid collapse back to zero.

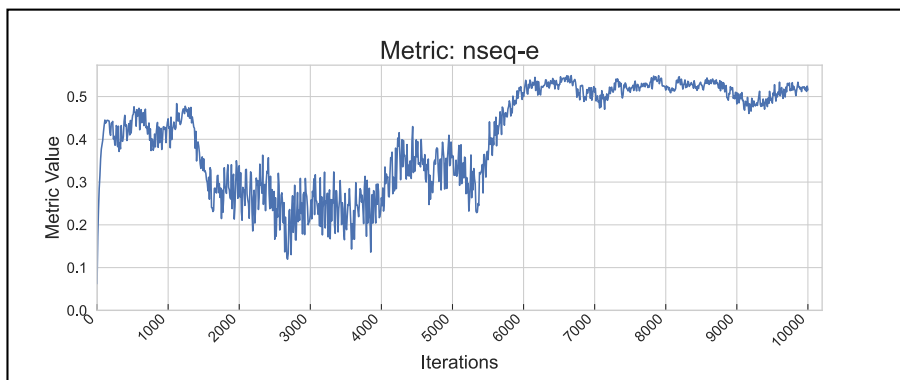


Figure 4.26 Complexity metric: Normalized sequence entropy. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.26 illustrates the process drift simulation results for the normalized sequence entropy (“nseq-e”). The curve demonstrates distinct phases of decline, recovery, and stabilization, combined with short-term fluctuations. At the start of the simulation, the metric begins at values between 0.4 and 0.45. During the first 2000 iterations, it shows oscillations but remains within this range. From iteration 2000 to about 5000, the metric declines noticeably, with values falling toward 0.2 and showing increased volatility in this phase. Around iteration 5500, the curve begins a marked upward movement, rising steadily and surpassing earlier levels. Between iterations 6000 and 8000, the metric stabilizes around 0.5, with only small oscillations. This stable plateau continues until the end of the simulation, where the values remain close to 0.5.

In summary, the curve of the time series for the normalized sequence entropy is characterized by an initial stable phase, a mid-simulation decline accompanied by high volatility, and a subsequent recovery leading to stable high levels in the later iterations.

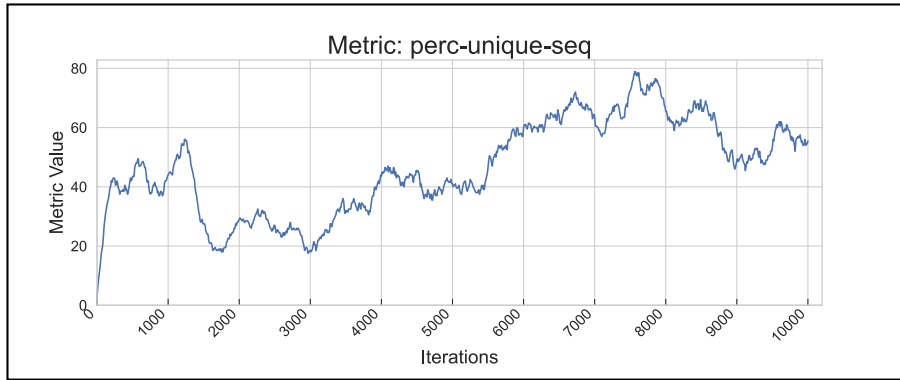


Figure 4.27 Complexity metric: Percentage unique sequences. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.27 illustrates the process drift simulation results for the percentage of unique sequences (“perc-unique-seq”). The curve shows recurring fluctuations, a general long-term upward tendency, and periods of local stabilization. At the start of the simulation, the metric increases sharply from near zero to values around 40–50 by iteration 1000. After this initial rise, the curve shows alternating peaks and troughs between 20 and 50 until approximately iteration 5000, reflecting short-term oscillations without a clear directional trend. From iteration 5500 onward, the curve demonstrates a more pronounced upward trajectory. The metric climbs steadily, reaching values above 70 near iteration 8000, which represents the maximum observed level. In the later part of the simulation (iterations 8500 - 10,000), the curve decreases slightly but stabilizes in the range between 55 and 65, with moderate fluctuations around this plateau.

In summary, the curve of the time series for the percentage of unique sequences metric is characterized by an initial sharp increase, a prolonged oscillatory phase, and a steady upward trend during the second half of the simulation that culminates in a high plateau toward the end of the simulation.

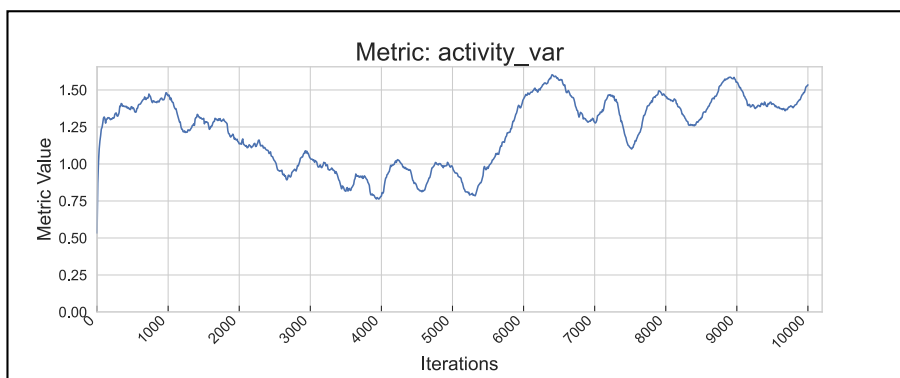


Figure 4.28 Complexity metric: Activity variation. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.28 illustrates the process drift simulation results for the activity variation (“activity_var”). The curve displays alternating phases of decline and growth, forming a wave-like pattern with moderate oscillations around an upward-shifting baseline. In the early iterations up until iteration 1000, the metric rises steadily from near 1.0 to values around 1.5. This is followed by a gradual decline between iterations 1000 and 4000, during which the metric drops below 1.0 at its lowest points. During iteration 4000 and 5000, the metric oscillates in a range between roughly 0.75 and 1.0. Beginning around iteration 5000, the curve reverses direction, showing a strong upward movement that peaks above 1.5 near iteration 7000. In the subsequent period between iterations 7000 and 10,000, the metric continues to oscillate, alternating between 1.3 and 1.6, without a clear long-term decline, maintaining relatively high levels throughout.

In summary, the curve of the time series for the activity variation metric is characterized by a wave-like dynamic: an early rise, a mid-simulation decline to lower values, and a strong recovery with sustained oscillations at higher levels toward the end of the simulation.

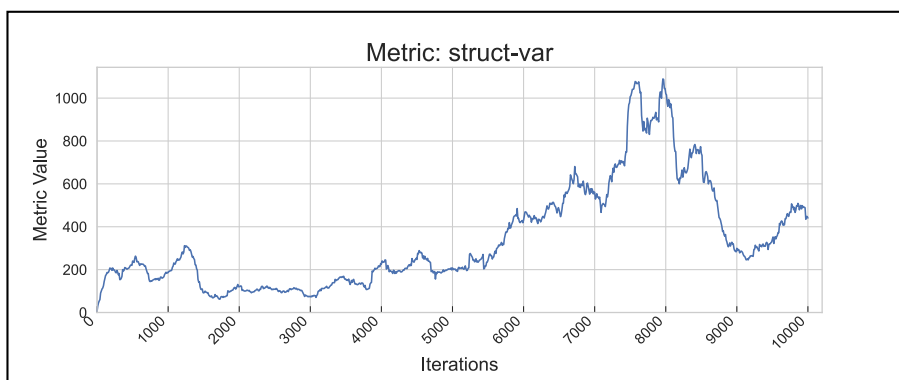


Figure 4.29 Complexity metric: Structural variety. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.29 illustrates the process drift simulation results for the structural variety (“struct-var”). The curve exhibits a combination of gradual long-term growth, pronounced peaks, and subsequent declines, giving it a multi-phase dynamic profile. In the early iterations, up to around iteration 1200, the metric rises from near zero to values around 300, with several small oscillations. Afterwards, between iteration 1200 and 1500, there is a pronounced decline back to metric levels under 100. Between iterations 1500 and 5500, the curve fluctuates within a relatively narrow band (100–300), reflecting a prolonged period of moderate stability. From iteration 5500 onward, the curve begins a pronounced upward trajectory, accelerating into a phase of strong growth. The metric reaches its highest values between iterations 7500 and 8200, peaking above 1000. This late-phase peak is followed by a noticeable decline between iterations 8200 and 9000,

bringing values back down to a range between 300 and 400. In the final part of the simulation, between iterations 9000 and 10,000, the curve rises again modestly, stabilizing at levels between 400 and 500.

In summary, the curve of the time series for the structural variety metric is characterized by an initial rise, a prolonged middle phase of moderate stability, a strong late growth culminating in a major peak or spike, and a subsequent decline with partial recovery at the end.

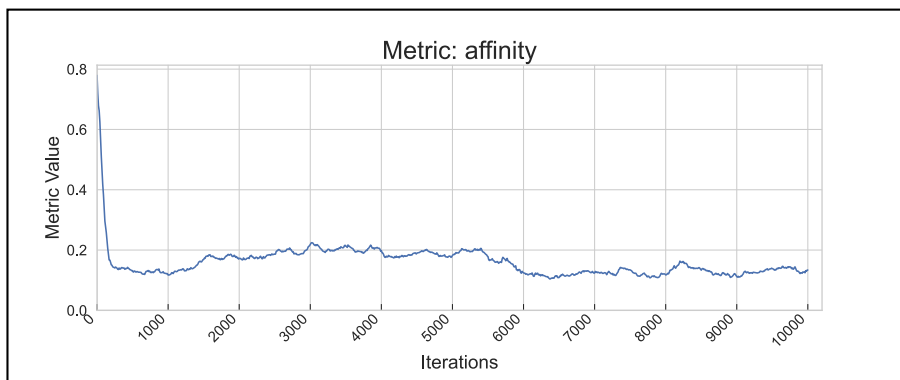


Figure 4.30 Complexity metric: Affinity. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.30 illustrates the process drift simulation results for the affinity (“affinity”). The curve is characterized by an early steep drop followed by a long period of stability at lower levels, with only minor oscillations. At the beginning of the simulation, the metric starts at a high value of around 0.8. Within the first 500 iterations, it decreases sharply, levelling off near a range between 0.1 and 0.2. This coincides with the simulation start, and represents a close to eight-fold reduction in the metric during these initial 500 iterations of the simulation. From iteration 1000 onward, the curve maintains this lower range, with small fluctuations but no major directional shifts, with only temporarily plateauing around levels of 0.2 between iterations 2000 and 5500. Across the remainder of the simulation, between iterations 5500 and 10,000, the metric remains stable, oscillating around values close to 0.15. While minor local peaks and dips occur, they do not alter the long-term flat trajectory.

In summary, the curve of the time series for the affinity metric is defined by a two-phase pattern: a rapid initial decline, followed by a prolonged period of stability at consistently lower levels.

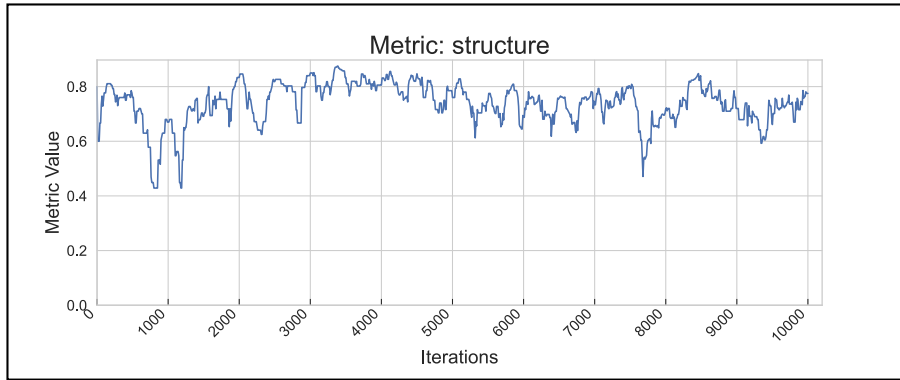


Figure 4.31 Complexity metric: Structure. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.31 illustrates the process drift simulation results for the structure (“structure”). The curve demonstrates a consistently oscillating pattern, with values remaining within a relatively narrow range throughout the simulation. At the start of the simulation, up until around iteration 1000, the metric fluctuates more strongly, with brief dips below 0.5 before recovering quickly to higher levels. From iteration 1500 onward, the curve stabilizes into a recurring oscillatory pattern, typically ranging between 0.65 and 0.85. Throughout the remainder of the simulation, for iterations 2000 to 10,000, the curve continues to oscillate frequently within this band. While no sustained upward or downward trend is visible, short-term peaks above 0.8 and dips near 0.6 appear regularly. One of these dips happens between iteration 7500 and 7700, with a rapid decline from metric values of 0.8 to below 0.5. However, this decline is again quickly recovered, with a counter-oscillation back to levels above 0.8 around iteration 8400.

In summary, the curve of the time series for the structure metric is characterized by high-frequency oscillations around a stable central band, with no major long-term shifts but persistent local variability.

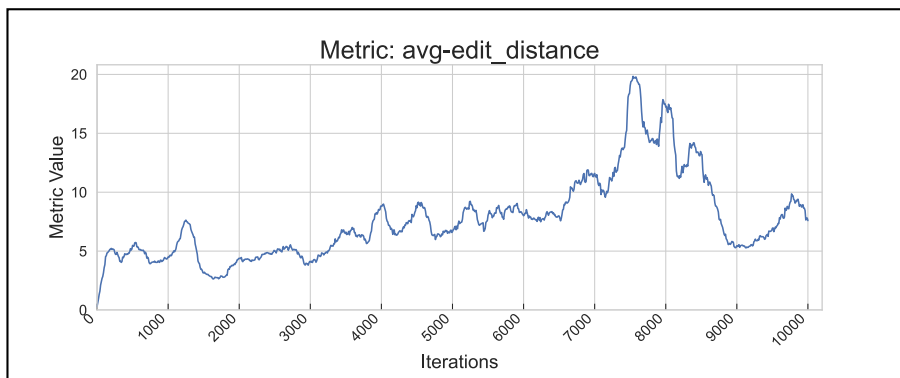


Figure 4.32 Complexity metric: Average edit distance. Process drift simulation measurement results for Mannhardt hospital event log.

Figure 4.32 illustrates the process drift simulation results for the average edit distance (“avg-edit_distance”). The curve displays a long-term upward tendency, punctuated by recurring oscillations and a pronounced late-phase peak, followed by a rapid decline back to lower levels. In the early iterations up to iteration 2000, the metric rises quickly from zero to values between 5 and 7, with visible oscillations. Between iterations 2000 and 5000, the curve alternates between smaller peaks and troughs, fluctuating mostly within a range of 4 to 9. From iteration 6000 onward, the metric enters a stronger growth phase, with values climbing steadily. Around iterations 7500 to 8200, the curve reaches its maximum, peaking close to 20, followed by another local maximum at levels above 17. After this peak, the metric declines sharply, stabilizing between 6 and 10 in the closing iterations, though with continued short-term oscillations.

In summary, the curve of the time series for the average edit distance metric is characterized by an initial rise, a prolonged oscillatory middle phase, a strong late surge culminating in a major peak, and a subsequent decline toward moderate values at the end of the simulation.

4.3.3 Simulation Results BPI 2011 Hospital Event Log

To provide further narrative examples of the process drift dynamics from a comparable background, with representative entropy patterns and insightful metric trends, curves are plotted for the simulation results time series for the same set of metrics as before, but for the BPI 2011 hospital event log. The overall count of simulation iterations remains unchanged.

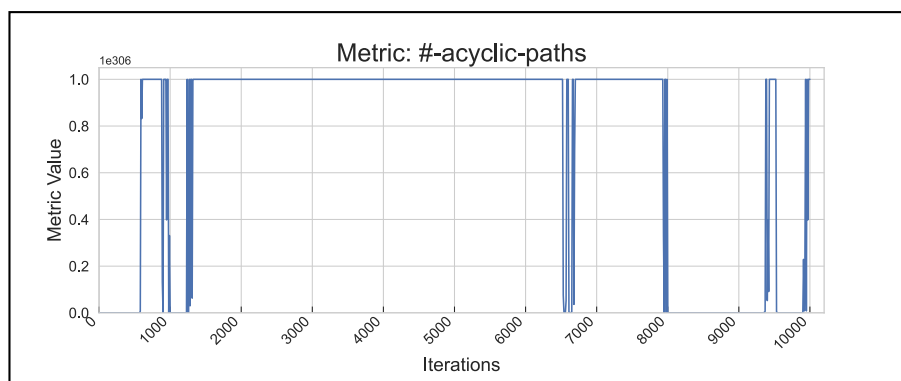


Figure 4.33 Complexity metric: Number of acyclic paths. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.33 illustrates the process drift simulation results for the number of acyclic paths (“#-acyclic-paths”). The curve is dominated by extreme spikes that appear intermittently, separated by long stretches of near-zero values or stable plateaus. In the early iterations,

up to around 1300, the metric shows several abrupt surges, reaching extremely large values on the order of 10^{306} , followed by immediate collapses back to near-zero, or by intermittent plateaus at the same level. After this phase, the curve remains at this extremely high level for a significant portion of the simulation window, until around iteration 6500. Similar extreme spikes reappear at several later points in the simulation, notably around iterations 6500, 8000, and 9000, and again near iteration 10,000. These peaks are highly localized, occurring suddenly and vanishing just as quickly, leaving the curve flat in the intervals between them.

In summary, the curve for the time series of the number of acyclic paths metric for this simulation is characterized by a highly irregular flat–spike pattern, with short-lived extreme peaks interspersed within long phases of zero values or plateaus, at maximum reaching values of 10^{306} .

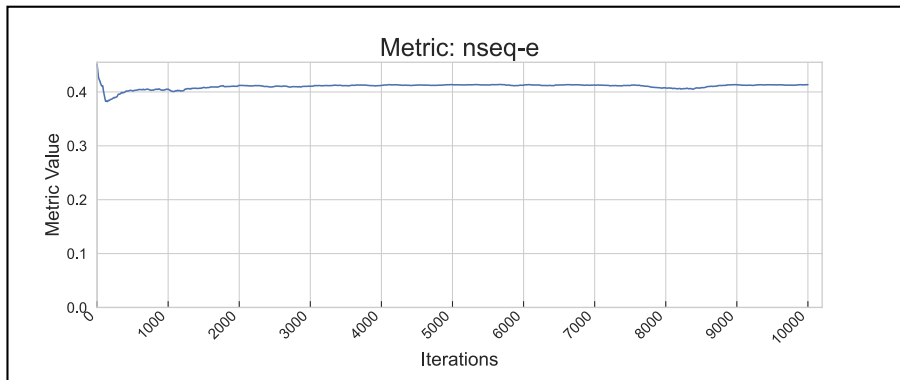


Figure 4.34 Complexity metric: Normalized sequence entropy. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.34 illustrates the process drift simulation results for the normalized sequence entropy (“nseq-e”). The curve exhibits an initial drop, and an afterwards stable trajectory with only minimal variation throughout the entire simulation. At the beginning, the metric starts at around 0.45, quickly dipping to levels below 0.4 before stabilizing. From iteration 500 onward, the curve levels off around values between 0.40 and 0.42. This stable pattern persists consistently across the remainder of the simulation, with only minor local oscillations visible but no significant upward or downward trends.

In summary, the curve of the time series for the normalized sequence entropy metric is characterized by long-term stability at a consistent level, showing minimal fluctuations and no major directional changes across the simulation window.

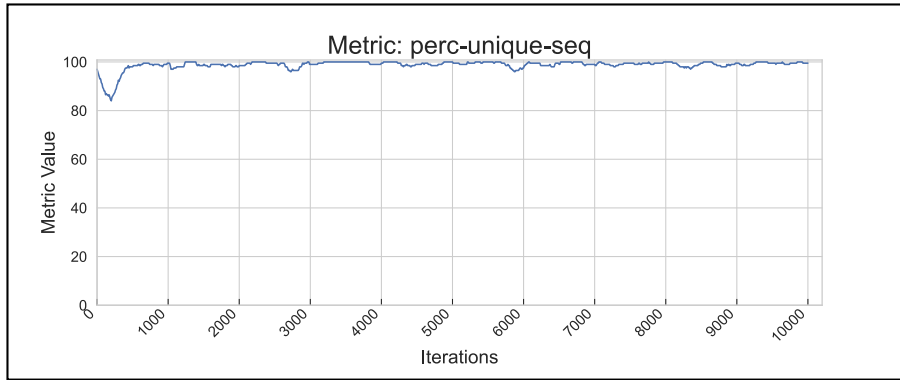


Figure 4.35 Complexity metric: Percentage unique sequences. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.35 illustrates the process drift simulation results for the percentage of unique sequences (“perc-unique-seq”). The curve is marked by persistently high values, with only minor fluctuations over the course of the simulation. At the start, the metric begins slightly below 90 % but quickly rises toward levels of 98 % to 100 % within the first 1000 iterations. From this point onward, the curve stabilizes, remaining consistently between 98 % and 100 %. Minor short-term dips appear occasionally, for example around iterations 2000 and 3000, but these are brief, and the metric rapidly returns to near-maximum levels. Across the later iterations, from iteration 5000 to 10,000, the curve continues to exhibit the same stable high-level behaviour, oscillating narrowly around 98 % to 100 % without significant downward or upward deviations.

In summary, the curve of the time series for the percentage of unique sequences metric is characterized by an early dip followed by long-term stability at near-maximum values, with only minor and short-lived dips throughout the simulation.

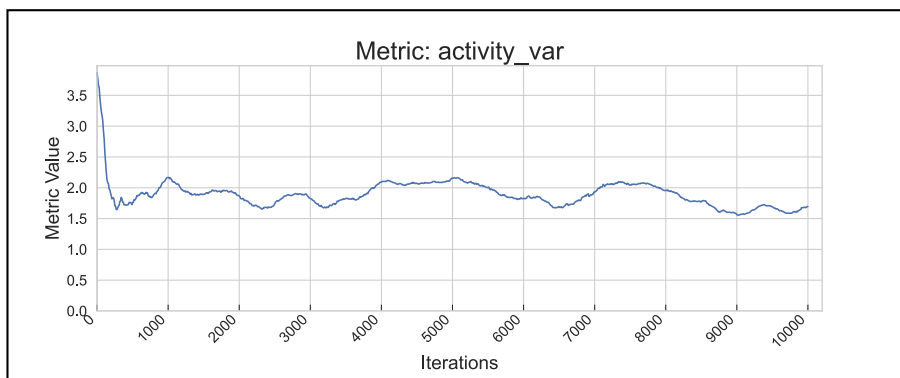


Figure 4.36 Complexity metric: Activity variation. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.36 illustrates the process drift simulation results for the activity variation (“activity_var”). The curve is characterized by an initial steep decline, followed by

oscillations around a relatively stable mid-level range, and a gradual downward tendency toward the end. At the start of the simulation, the metric begins at a high metric value above 3.5 but drops sharply during the first 300 iterations, stabilizing near 1.8 to 2.0. From iterations 300 to 5000, the curve exhibits oscillatory behaviour, with repeated rises and falls within this range, showing no strong long-term directional trend. After iteration 5000, the oscillations persist but with a subtle downward drift. By the final iterations, the curve stabilizes closer to values around 1.5 to 1.7, marking the lowest levels since the early sharp drop.

In summary, the curve of the time series for the activity variation metric is characterized by an early steep decline, a prolonged oscillatory middle phase, and a late slight downward drift, settling into a lower range by the end of the simulation.

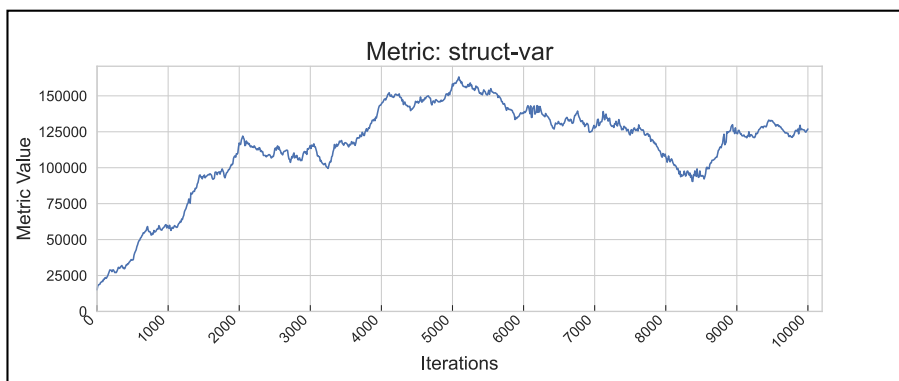


Figure 4.37 Complexity metric: Structural variety. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.37 illustrates the process drift simulation results for the structural variety (“struct-var”). The curve exhibits a strong early growth phase, a mid-simulation peak, and subsequent fluctuations with a subsequent decline with a late partial recovery. At the start of the simulation, the metric begins at values around 20,000 and rises steadily, reaching approximately 100,000 by iteration 2000. Between iterations 2000 and 5000, the curve continues to increase steadily, peaking at values above 150,000, which represents the highest sustained level of the entire simulation. Notably, there are no sudden peaks or spikes in the metric value. After this peak, the metric shows a gradual downward trend with fluctuations, stabilizing near 120,000 by iteration 7000. Around iteration 8500, the curve drops more sharply, reaching a local minimum slightly below 100,000. In the final iterations, the curve recovers partially, stabilizing between 120,000 and 130,000.

In summary, the curve of the time series for the structural variety metric is characterized by strong early growth, a mid-simulation peak, a decline with fluctuations, and a late partial recovery, maintaining relatively high values throughout the simulation.

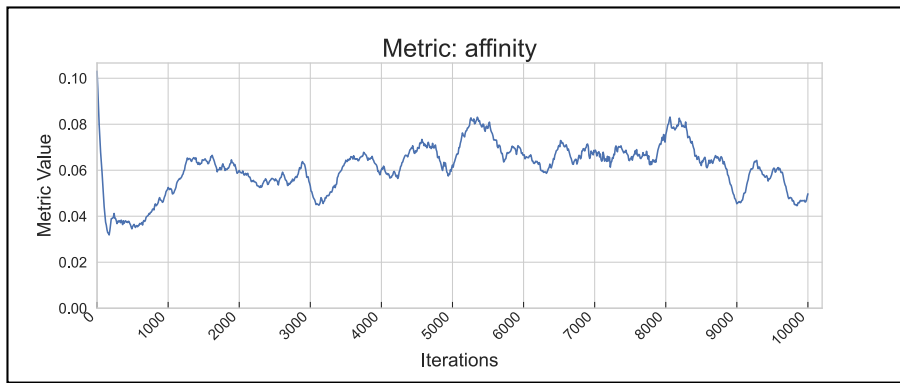


Figure 4.38 Complexity metric: Affinity. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.38 illustrates the process drift simulation results for the affinity (“affinity”). The curve demonstrates a significant early decline, relative recovery, followed by moderate fluctuations around a relatively narrow band, with phases of gradual increase followed by stabilization and decline. At the start of the simulation, the metric begins at a relatively high value near 0.10 but quickly drops below 0.05 within the first 200 iterations. After this early decline, the curve rises again, reaching values between 0.06 and 0.07 around iteration 1500. Between iterations 2000 and 7000, the metric maintains a generally stable profile with oscillations, typically fluctuating between 0.05 and 0.08. Several local peaks appear during this phase, including two sharper rises around iterations 5500 and 8000, both exceeding 0.08. In the final iterations, between iterations 9000 and 10,000, the curve trends downward, falling back toward values around 0.05, marking the lowest levels since the early decline.

In summary, the curve of the time series for the affinity metric is characterized by an initial drop, a long mid-simulation phase of oscillatory stability with intermittent peaks, and a final decline toward lower values.

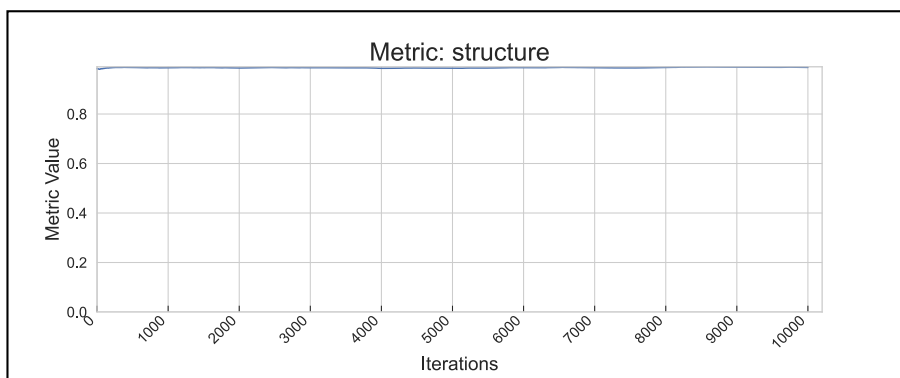


Figure 4.39 Complexity metric: Structure. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.39 illustrates the process drift simulation results for the structure (“structure”). The curve remains virtually constant throughout the entire simulation, with values close to 1.0. From the beginning to the end of the simulation, for iterations 0 to 10,000, the metric shows no visible upward or downward trends. Only minimal and barely perceptible variations can be observed, but the overall trajectory is essentially flat.

In summary, the curve of the time series for the structure metric is characterized by exceptional stability, maintaining values close to the upper boundary of its range across the full simulation without meaningful fluctuations.

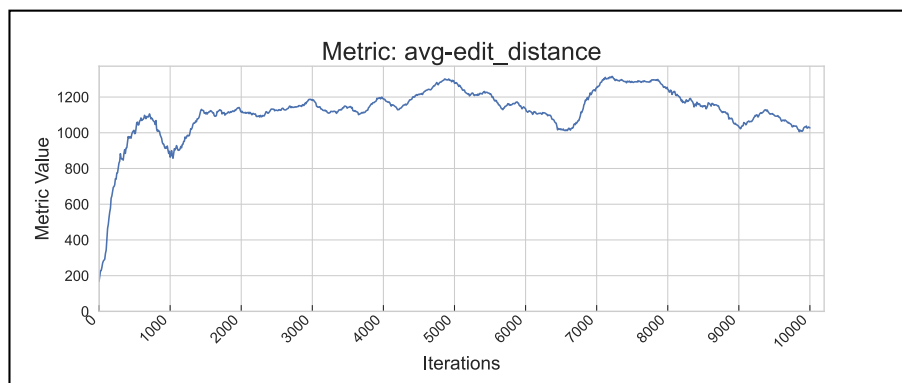


Figure 4.40 Complexity metric: Average edit distance. Process drift simulation measurement results for BPI 2011 hospital event log.

Figure 4.40 illustrates the process drift simulation results for the average edit distance (“avg-edit_distance”). The curve shows a rapid early rise, a prolonged phase of stability with moderate oscillations, and smaller fluctuations toward the end. In the initial iterations up to iteration 1000, the metric increases sharply from values around 200 to values around 1100, followed by a short dip to around 900 and quick recovery back to levels of 1100. From iteration 1500 onward, the curve stabilizes within the range of 1000–1200, with periodic oscillations. Between iterations 4000 and 8000, the metric reaches its highest sustained levels, with several peaks exceeding 1200. This is followed by a gradual downward shift beginning after iteration 8000, where the curve trends toward values between 950 and 1050 by the end of the simulation.

In summary, the curve of the time series for the average edit distance is characterized by an early steep rise, a long mid-simulation plateau with oscillations around high values, and a late slight decline that stabilizes at moderately lower levels.

5 Discussion

5.1 Summary of Findings

In the empirical, event-log based analysis of various process complexity metrics, the relationships between these metrics, their dynamics over time, and their behaviour in the context of and sensitivity to process drift are examined. The findings highlight multiple key insights, in particular that process complexity metrics exhibit complex, diverse, and sometimes divergent patterns, both over time in real-world event logs, and in response to simulated process drift.

The statistical correlation analysis reveals that a multitude of complexity metrics exhibit strong, statistically significant correlations, while others exhibit only weak or neutral correlations. The Spearman correlation yields the highest number of statistically significant correlations, given a strict yet commonly used p-value threshold of 0.01, which is expected given that the Spearman method is less sensitive to outliers and does not require linear relationships between the variables. As such, the Spearman method is the preferred method for the correlation analysis.

Apart from these methodological insights, the correlation analysis reveals critical characteristics of the relationships between the process complexity metrics in scope of the thesis. In line with the basic properties of the correlations, the complexity metrics exhibit strong positive, neutral or ambiguous, and strong negative correlations. Strong positive correlations are found for the relationships between the basic entropy metrics, such as the variant entropy (“var-e”) and the sequence entropy (“seq-e”) and the normalized entropy metrics such as the normalized variant entropy (“nvar-e”) or the normalized sequence entropy (Nseq-e”), and between the enriched entropy metrics, such as between the enriched variant entropy (“en-var-e”) and the enriched sequence entropy (“en-seq-e”). Additionally, the analysis finds strong positive correlations between among a set of complexity metrics, encompassing the variant entropy (“var-e”), the sequence entropy (“seq-e”), the lempel-ziv complexity (“lempel-ziv”), the number of ties (“#-ties”), the number of events (“#-e”), the structural variety (“struct-var”), the number of acyclic paths (“#-acyclic-paths”) and the number of unique sequences (“#-unique-seq”). These metrics all share strong positive correlations among each other, forming a cluster of metrics that appear to be uniquely connected and thus correlated. Additionally, there is another such cluster, sharing strong positive correlations among each other, consisting of the percentage of unique sequences (“perc-unique-seq”), the deviation from random (“dev_random”), the order variation (“order_var”), the activity variation (“activity_var”), the average number of distinct events (“avg-distinct-e”) and the average sequence length

(“avg-seq-len”). This cluster also exhibits strong to moderately strong positive correlations with the structure (“structure”). Overall, these two clusters seem to represent two distinct groups of complexity metrics, that share similarities and dependencies among themselves, leading to similar dynamics and behaviours when applied to event log datasets. These similarities and dependencies likely arise from shared computational approaches and methods, representing similar perspectives on process complexities, unless there is another set of variables that affect each metric in the cluster, acting as confounders in this analysis. In turn, both clusters exhibit weak, neutral, or even negative correlations vis-à-vis each other, thereby providing different perspectives on process complexity.

Other metrics similarly exhibit neutral correlations, such as between the affinity (“affinity”) and the lempel-ziv complexity (“lempel-ziv”) or the normalized sequence entropy (“nseq-e”), or between the normalized variant entropy (“nvar-e”) and the number of events (“#-e”) or the maximum sequence length (“max-seq-len”). Furthermore, similar neutral correlations are also exhibited between the lempel-ziv complexity (“lempel-ziv”) and the order variation (“order_var”) and the affinity (“affinity”), or between the structural variety (“struct-var”) and the order variation (“order_var”) and affinity (“affinity”). These neutral correlations highlight another distinct pattern among the process complexity metrics, with metrics that are invariant to each other, without shared dependencies or other influences. These metrics thus provide nearly non-overlapping perspectives on process complexity, with distinct lenses and computational approaches that increase their combined informational value for process mining applications and process complexity research alike. Similarly, metrics that exhibit only weak and near-neutral correlations, be they positive or negative, do not yield clear insights into their respective behaviour, indicating an ambiguous relationship that might be likewise caused by distinct perspectives on process complexity, or which might require closer case-by-case analyses due to event-log-specific effects from the data distribution or the data characteristics.

Lastly, there is another group of metrics that highlight a third key pattern as part of the correlation analysis. The strongly negatively correlated metrics, such as seen between the normalized variant entropy (“nvar-e”) and the percentage of unique sequences (“perc-unique-seq”) and the minimum sequence length (“min-seq-len”), or between the percentage of unique sequences (“perc-unique-seq”) and the number of unique sequences (“#-unique-seq”) or the average time difference between events (“avg-td-e”), or the average sequence length (“avg-seq-len”) and the average time difference between events (“avg-td-e”), represent adverse and opposing metric behaviours. These opposing metrics exhibit inverse dynamics, reacting in opposite directions for the same influence. This

effect is likely caused shared underlying dependencies, as can be seen with the strongly positively correlated metrics, combined with inverse calculation approaches that then lead to the inverse dynamics. These diverging approaches then represent different perspectives on process complexity, yet for the same underlying process characteristics, in essence representing opposing perspectives on what constitutes process complexity.

The findings of the empirical temporal dynamics analysis on the selected Mannhardt and BPI 2011 hospital event logs reveal multiple shared patterns across the range of complexity metrics. The curves highlighted in the analysis reveal patterns consisting of sharp and dramatic spikes or bursts, mostly stable oscillations, stable long-time trends, and late-phase surges or declines, either isolated or in combination. Metrics such as the number of acyclic paths (“#-acyclic-paths”), the structural variety (“struct-var”) or the average edit distance (“avg-edit_distance”) exhibit a pattern characteristics by significant spikes or bursts in the complexity metrics value, either reoccurring in the case of the number of acyclic paths and the average edit distance, or with a single occurrence as is the case for the structural variety. For the number of acyclic paths, the spike with the largest magnitude occurs during the second quarter of 2013 for the Mannhardt hospital event log, and during the first quarter of 2005 for the BPI 2011 hospital event log. The spike in the average edit distance metric with the largest magnitude occurs during the first quarter of 2007, and only in the BPI 2011 hospital event log, while the spike in the structural variety occurs only once for either event log, at the end of the first quarter 2013 for the Mannhardt hospital event log, and during the first quarter for the BPI 2011 hospital event log. These spikes or burst indicate sudden changes or shifts in the composition of the sublogs for which these metrics were calculated. These may then have outsized effect on individual complexity metrics, dependent on the computation and on the dependencies of the respective metric. In this context, the structure (“structure”) exhibits another noteworthy yet conflicting pattern, when viewed in conjunction with the structural variety. While the structural variety metric peaks in the Mannhardt hospital event log, the structure metric exhibits a sudden dip, falling to its lowest level over the entire observation period. This behaviour reinforces the finding of conflicting complexity metrics from the correlation analysis, highlighting the opposing dynamics at play here.

The pattern of relatively stable oscillations around a narrow metric range is exhibited by the normalized sequence entropy (“nseq-e”), the percentage of unique sequences (“perc-unique-seq”), the activity variation (“activity_var”), the affinity (“affinity”), and the structure (“structure”) complexity metric. This pattern is present in both event logs chosen for the narrative examples and closely matched by a majority of other temporal dynamics from the remaining event logs. These stable patterns reflect that the respective metric were not affected by changes in the underlying sublogs, hinting at independence from the

respective factors or variables, or at robust calculation methods that are insensitive to sudden yet intermittent change. Similarly, the structural variety (“struct-var”) exhibits a similarly stable curve with only minor oscillations, but with a distinct, stable long-term downwards trend across the entire observation period, and across both event logs chosen for the narrative examples. This pattern is also confirmed by the Mann-Kendall test for both event log datasets, and likewise present in multiple other event logs. This stable trend pattern likely reflects a trend in the underlying sublogs, which may stem from the overall distribution of the event log dataset or from an underlying, continuous dynamic in the event log dataset, such as an underlying process drift that slowly but steadily changes the event log characteristics and composition.

Finally, the pattern of late-phase surges or declines is exhibited by the percentage of unique sequences (“perc-unique-seq”), the activity variation (“activity_var”), the affinity (“affinity”), and the average edit distance (“avg-edit_distance”), each time in the last segments of the observation periods. However, this pattern is less intuitive than those mentioned before, with diverging patterns across event log datasets and significant variations in the patterns strength. The percentage of unique sequences surges only in the last segment of the Mannhardt hospital event log, which is also true for the average edit distance. Both do not show the same behaviour for the BPI 2011 hospital event log. The activity variation in turn exhibits a decline for both datasets, but while the decline for the Mannhardt hospital event log is significant and multitudes larger than previous oscillations, the decline in the BPI 2011 hospital event log is only minor in comparison. Additionally, the affinity metric surges in the last segment of the Mannhardt hospital event log, while declining in the BPI 2011 hospital event log. These late-phase surges and declines are likely to be caused by the distribution of the event log datasets in combination with the bracketing of traces for the sliding window approach used for the temporal dynamics analysis. Due to this approach, sublogs at the beginning or the end of the event log dataset can exhibit drastically different characteristics from other sublogs, for example with regard to the number of traces in the respective sublog. As such, these represent edge-cases in the analysis of event logs, which might cause these drastic late-phase changes in some of the complexity metrics that are sensitive, due to computation or dependencies, to such changes. Other factors that might cause such changes could be sudden process changes in the respective observation period.

The process drift simulation, in turn, highlights and reinforces the key points raised by the analysis of the temporal dynamics. The findings again highlight different shared patterns, which consist of sudden spikes or bursts, oscillatory and wave-like patterns, phases of growth, decline, or recovery, and sudden surges or declines at the fringe phases of the observation period. Once again, the spike- or burst-pattern is particularly

noteworthy with the number of acyclic paths metric (“#-acyclic-paths”), appearing in both datasets. While the spike occurs once for the Mannhardt hospital event log around iteration 8000, it appears multiple times for the BPI 2011 hospital event log, partly followed by a persistent plateau phase. These plateaus for the BPI 2011 are caused by technical limitations in conjunction with the computation of the number of acyclic paths, which utilizes exponential variables derived from event log properties, thereby exceeding the bounds of the technical data type in these cases. The actual, unbounded metric is thus still expected to exhibit the same pattern visible for the Mannhardt hospital event log and other event log datasets. Similar, yet smaller peaks with a lower magnitude are visible for the structural variety (“struct-var”) and the average edit distance (“avg-edit_distance”), but only for the Mannhardt hospital event log, both appearing close to iteration 7500. Given the window size of 200 traces, these two peaks are thus only linked to the peak exhibited by the number of acyclic paths by a subset of traces. In turn, the latter two peaks coincide with local maxima of the percentage of unique sequences (“perc-unique-seq”) and the average sequence length (“avg-sequence-length”). The number of acyclic paths in turn coincides with a subsequent local maximum of the average sequence length (“avg-seq-len”) and an intermittent dip of the number of event types (“#-et”), which, combined with the exponential function utilizing the number of event types and the number of direct follows relationships between event types, which is prone to increase non-linearly, explains the mechanism behind the sudden peaks and bursts of the number of acyclic paths metric.

The oscillatory patterns on the other hand are exhibited by metrics such as the structure (“structure”), and the normalized sequence entropy (“nseq-e”), as well as in some segments by the percentage of unique sequences (“perc-unique-seq”), the activity variation (“activity_var”), the structural variety (“struct-var”), and the average edit distance (“avg-edit_distance”). For the former, this pattern translates into a simulation-long, oscillating, but otherwise stable curve, with a narrow range between peaks and troughs. This pattern is visible for both narrative example datasets, with diminishing oscillations approaching a constant curve for the BPI 2011 hospital event log. For the latter, this oscillatory pattern is only present in segments of the overall simulation observation period, such as in the first half of the simulation, until roughly iteration 5000, for the Mannhardt hospital event log, or for larger segments in the BPI 2011 hospital event log. The more stable oscillatory pattern hints at less sensitive metrics, as seen before with the temporal dynamics, with the minor difference of the underlying iterative change as implemented for the process drift simulation. As such, these metrics demonstrate reduced sensitivity to incremental process drift through random variations in the direct-follows relationships of event types in the presence of fixed event log sizes and fixed maximum trace lengths. Building on this, the latter metrics exhibit phases of growth,

decline, and recovery outside their respective oscillatory patterns. These phases are particularly noteworthy for the structural variety, which first exhibits the stable oscillatory pattern, followed by a persistent growth phase, a short minor peak as described before, and decline phase with a subsequent partial recovery, all for the Mannhardt hospital event log. For the BPI 2011 hospital event log, it exhibits a substantial growth phase, followed some stable oscillations and then a lasting decline with a subsequent recovery. Other metrics, such as the activity variation for the Mannhardt hospital event log, or the average edit distance for both narrative example event logs, exhibit similarly phased patterns. These phases signal the presence of underlying trends in the event log characteristics, similar to the long-term trends found in the temporal dynamics of the structural variety, while also highlighting that the identified metrics are sensitive to such changes, in contrast to those metrics that do not share these phase patterns.

Lastly, the process drift simulation results also show patterns consisting of rapid surges or declines at the beginning of the process drift simulation, mirroring those found during the temporal dynamics analysis at the end of the observation period. These are particularly apparent in the curves of the affinity (“affinity”) metric, with a rapid decline at the start of the simulation for both of the narrative example event logs, during the first roughly 200 iterations of the simulation. However, this pattern is also visible for the normalized sequence entropy (“nseq-e”), with a decline at the start of the simulation, the percentage of unique sequences (“perc-unique-seq”), with a surge at the start of the simulation, the activity variation (“activity_var”), with a decline at the start of the simulation, and the average edit distance (“avg-edit_distance”), with a rapid surge at the start of the simulation. This pattern is not exhibited by all complexity metrics chosen for the narrative examples, and likewise not exhibited by all metrics in scope of the thesis, as can be seen in the appendix. These early fluctuations emphasize the high sensitivity of some of the metrics to the initial stages of process drift, particularly when anchored in real-world log data. Accordingly, the metrics highlighted before both serve as early signals for incremental change in the underlying process, while also necessitating special caution when relying solely on these metrics to measure and assess process complexity, as even relatively small variations can lead to outsized impacts on these metrics. It thus also demonstrates that different metrics respond non-linearly and asymmetrically to process drift. These findings reinforce the value of using complementary metrics when analysing process complexity under changing conditions.

Combining these key findings of the three analyses grants a comprehensive perspective on the intricate and multi-faceted relationships between the complexity metrics, reinforcing the notion that process complexity itself is a multi-faceted concept, that requires different lenses to capture all its aspects. In particular, these findings enable

making sense of the dynamics observed in the event logs and in the process drift simulation and theorizing about the underlying causes and potential structures within process complexity. By grouping and clustering the relationships between the process complexity metrics and linking these clusters to the observed temporal dynamics and drift dynamics, several distinct patterns become apparent.

First of all, the pattern consisting of singular or reoccurring spikes or bursts in the respective process complexity metric may represent *tipping points*. The sudden spikes or bursts around these tipping points, such as seen with the number of acyclic paths metric, or to a smaller degree with the structural variety or the average edit distance metrics, reflect recent findings of process simulations in the context of process complexity measurement under the condition of process drift (B. T. Pentland, Liu, et al., 2020; B. T. Pentland et al., 2021). Similarly, they likewise mirror the sudden appearances of gradients during process drift detection, either with computer vision based method (Kraus & van der Aa, 2024; Yeshchenko et al., 2022), or with transition matrices (Zhang et al., 2025; Zheng et al., 2017). These sudden surges, culminating in spikes or bursts of complexity that are sometimes multitudes larger than the previous oscillations in the metrics, may represent tipping points for the respective metric, where incremental changes suddenly lead to outsized impacts on the metric, thus representing critical points to analyse and understand the dynamics of process complexity.

Secondly, the diverging, inverse dynamics of metrics may represent *trade-offs* between these metrics when measuring process complexity. These dynamics, found as part of the correlation analysis in the form of strongly negatively correlated metrics, as part of the temporal dynamics in the diverging sudden surges or declines at the end of the observation period, and as part of the process drift simulation in the rapid surge or decline at the start of the simulation, show that certain metrics react inversely to the same exogenous influences. As stated before, these inverse behaviours may represent different perspectives on process complexity that form the basis of the respective metrics. This naturally leads to the notion of trade-offs when utilizing these metrics to measure process complexity. While changes, such as those caused by process drift, may cause one metric to surge, another may decline in kind. Integrating metrics that exhibit such trade-offs both provides insights into such situations and provides a more comprehensive lens on process complexity, providing context that may indicate that the underlying process complexity has not actually increased or decreased, but rather shifted from one aspect to another.

Thirdly, the correlation analysis in particular highlights the existence of uncorrelated relationships between metrics, potentially representing *invariant* process complexity metrics. Such invariant metrics are characterized by neutral or weak correlations with

other complexity metrics and exhibit distinct patterns in the temporal dynamics analysis. These patterns are stable, and stationary with no clear detectable trend, resulting in near-constant curves, highlighting that these metrics are uncorrelated to metrics with pronounced dynamics. Applied to the process drift simulation results, this concept can be extended to define *drift invariant* process complexity metrics, which likewise exhibit stationary patterns without detectable trend under the condition of incremental drift, as simulated during the process drift simulation. Such metrics appear unresponsive and insensitive to process changes, thus limiting their ability to signal process change.

Another theoretical caveat of the correlation analysis is the existence of distinct groups of metrics exhibit strong positive correlations vis-à-vis each other. These groups may represent *clusters* of process complexity metrics, with strong intra-cluster correlations and dependencies, and low inter-cluster correlations and dependencies. Such clusters, if existent, could explain the pattern of shared process dynamics and process drift dynamics across the process complexity metrics, and would further provide structure for the evaluation and application of process complexity metrics.

Lastly, the analyses reveal subtle yet distinct differences between the patterns found as part of the temporal dynamics analysis and the process drift simulation. While the empirical patterns during the temporal dynamics analysis are dominated by relatively stable patterns, with limited oscillations and rare long-term trends, and some sudden bursts, surges, or declines, the patterns found during the process drift simulation consist of more frequent and more volatile oscillations and wave-like patterns. In particular, these patterns seem to represent distinct cycles of growth, peak, decline, and recovery. These increased dynamics may directly represent the influence of the induced incremental process drift, leading to increased dynamics in the process complexity metrics compared to lower levels of drift in the real-world event log datasets, and seem to become visible in the structure of the patterns revealed by the analyses.

5.2 Theoretical Contributions

As part of the empirical, event-log-based analysis of various process complexity metrics, the relationships among these metrics, their temporal dynamics, and their sensitivity to drift were examined. The findings highlight that process complexity metrics exhibit complex and sometimes divergent behaviours, both over time and in response to process drift. While some metrics exhibit linear or constant trends, others show non-linear dynamics, including sudden bursts of complexity in the form of tipping points. These differences, along with partially matching and diverging temporal patterns, suggest that the complexity metrics are influenced by distinct process characteristics and capture different facets of process behaviour. Regarding process drift, the findings indicate that

the complexity metrics vary in their sensitivity to incremental changes in the process, affecting both the extent to which drift impacts the metrics and their suitability for detecting such drift. Across the three analyses, the findings provide a theoretical structure for the process complexity dynamics, conceptually distinguishing between *tipping points*, *trade-offs*, *invariant* and *drift-invariant* metrics, and finding *clusters* of metrics with diverging behaviours and relationships. Overall, no previous studies elucidate the relationships and the temporal and process drift dynamics of process complexity metrics with the same scope and breadth as this thesis. In light of these findings, this thesis makes three central contributions.

First, the thesis offers a comprehensive correlation analysis of the interrelationships among process complexity metrics, expanding existing research (Augusto et al., 2022; Vidgof & Mendling, 2023; Vidgof et al., 2023) by incorporating novel metrics from adjacent fields. The correlation analysis substantiates the claim that process complexity is inherently multi-faceted (Lauterbach et al., 2020), with different metrics and in particular with different clusters of metrics providing distinct and complementary lenses for understanding and measuring complexity. These differentiated perspectives not only deepen the understanding of what specific aspects of complexity individual metrics capture but also enable the identification of distinct patterns among the dynamics exhibited by the complexity metrics. These include highly correlated interdependencies among clusters, non-correlated invariant metrics, and trade-offs between other metrics that suggest shifts across dimensions of complexity, whereby a decrease in one metric may coincide with an increase in another, suggesting that complexity is not necessarily reduced but rather redistributed.

Second, the thesis provides novel insights into the temporal dynamics exhibited by process complexity metrics, based on real-world event log data. The temporal dynamics analysis utilizes a sliding window approach combined with complexity measurements, plotting the curves of the complexity metrics over time across the sliding windows. This expands on current research by integrating a wide variety of metrics instead of focussing on singular metrics (Grisold et al., 2022; Hærem et al., 2015; Kim et al., 2025; Schreiber & Abbad-Andaloussi, 2024; Wurm et al., 2021), and further expands the scope of existing research by integrating a wide variety of real-world, empirical event logs for the correlation analysis, whereas previous studies often focussed on single real-world event logs or synthetic event logs (Cremerius et al., 2024; Cremerius & Weske, 2022; B. T. Pentland et al., 2024; Wurm et al., 2021). While prior work has often employed single complexity metrics as proxies to operationalize process dynamics (Franzoi et al., 2024; Goh & Pentland, 2019; B. T. Pentland, Liu, et al., 2020), the analysis reveals that different metrics follow distinct temporal trajectories, with partially matching or diverging

patterns. This underscores the importance of examining complexity metrics both individually and in relation to each other over time (Lauterbach et al., 2020). As such, the findings open up new avenues for identifying potential tipping points or detecting temporal trade-offs between complexity metrics, which can help future research in BPM, information systems, and organization studies to engage in more nuanced examination and theorizing of the dynamics of processes or organizational routines.

Third, the thesis examines how different process complexity metrics respond to simulated process drift by applying the simulation model proposed by Pentland et al. (B. T. Pentland, Liu, et al., 2020) to event log data, while expanding this model by integrating empirical data in the form of real-world event logs, and implementing it for the full range of complexity metrics selected for this thesis. This thesis thus expands current research, which previously focussed on singular metrics, singular datasets, or only on synthetic datasets (Miller et al., 2012; B. T. Pentland, Liu, et al., 2020; B. T. Pentland et al., 2021; Vidgof, 2025). The results confirm Pentland et al.'s finding that the number of acyclic paths can exhibit sudden bursts of complexity, even for real-world event log data. Interestingly, similar patterns (i.e., peaking at the same simulation point) also emerge for other metrics, such as the structural variety. However, the magnitude of variation is significantly lower for other metrics. At the same time, some process complexity metrics (e.g., average affinity, normalized sequence entropy) remain largely unaffected, showing minimal change throughout the simulated drift. These findings highlight that process complexity metrics differ substantially in their sensitivity to incremental change. This variation is crucial to consider in both empirical investigations and theoretical developments related to process dynamics in the future. Going further, the results of the process drift simulation further highlight the context-sensitivity of such simulations, providing another lens to make sense of the results of previous process simulation research (Miller et al., 2012; B. T. Pentland, Liu, et al., 2020; B. T. Pentland et al., 2021; Vidgof, 2025). Furthermore, the findings also confirm the findings of (B. T. Pentland et al., 2021) B. T. Pentland et al., (2021) that incremental change can drastically change the configuration of processes, as measured by the process complexity.

All in all, the findings underscore that there is no one-size-fits-all approach to measuring process complexity. The choice of metric is likely context-dependent, with each measure offering distinct strengths and limitations. A nuanced understanding of what each metric captures and how it behaves over time or under change is essential for conducting effective analyses of process dynamics and for developing robust theoretical insights. In the same vein, process complexity metrics require context to facilitate thorough analyses and simulations, and select appropriate metrics, but can similarly provide context to analyse and evaluate process dynamics and change patterns.

5.3 Practical Implications

The findings of the analyses that are part of this thesis also have direct implications for the practical implementation and integration of process complexity metrics in organizational settings or in academia. First of all, the multi-faceted nature of process complexity makes clear that no single metric can provide a clear picture of process complexity in organizational settings. Instead, the combination of multiple metrics that represent different clusters of metrics, with diverging relationships and trade-offs, provides a more comprehensive picture of the underlying aspects of process complexity. Through the integration multiple perspectives, data mining applications can more closely monitor and analyse process complexity, helping process mining specialists and analysts to react appropriately to process change and to more effectively manage processes.

Furthermore, process complexity context, for example from understanding the relationship between utilized complexity metrics and plotting these metrics over time, may provide additional insights that enable them to interpret, integrate, and utilize these process complexity metrics for their day-to-day activities. The choice of process complexity metrics is highly context-dependent. While process mining results in general require context of the organizational setting for sensemaking (Franzoi et al., 2023; Grisold, van der Aa, et al., 2024), the behaviours and dynamics of process complexity metrics in particular are heavily influenced by the characteristics of the event log, such as by the data distribution, process characteristics, or event attributes. Similarly, process complexity metrics can provide context for analyses and evaluations of process dynamics and change patterns, enabling the utilization of another level of analysis.

Furthermore, the scalability of these complexity metrics, for use on large data sets in the context of big data and real-world event logs, is also particularly relevant (Kriegel et al., 2017). In this context, the variation-based complexity metrics that use Levenshtein-distance based calculation method suffer from significant issues for sufficiently large event logs, as the calculations exhibit exponentially increasing memory requirements. Even for the datasets in scope of this thesis, this leads serious crashes that prevent the effective integration of these metrics, unless workarounds are applied. To this end, the downsampling of event logs proves to be an effective tool, which may be applied for sufficiently large datalogs where such sampling does not significantly change the underlying trace distribution.

The thesis also provides further insights for process drift simulations in research or for practical applications, when utilizing the extended approach of B. T. Pentland, Liu, et al. (2020). The results of the process drift simulation highlight the context-sensitivity of such simulations, indicating that the choice of simulation parameters is critical for the validity

and interpretability of the results. To summarize the process drift simulation results, the drastic dynamics at the start of the process drift simulation indicate that the settings of the simulation do not necessarily reflect real-world process drift. While the effects on the dynamics of the metrics of the BPI 2011 hospital are limited, the dynamics of the Mannhardt hospital event log drastically change at the start of the simulation, indicating a significant change from the initial dataset distribution. While the BPI 2011 hospital event log had a very high share of unique sequences across the entire event log dataset, the Mannhardt hospital event log had a very low, near-zero share of such unique sequences. Combined with the iterative process drift implementation of the simulation, which introduces variation in the transition matrix between event types, the Mannhardt hospital event log appears much more sensitive to this type of process drift. As such, choosing adequate and reliable parameters for process drift simulation is both extremely important for valid results, but also requires deep contextual understanding of the complexity metrics in scope, and of the event log dataset on which the simulation is supposed to be run.

5.4 Limitations and Future Research Opportunities

Apart from the contributions and practical implications, there were also clear limitations due to technical issues and the scope of the thesis. As stated before, the computational complexity of some of the distance-based metrics requires downsampling to provide a workaround. While applicable to the datasets in scope of the thesis, such workarounds can potentially distort the underlying distribution, reducing the validity of the findings. Additionally, the selection of complexity metrics is reduced to quantitative metrics, thereby not including the probability of action pairs (Grisold et al., 2022), which would have provided another qualitative, transition-matrix-based perspective on direct-follows relationships. Similarly, technical issues with the pattern-based metrics of the repertoire and the routinization likewise prevented their inclusion in this thesis. Another key limitation is the selection of narrative examples for the temporal dynamics analysis and the process drift simulation results. While an exhaustive analysis and explanation of findings across all metrics and ideally all datasets would have been preferable, the scope of the thesis again required a limited selection of examples to provide narrative structure. Similarly, the analysis of the correlation results could involve a statistical clustering analysis to systematically determine clusters of process complexity metrics.

The process drift simulation likewise faced limitations. As stated before, choosing adequate and reliable parameters for process drift simulation is both extremely important for valid results, but also requires deep contextual understanding of the complexity metrics in scope, and of the event log dataset on which the simulation is supposed to be

run. This thesis could only evaluate a limited selection of process drift simulation runs, highlighting only one of these runs with parameters that were previously used in influential studies using this approach (B. T. Pentland, Liu, et al., 2020; B. T. Pentland et al., 2021) to provide comparable results. In the future, similar analyses and evaluations should both explore a wider selection of process drift simulation parameters, and ideally differentiate between event log datasets while choosing adequate parameters to account for different context requirements.

Overall, there is ample potential for further research into process complexity and complexity dynamics, both over time and in the presence of process drift, by applying the analyses of this thesis to wider sets of event logs and complexity metrics. Additionally, the depth of the analyses can be further improved upon, by running more granular analyses. The sheer breadth of temporal dynamics and process drift simulation results, combined with the potential of variation in the simulation parameters, yields large amounts of data that lend themselves to in-depth, granular studies. The application of additional aggregate statistics to quantify overarching trends and dynamics would further support the validity of the results of such studies. While the scope as to be limited, further research could extend this work, utilizing these in-depth, granular analyses.

On the other hand, the thesis highlights several additional key fields of interest in the context of process complexity research, that are promising avenues for future studies in the field. One of these research opportunities is the interplay between organizational context, sensemaking, process dynamics, and process complexity. While there are recent contributions that focus on sensemaking in process mining (Grisold, van der Aa, et al., 2024) and process dynamics (Franzoi et al., 2023), the application and evaluation of these findings onto the results of this thesis in particular and process complexity in general still holds potential to generate critical insights, as stated in the previous sections. Another avenue can be found in the topic of process drift and drift detection, where the implications of the thesis regarding process complexity dynamics, and the conceptual structures in the temporal and process drift patterns, may prove critical to understand and detect process drift. A similar opportunity has been raised by Vidgof (2025), noting that process complexity metrics may be used to detect and analyse process drift. Additionally, other recent process complexity metrics can and should be integrated in future evaluations of process complexity. This includes those metrics included in this thesis, such as the repertoire and routinization (Hansson et al., 2023) or the probability of action pairs (Grisold et al., 2022). Other recent metrics of interest are the path coherence, which utilizes event attributes to integrate the data perspective in addition to the control-flow perspective (Kim et al., 2025), or the application of topological data analysis to event logs to capture action patterns (Zhang et al., 2025).

6 Conclusion

Processes are the core of how organizations function, adapt, and create value, and are likewise inherently dynamic, changing over time due to both endogenous adaptations and exogenous influences. As such, it has become increasingly important to find ways to systematically observe, measure, and understand these dynamics, where one promising approach for capturing such process dynamics lies in analysing the structural characteristics of processes using quantitative metric, particularly those that measure process complexity, which refers to features such as variability, intricacy, or unpredictability in how processes unfold. However, there is currently a lack of understanding how these process complexity metrics relate to each other, which dynamics they exhibit over time, and how they react to process drift. To address this gap, this thesis answers the three research questions, by systematically analysing a variety of well-established and recent process complexity measures for multiple event log datasets, employing a correlation analysis, a temporal dynamics analysis, and a process drift simulation.

The findings highlight that process complexity metrics exhibit complex and sometimes divergent behaviours, both over time and in response to process drift. While some metrics exhibit linear or constant trends, others show non-linear dynamics, including sudden bursts of complexity in the form of tipping points. These differences suggest that complexity metrics are influenced by distinct process characteristics and capture different facets of process behaviour. Regarding process drift, the findings indicate that the complexity metrics vary in their sensitivity to incremental changes in the process, affecting both the extent to which drift impacts the metrics and their suitability for detecting such drift. Across the three analyses, the findings provide a theoretical structure for the process complexity dynamics, conceptually distinguishing between tipping points, trade-offs, invariant and drift-invariant metrics, and finding clusters of metrics with diverging behaviours and relationships. The thesis expands on previous simulation approaches, applying them to multiple complexity metrics and real-world event logs. Additionally, the findings likewise indicate future research potential in this approach, as well as in the application of process complexity for process drift detection.

All in all, the findings of this thesis underscore that there is no one-size-fits-all approach to measuring process complexity. The choice of metric is likely context-dependent, with each measure offering distinct strengths and limitations. A nuanced understanding of what each metric captures and how it behaves over time or under change is essential for conducting effective analyses of process dynamics and for developing robust theoretical insights.

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Appendix

A Correlation Analysis Results

Correlation Analysis Mannhardt Hospital Event Log

To provide a more granular perspective on the results of the correlation analysis, the results for the Mannhardt hospital event log are displayed in tables B.1, B.2, and B.3. Since the differences between both methods are minor, and Spearman is less sensitive to outliers in reduced sampling sizes, only the Spearman correlations are displayed for the single dataset results. Correlations are calculated for p-values of 0.05 and 0.01, with correlations with a p-value above or equal to the threshold of 0.05 highlighted in bold and those with a p-value above or equal to 0.01 but below 0.05 underscored, thereby ensuring highly significant results.

While a majority of correlations are consistent with the Spearman correlations calculated across all datasets, there are also key differences specific to the Mannhardt hospital event log. First, there is a reduction in the overall number of statistically significant correlations, as would be expected from a reduced sample size. Secondly, there are no correlation coefficients for the minimum sequence length (“min-seq-len”), as these values could not be calculated. The minimum sequence length for the Mannhardt hospital event log takes the value of one for all windows into which the event log is split, as there are numerous traces with the length of one event. Such continuous vectors prevent the correct calculation of the Pearson or Spearman correlation, representing a non-defined operation. As such, all such correlations can be ignored. Thirdly, the normalized variant entropy (“nvar-e”) is exhibiting mostly non-significant correlations.

All correlation measurements, including for all datasets and both Pearson and Spearman methods, are available in the Github repo, found here: https://github.com/MrWaitomo/WWU_processcomplexity

	var-e	nvar-e	seq-e	nseq-e	en-var-e	en-nvar-e	en-seq-e	en-nseq-e	#-ties
nvar-e	-0.05								
seq-e	0.80	0.02							
nseq-e	0.27	0.09	0.62						
en-var-e	0.91	-0.05	0.83	<u>0.20</u>					
en-nvar-e	-0.30	0.12	-0.14	<u>-0.19</u>	<u>-0.19</u>				
en-seq-e	0.90	-0.05	0.85	0.21	0.99	-0.14			
en-nseq-e	0.56	0.13	0.49	0.28	0.54	-0.13	0.52		
#-ties	0.99	-0.01	0.79	0.27	0.90	-0.33	0.90	0.56	
lempel-ziv	0.90	-0.04	0.81	0.21	0.95	-0.07	0.96	0.58	0.88
#-unique-seq	0.98	0.06	0.79	0.26	0.91	-0.31	0.90	0.53	0.98
perc-unique-seq	0.77	0.04	0.50	0.27	0.56	-0.51	0.54	0.50	0.79
#-e	0.90	-0.05	0.85	0.21	0.99	-0.14	1.00	0.49	0.89
#-et	0.37	<u>0.19</u>	0.35	0.14	0.36	-0.05	0.37	0.48	0.35
#-seq	0.81	-0.02	0.81	<u>0.20</u>	0.93	0.01	0.95	0.37	0.79
min-seq-len									
max-seq-len	0.73	-0.43	0.58	0.24	0.65	-0.38	0.65	0.46	0.71
avg-seq-len	0.77	-0.11	0.63	0.27	0.72	-0.51	0.71	0.61	0.78
avg-td-e	0.32	0.08	0.28	0.27	<u>0.19</u>	-0.03	0.21	0.37	0.32
dev_random	<u>0.19</u>	0.06	0.00	<u>0.18</u>	0.01	-0.44	-0.01	-0.09	0.22
avg-edit-distance	-0.72	-0.04	-0.78	<u>-0.18</u>	-0.87	-0.19	-0.90	-0.39	-0.70
struct-var	0.98	-0.12	0.79	0.26	0.90	-0.31	0.90	0.52	0.97
order_var	0.54	0.08	0.50	0.27	0.52	-0.33	0.52	0.66	0.54
activity_var	0.70	0.08	0.49	0.33	0.53	-0.56	0.51	0.58	0.73
#-acyclic-paths	0.86	-0.09	0.74	0.27	0.79	-0.09	0.80	0.57	0.82
avg-distinct-e	0.67	-0.02	0.60	0.28	0.65	-0.40	0.65	0.64	0.67
structure	-0.79	<u>0.18</u>	-0.68	-0.29	-0.72	0.15	-0.73	-0.48	-0.77
affinity	0.13	-0.31	0.12	-0.09	0.20	0.11	<u>0.20</u>	0.13	0.12

Table B.1 Spearman correlations of complexity metrics for Mannhardt hospital event log - part 1.

	lempel- ziv	#-unique- seq	perc- unique-seq	#-e	#-et	#-seq	min-seq- len	max-seq- len	avg-seq- len
#-unique-seq	0.88								
perc-unique-seq	0.55	0.79							
#-e	0.96	0.90	0.53						
#-et	0.38	0.35	<u>0.19</u>	0.37					
#-seq	0.90	0.81	0.38	0.95	0.38				
min-seq-len									
max-seq-len	0.67	0.65	0.53	0.65	0.34	0.59			
avg-seq-len	0.71	0.73	0.73	0.70	0.24	0.52		0.64	
avg-td-e	0.31	0.27	0.24	0.21	<u>0.19</u>	0.26		0.40	0.29
dev_random	-0.04	0.22	0.29	-0.01	-0.30	-0.08		0.10	0.13
avg-edit-distance	-0.85	-0.71	-0.25	-0.90	-0.44	-0.97		-0.50	-0.43
struct-var	0.88	0.95	0.74	0.90	0.36	0.81		0.78	0.76
order_var	0.51	0.51	0.53	0.51	0.40	0.34		0.41	0.80
activity_var	0.50	0.71	0.81	0.50	0.32	0.35		0.52	0.74
#-acyclic-paths	0.85	0.80	0.57	0.80	0.49	0.75		0.72	0.66
avg-distinct-e	0.65	0.63	0.62	0.64	0.35	0.46		0.54	0.95
structure	-0.77	-0.75	-0.64	-0.73	-0.08	-0.65		-0.67	-0.68
affinity	<u>0.18</u>	0.07	0.11	<u>0.19</u>	-0.04	0.06		0.05	0.33

Table B.2 Spearman correlations of complexity metrics for Mannhardt hospital event log - part 2.

	avg-td-e	dev_rand om	avg-edit- distance	struc t-var	order var	activity var	#-acyclic- paths	avg- distinct-e	structure
dev_random	0.04								
avg-edit-distance	-0.23	0.23							
struct-var	0.31	<u>0.18</u>	-0.72						
order_var	<u>0.16</u>	<u>-0.18</u>	-0.38	0.53					
activity_var	0.32	0.37	-0.26	0.70	0.69				
#-acyclic-paths	0.37	-0.05	-0.72	0.86	0.53	0.56			
avg-distinct-e	0.23	-0.05	-0.45	0.67	0.91	0.69	0.64		
structure	-0.39	-0.05	0.60	-0.81	-0.48	-0.55	-0.87	-0.61	
affinity	-0.42	-0.34	-0.14	0.14	0.41	0.00	<u>0.20</u>	0.44	-0.20

Table B.3 Spearman correlations of complexity metrics for Mannhardt hospital event log - part 3.

Correlation Analysis BPI 2011 Hospital Event Log

To provide another granular perspective on the correlation analysis results, with a similar real-world setting and thus comparability, the results for the BPI 2011 hospital event log are displayed in **tables B.4, B.5, and B.6**. Since the differences between both methods are minor, and Spearman is less sensitive to outliers in reduced sampling sizes, only the Spearman correlations are displayed for the single dataset results. Correlations are calculated for p-values of 0.05 and 0.01, with correlations with a p-value above or equal to the threshold of 0.05 highlighted in bold and those with a p-value above or equal to 0.01 but below 0.05 underscored, thereby ensuring highly significant results.

	var-e	nvar-e	seq-e	nseq-e	en-var-e	en-nvar-e	en-seq-e	en-nseq-e	#-ties
nvar-e	0.10								
seq-e	1.00	0.11							
nseq-e	0.22	0.97	0.24						
en-var-e	1.00	0.12	1.00	0.25					
en-nvar-e	0.11	0.99	0.13	0.96	0.14				
en-seq-e	1.00	0.12	1.00	0.25	1.00	0.14			
en-nseq-e	0.11	0.99	0.13	0.96	0.14	1.00	0.14		
#-ties	0.98	-0.04	0.98	0.08	0.98	-0.03	0.98	-0.03	
lempel-ziv	0.99	0.00	0.99	0.12	0.98	0.01	0.98	0.01	0.99
#-unique-seq	0.81	0.46	0.82	0.58	0.82	0.51	0.82	0.51	0.72
perc-unique-seq	-0.26	<u>-0.17</u>	-0.26	-0.25	-0.26	-0.23	-0.26	-0.23	<u>-0.19</u>
#-e	0.98	-0.04	0.98	0.09	0.98	-0.03	0.98	-0.03	1.00
#-et	0.93	0.11	0.94	0.25	0.93	0.14	0.93	0.14	0.91
#-seq	0.79	0.46	0.80	0.58	0.81	0.50	0.81	0.50	0.71
min-seq-len	-0.26	-0.10	-0.26	-0.13	-0.27	-0.12	-0.27	-0.12	-0.23
max-seq-len	0.73	-0.44	0.72	-0.33	0.72	-0.42	0.72	-0.42	0.82
avg-seq-len	0.42	-0.64	0.41	-0.58	0.40	-0.67	0.40	-0.67	0.54
avg-td-e	0.36	<u>0.19</u>	0.36	0.24	0.36	<u>0.20</u>	0.36	<u>0.20</u>	0.33
dev_random	0.70	<u>0.18</u>	0.70	0.31	0.70	0.22	0.70	0.22	0.65
avg-edit-distance	-0.41	-0.74	-0.42	-0.77	-0.43	-0.79	-0.43	-0.79	-0.29
struct-var	0.97	-0.04	0.97	0.09	0.96	-0.02	0.96	-0.02	0.98
order_var	-0.44	0.21	-0.44	0.13	-0.44	0.20	-0.44	0.20	-0.48
activity_var	0.74	0.21	0.74	0.34	0.75	0.25	0.75	0.25	0.69
#-acyclic-paths	0.97	0.04	0.97	<u>0.17</u>	0.97	0.06	0.97	0.06	0.96
avg-distinct-e	-0.01	-0.51	-0.02	-0.50	-0.03	-0.57	-0.03	-0.57	0.08
structure	0.70	0.25	0.71	0.36	0.71	0.27	0.71	0.27	0.66
affinity	-0.68	-0.14	-0.67	-0.25	-0.67	-0.15	-0.67	-0.15	-0.65

Table B.4 Spearman correlations of complexity metrics for BPI 2011 hospital event log - part 1.

	lempel- ziv	#-unique- seq	perc- unique-seq	#-e	#-et	#-seq	min-seq- len	max-seq- len	avg-seq- len
#-unique-seq	0.77								
perc-unique-seq	-0.23	-0.49							
#-e	0.99	0.73	<u>-0.20</u>						
#-et	0.94	0.85	-0.35	0.91					
#-seq	0.76	1.00	-0.55	0.72	0.84				
min-seq-len	-0.24	-0.43	0.35	-0.23	-0.29	-0.44			
max-seq-len	0.79	0.42	-0.11	0.81	0.67	0.42	-0.21		
avg-seq-len	0.48	-0.13	0.32	0.53	0.28	-0.15	0.13	0.68	
avg-td-e	0.33	0.36	<u>-0.16</u>	0.33	0.30	0.36	<u>-0.18</u>	<u>0.18</u>	0.00
dev_random	0.71	0.79	-0.45	0.65	0.82	0.79	-0.20	0.42	0.01
avg-edit-distance	-0.35	-0.82	0.52	-0.30	-0.50	-0.83	0.36	0.03	0.60
struct-var	0.99	0.74	-0.22	0.98	0.91	0.73	-0.25	0.81	0.50
order_var	-0.47	-0.26	<u>0.18</u>	-0.48	-0.39	-0.26	0.10	-0.52	-0.37
activity_var	0.74	0.83	-0.48	0.69	0.87	0.83	-0.28	0.44	0.00
#-acyclic-paths	0.98	0.80	-0.29	0.97	0.97	0.79	-0.26	0.74	0.40
avg-distinct-e	0.02	-0.51	0.59	0.07	<u>-0.15</u>	-0.54	0.37	0.24	0.81
structure	0.70	0.77	-0.39	0.66	0.88	0.77	-0.29	0.40	-0.01
affinity	-0.68	-0.71	0.35	-0.65	-0.70	-0.69	0.41	-0.48	-0.09

Table B.5 Spearman correlations of complexity metrics for BPI 2011 hospital event log - part 2.

	avg-td-e	dev_rand om	avg-edit- distance	struc t-var	order var	activity var	#-acyclic- paths	avg- distinct-e	structure
dev_random	0.28								
avg-edit-distance	-0.32	-0.57							
struct-var	0.32	0.68	-0.31						
order_var	-0.07	-0.27	-0.05	-0.49					
activity_var	0.27	0.94	-0.62	0.72	-0.33				
#-acyclic-paths	0.32	0.78	-0.41	0.97	-0.43	0.82			
avg-distinct-e	<u>-0.19</u>	-0.34	0.81	0.03	-0.10	-0.37	-0.06		
structure	0.20	0.75	-0.57	0.66	-0.25	0.83	0.75	-0.31	
affinity	-0.33	-0.64	0.44	-0.67	0.29	-0.68	-0.72	0.32	-0.56

Table B.6 Spearman correlations of complexity metrics for BPI 2011 hospital event log - part 3.

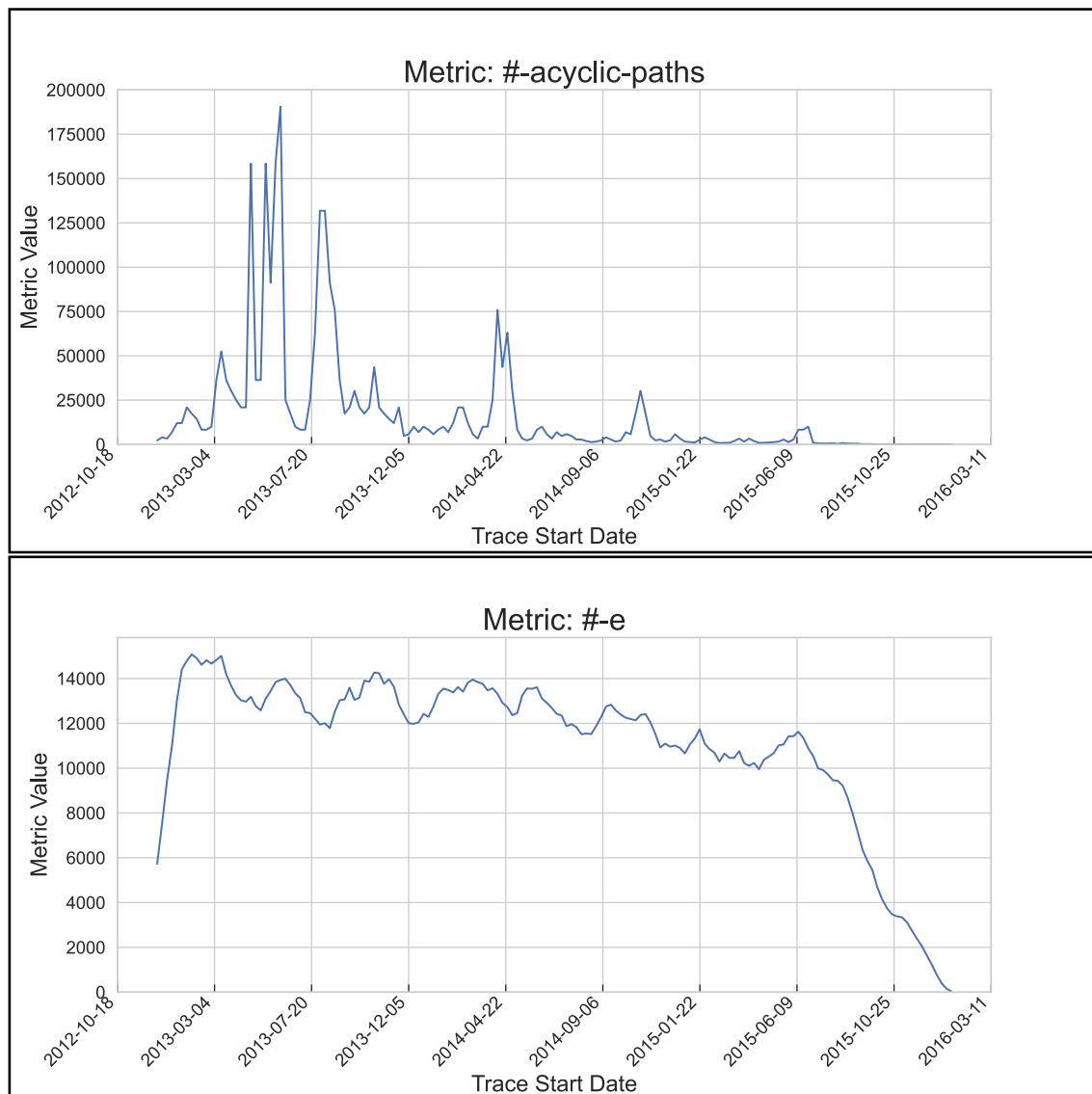
All correlation measurements, including for all datasets and both Pearson and Spearman methods, are available in the Github repo, found here:
https://github.com/MrWaitomo/WWU_processcomplexity

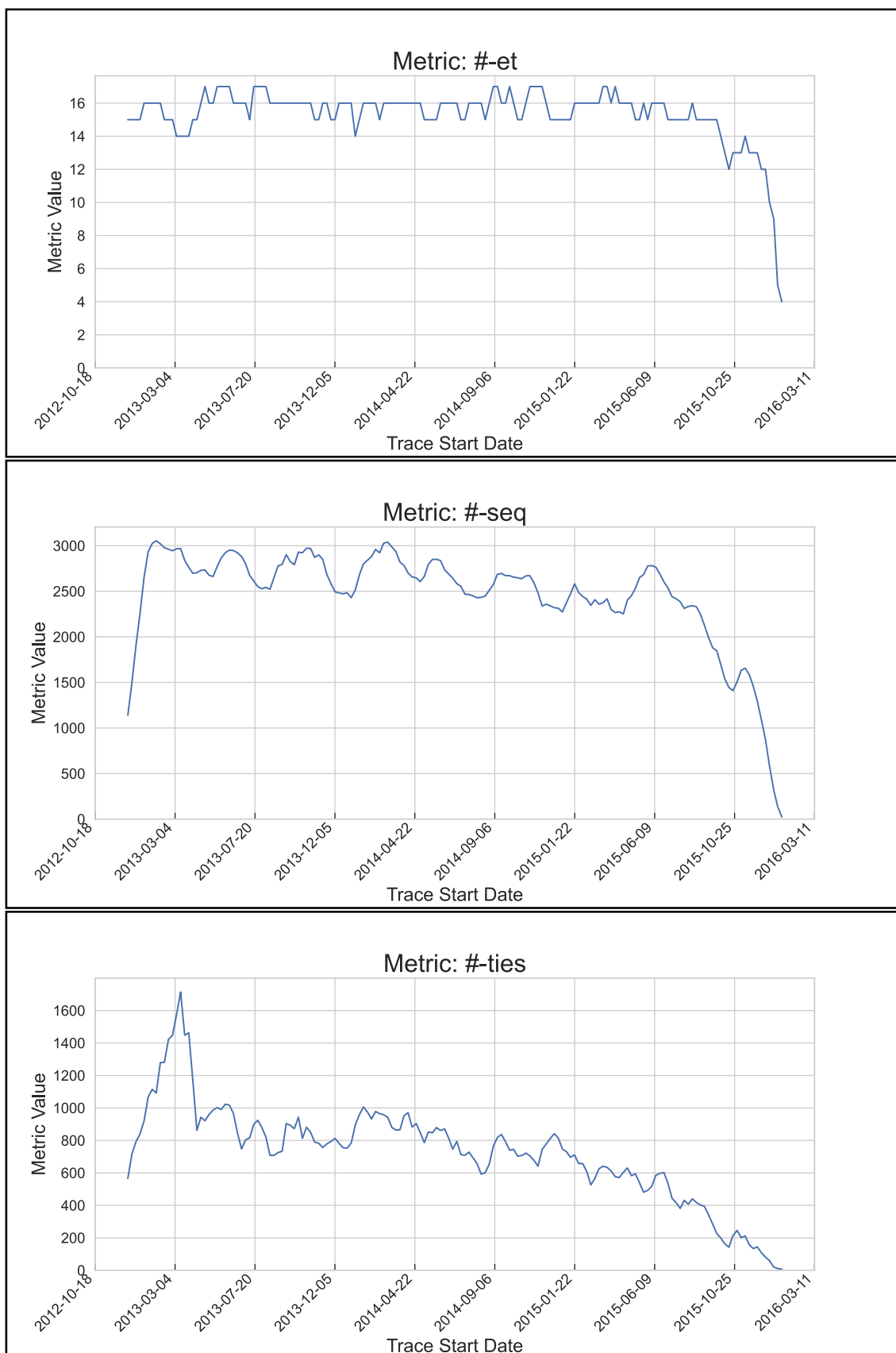
B Temporal Dynamics Additional Figures or Tables

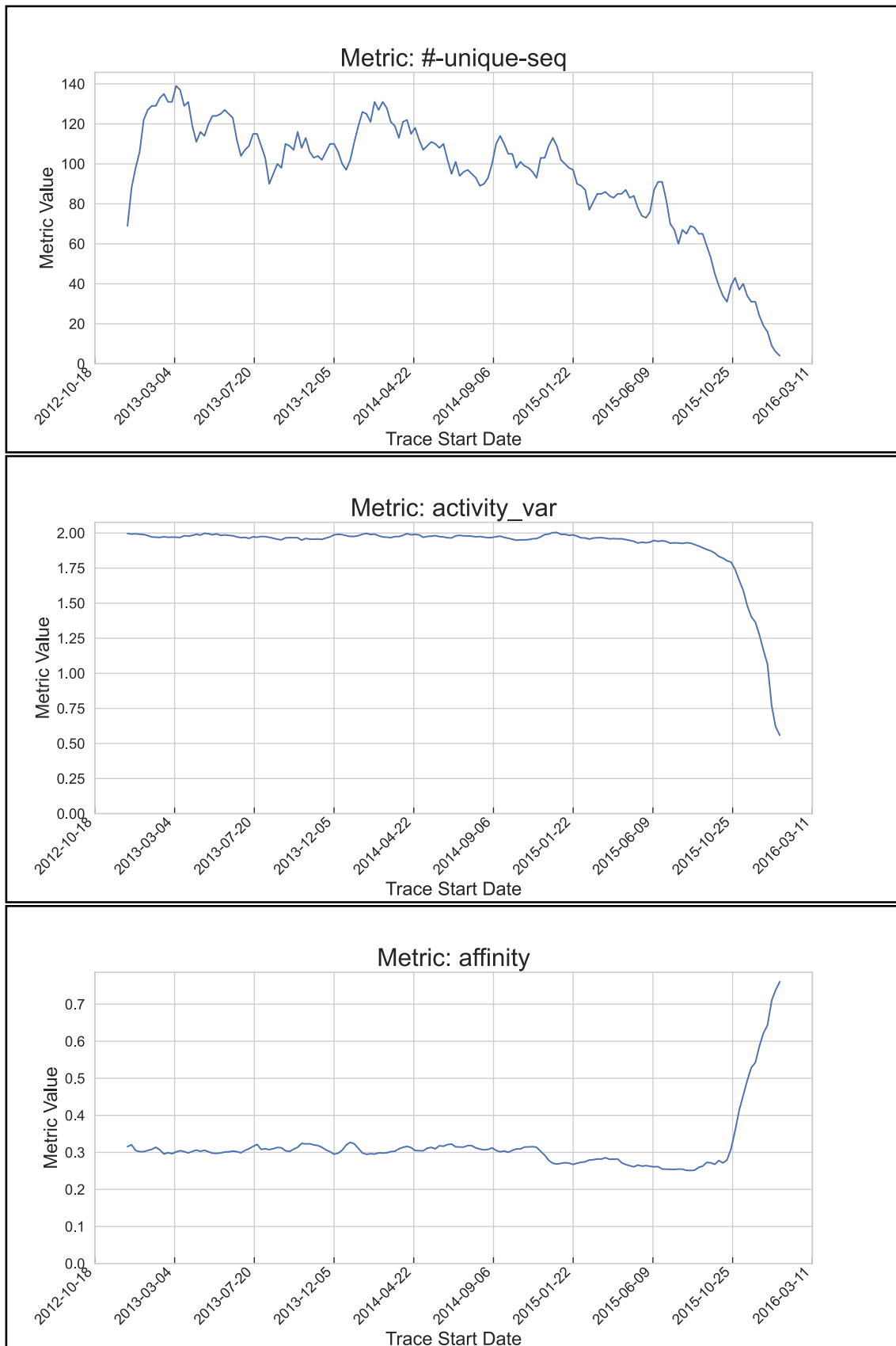
In the following section, all figures showing the plots of the process complexity metrics for the respective datasets can be found. Each metric is shown in one respective figure. All other figures can be found in the GitHub Repo:

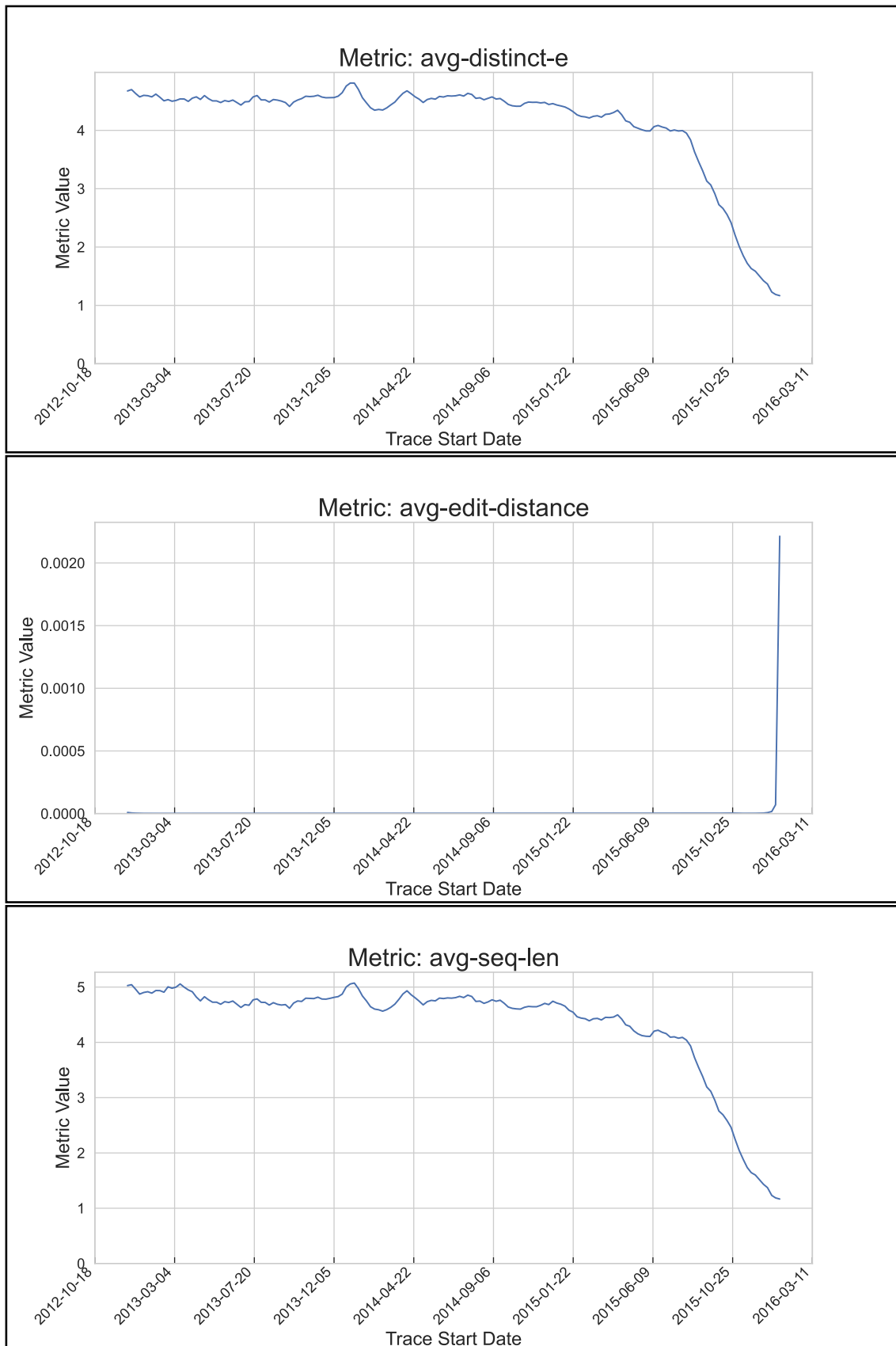
https://github.com/MrWaitomo/WWU_processcomplexity

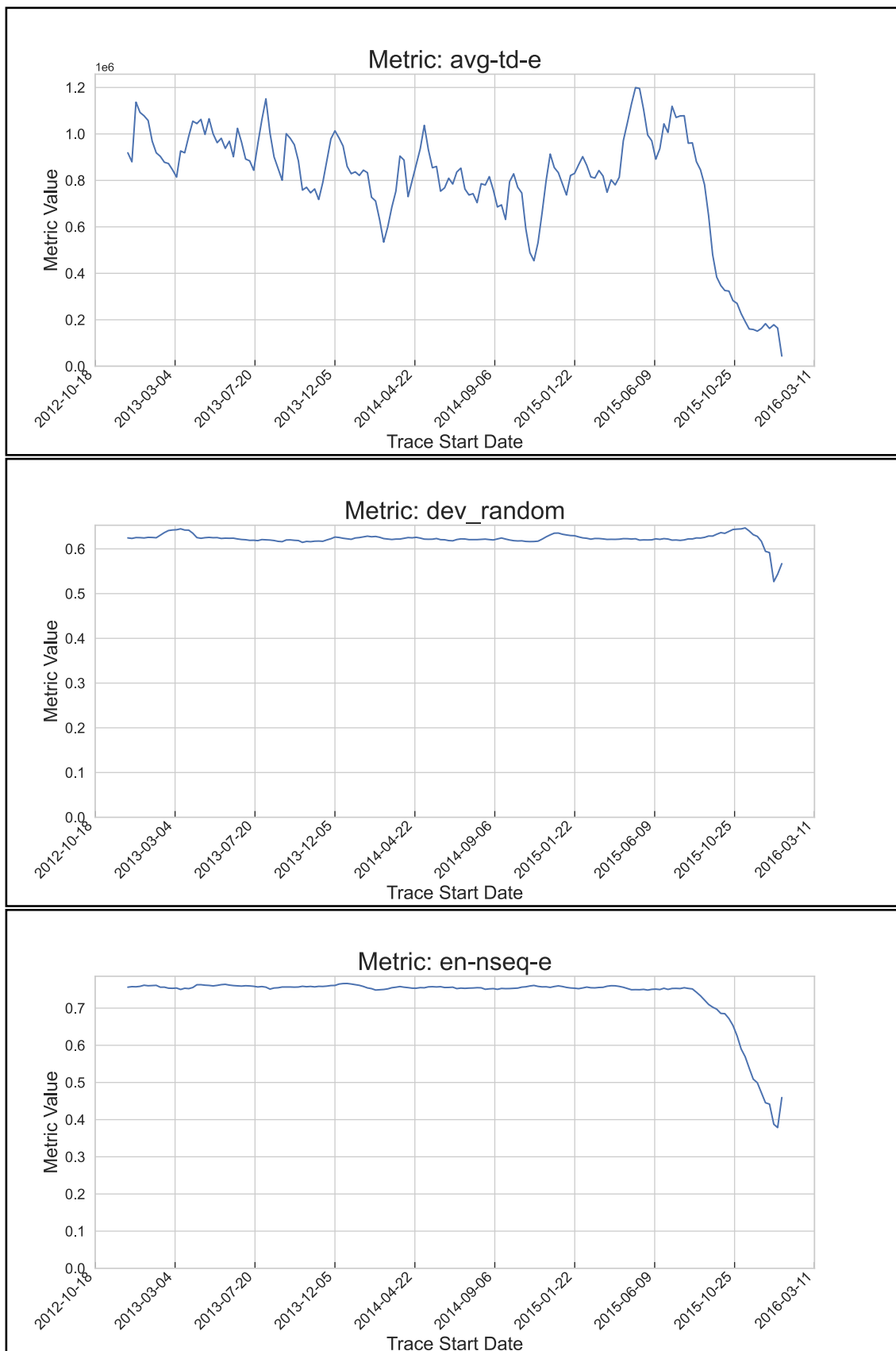
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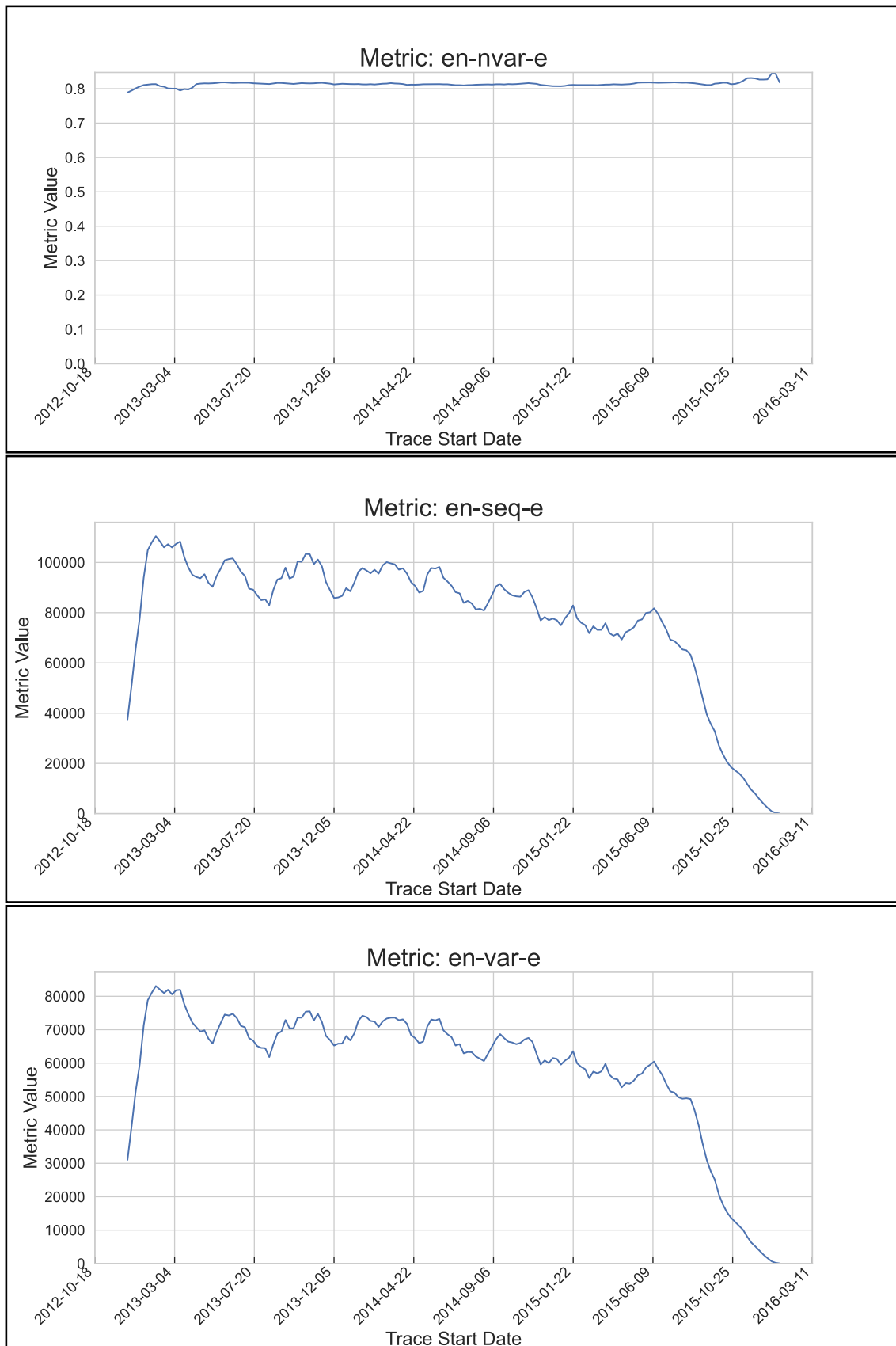


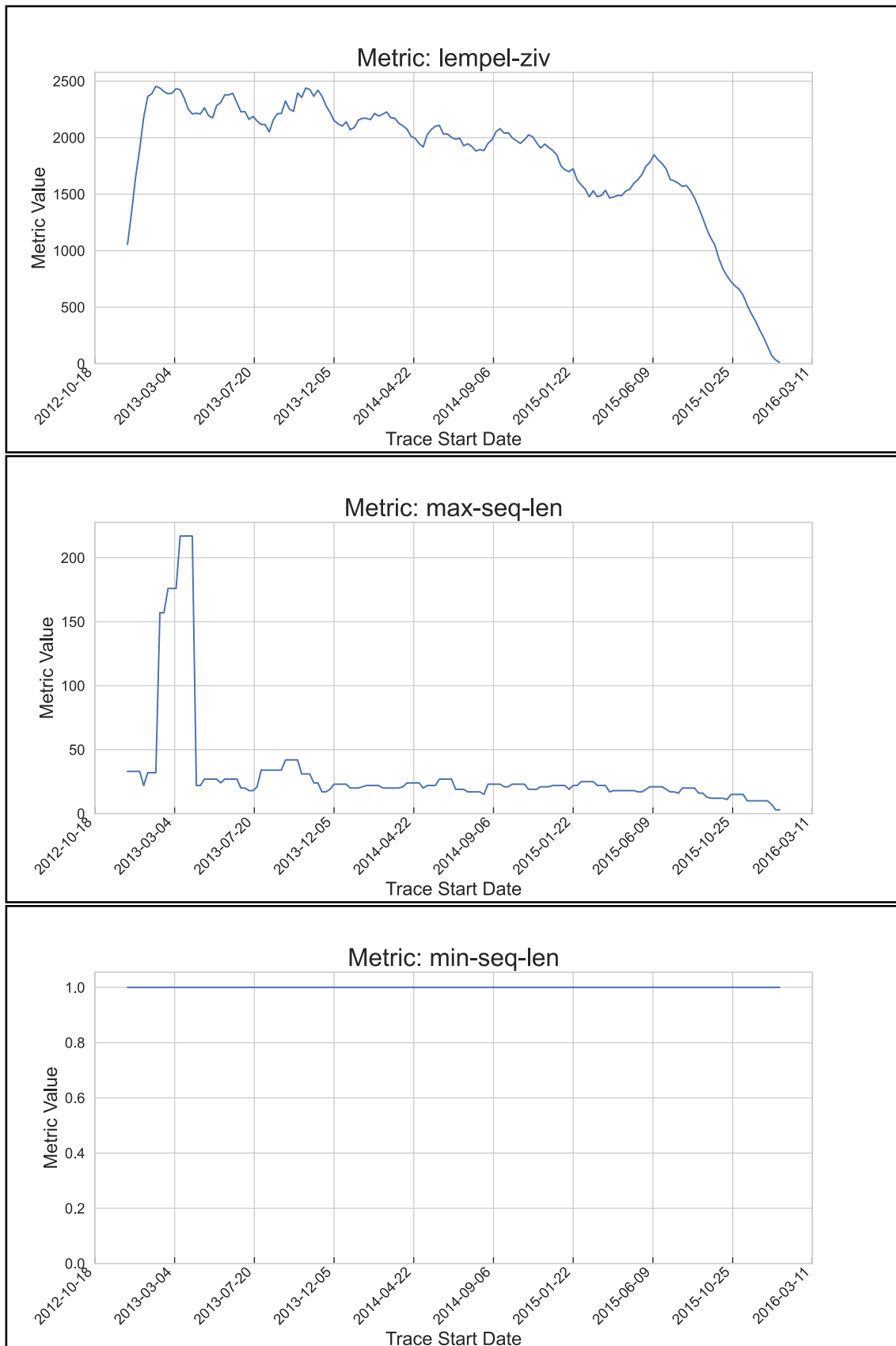


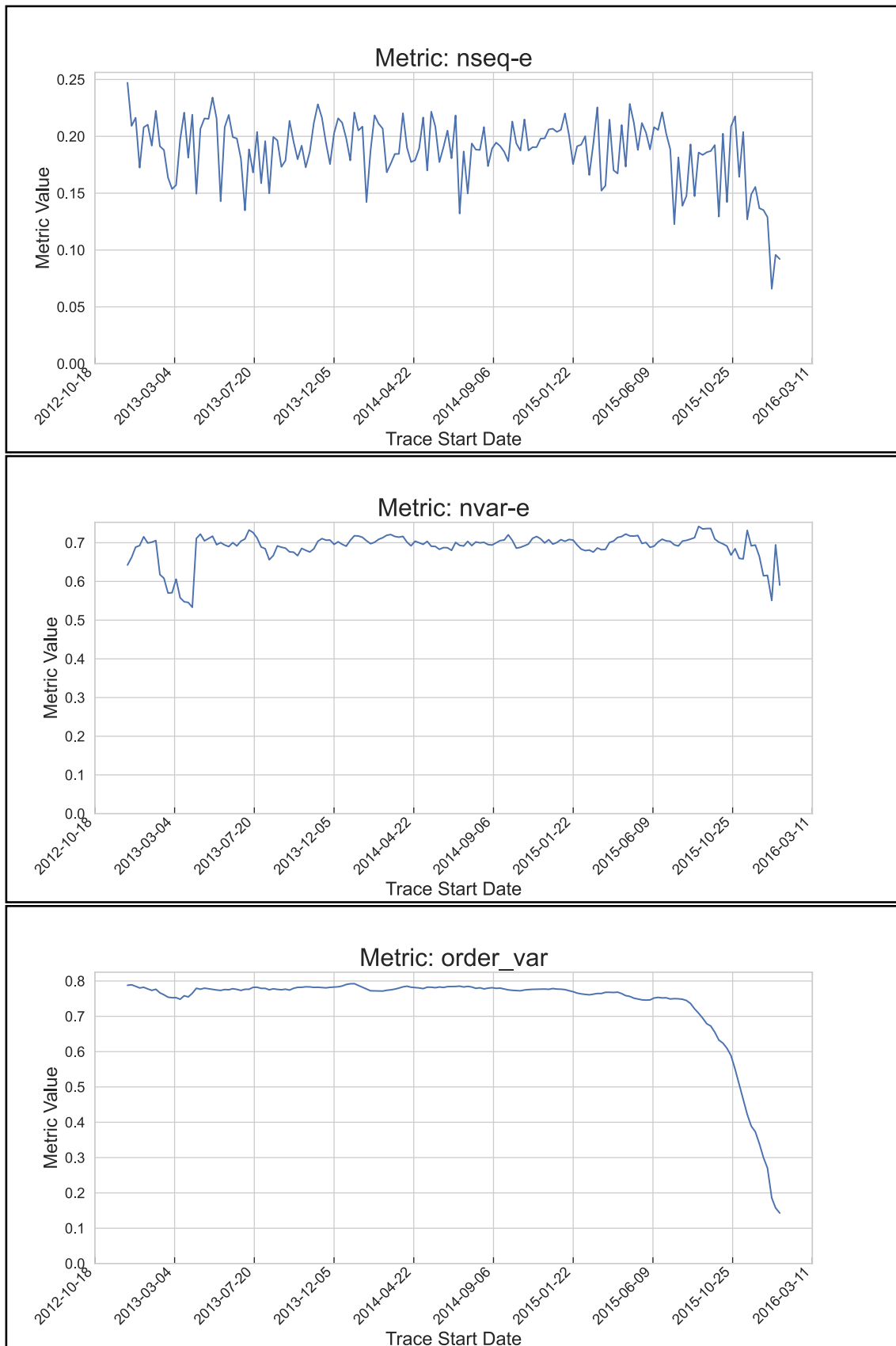


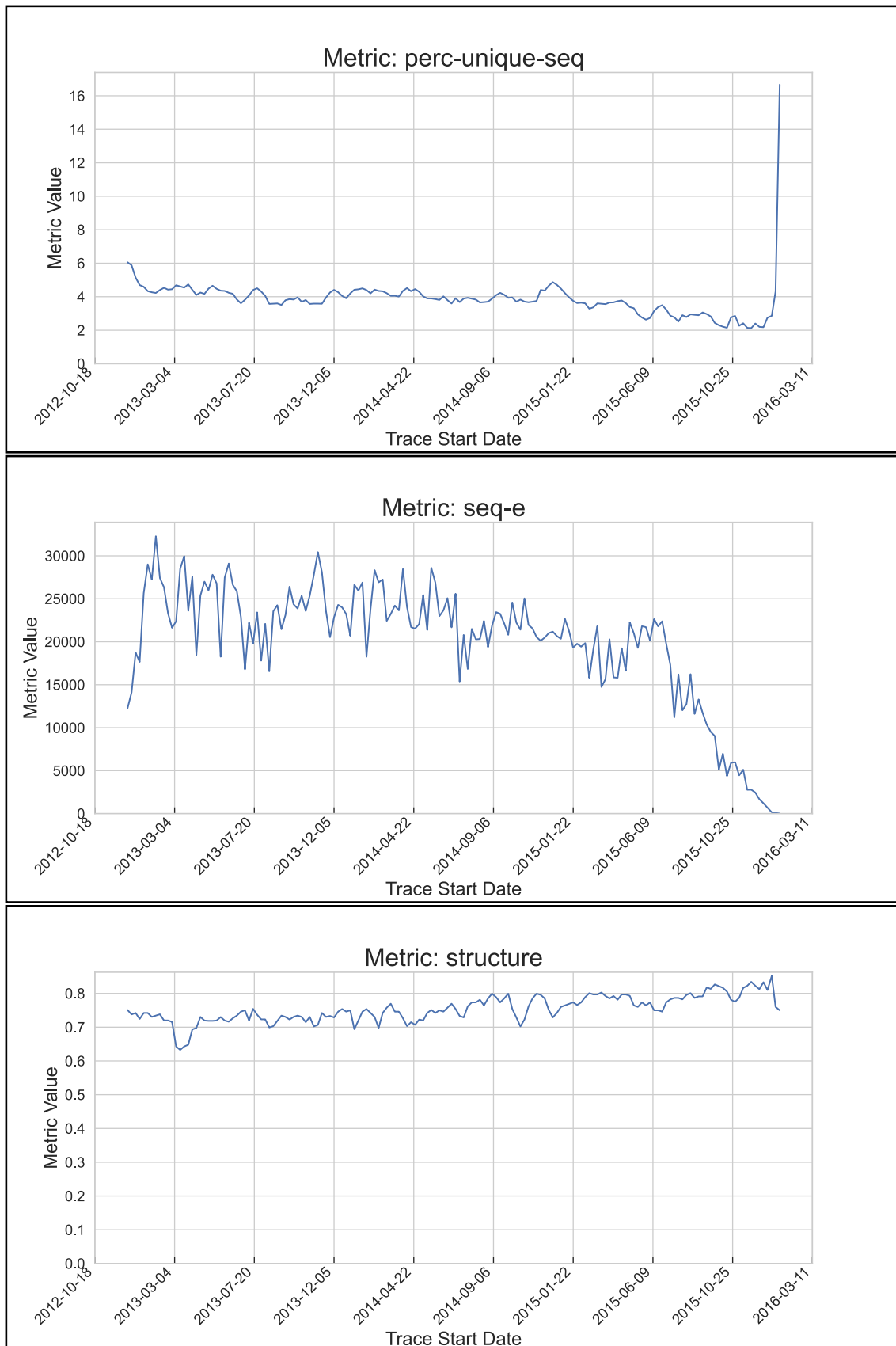


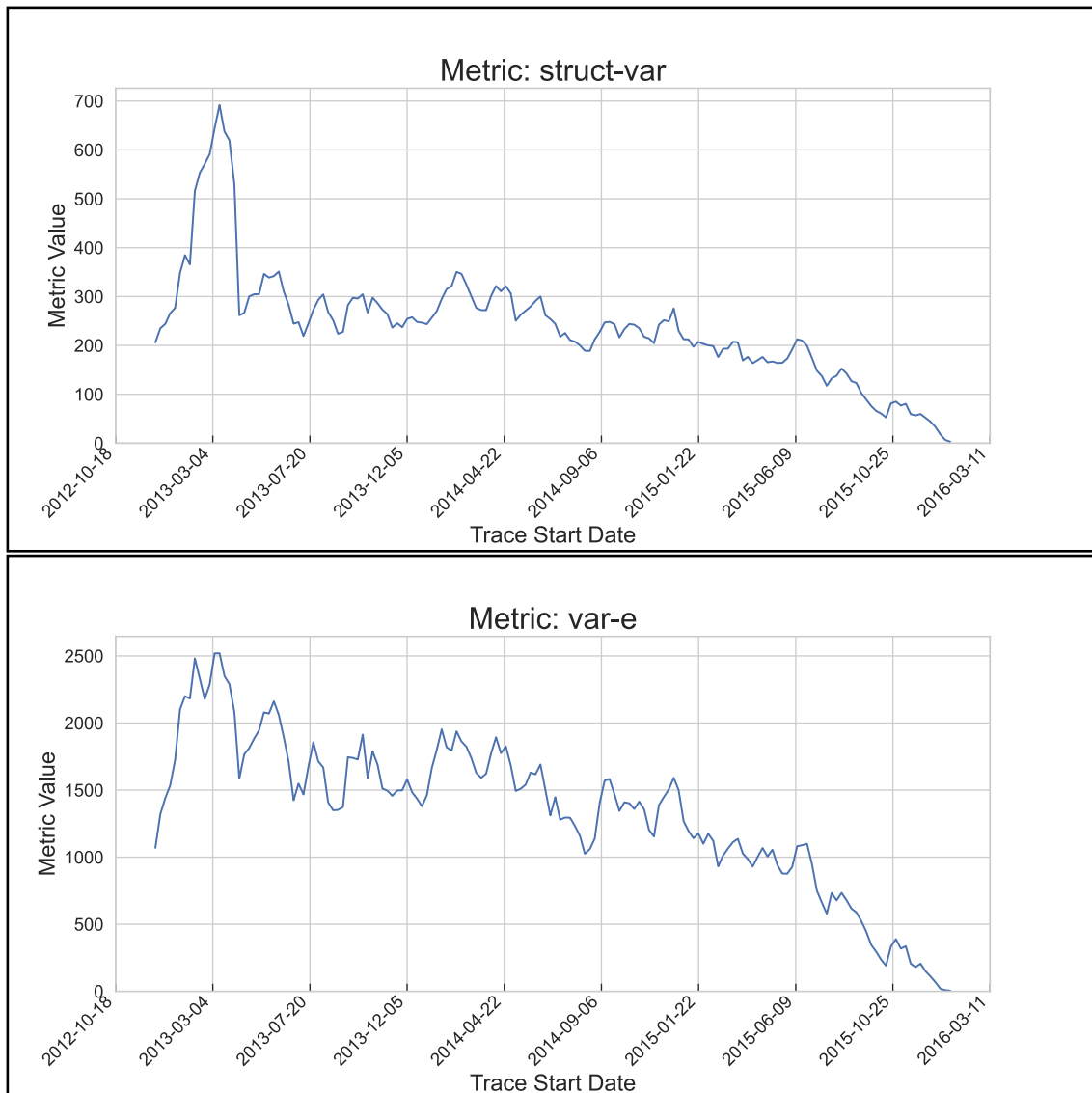


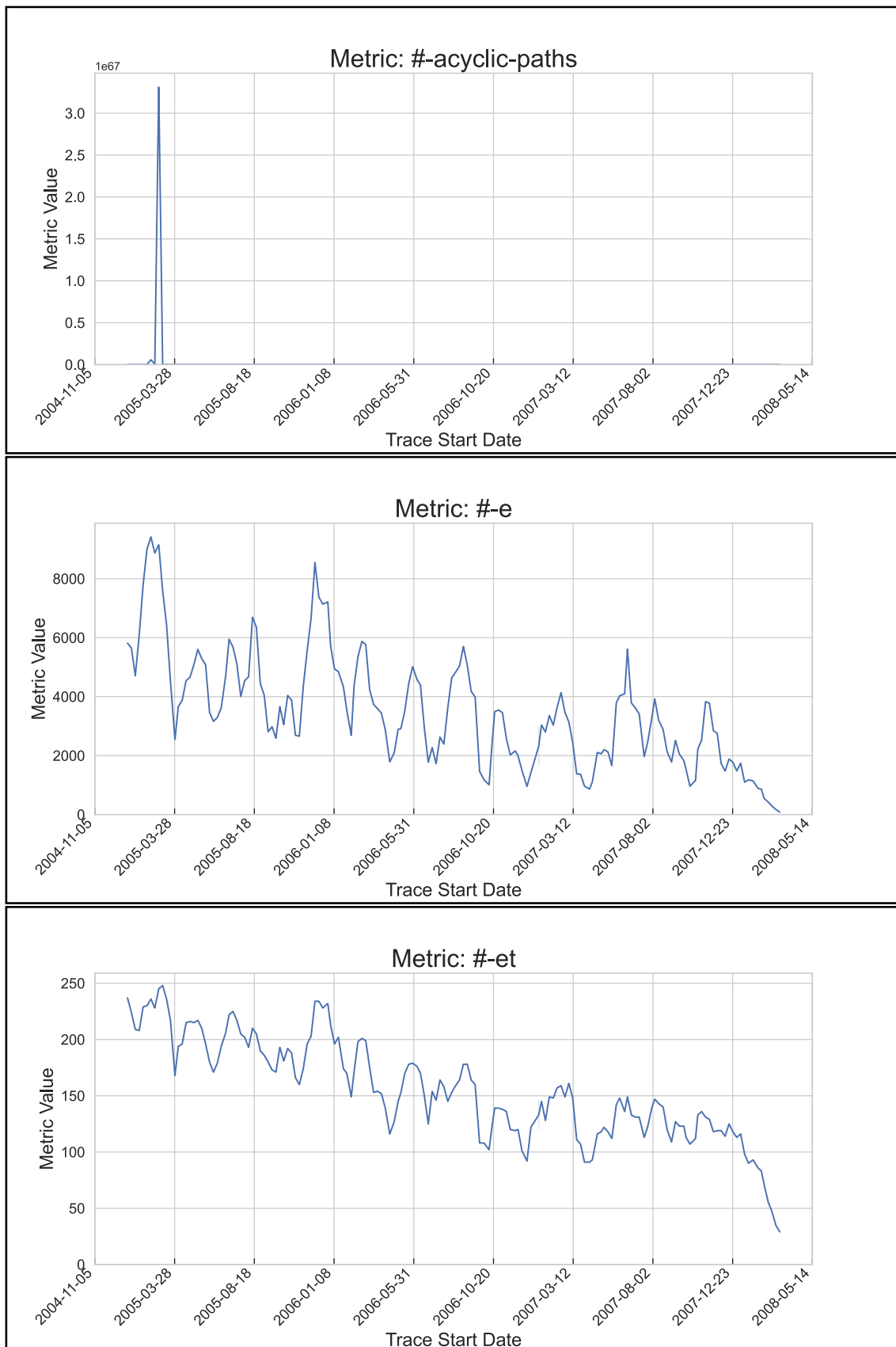


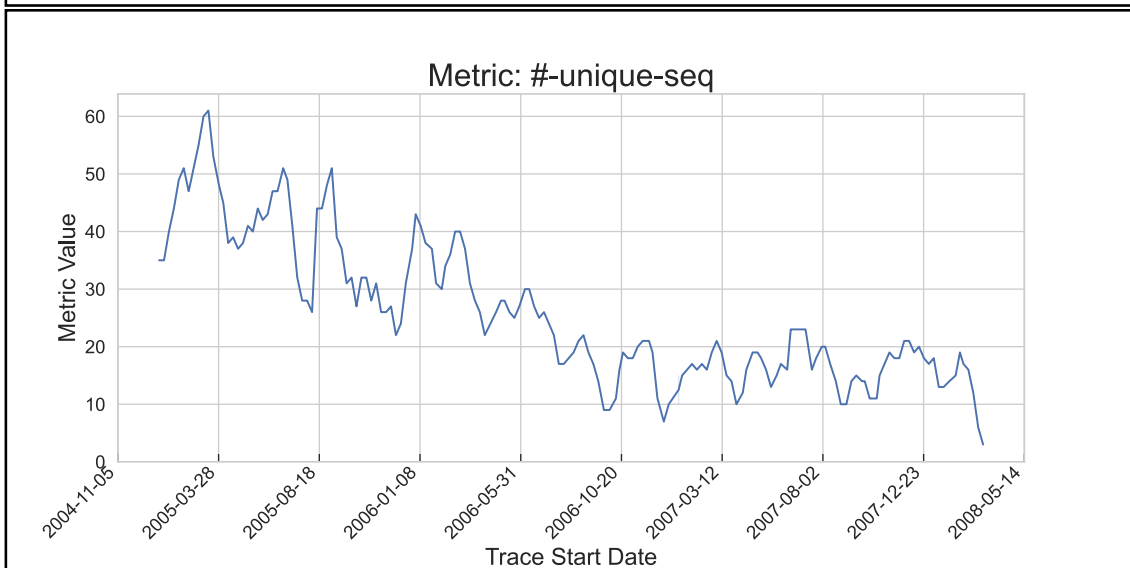
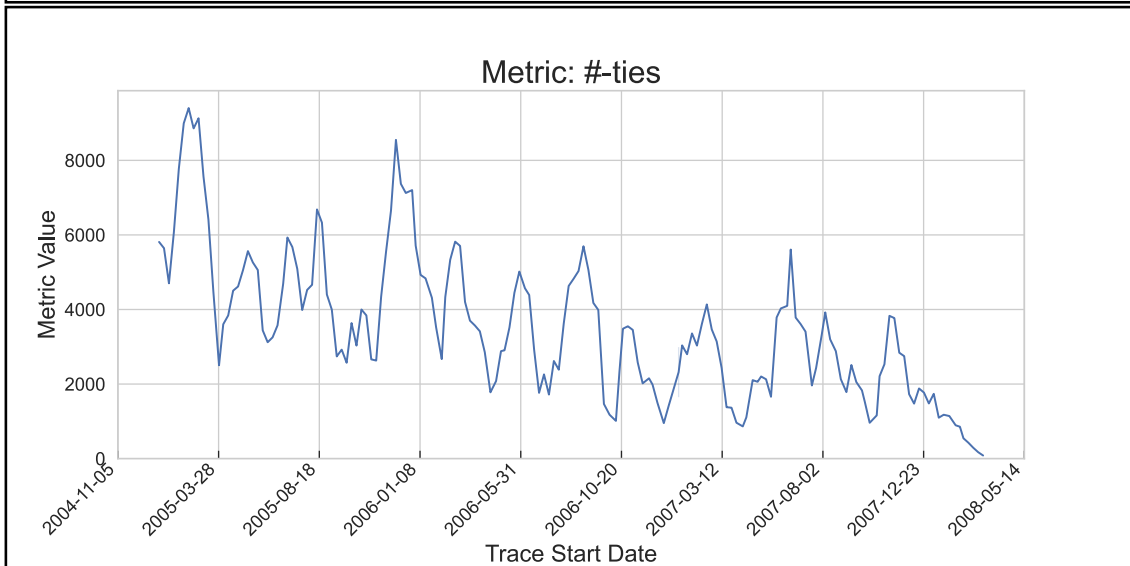
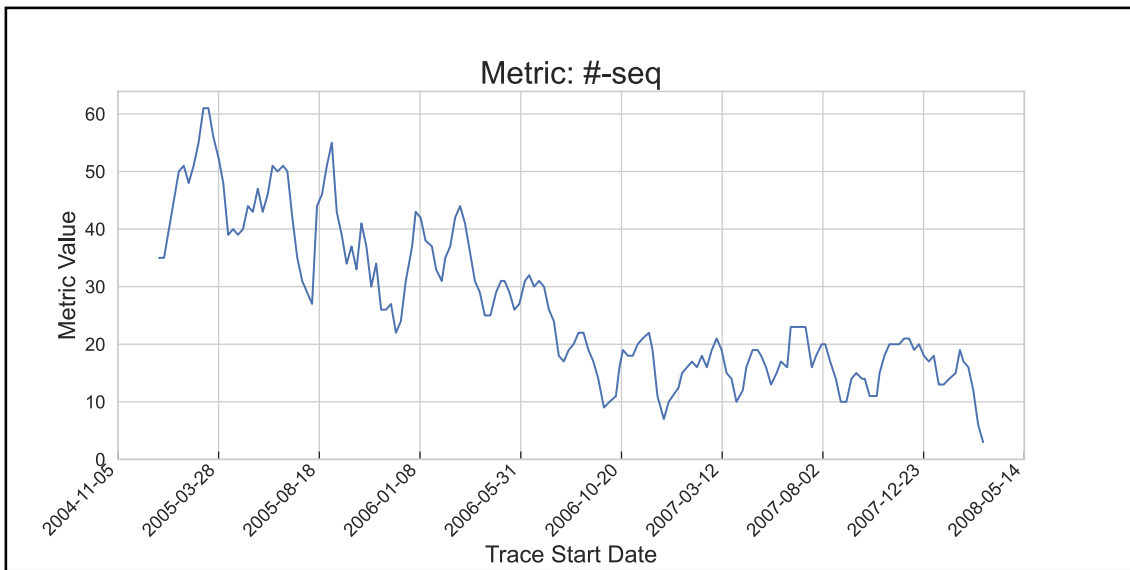


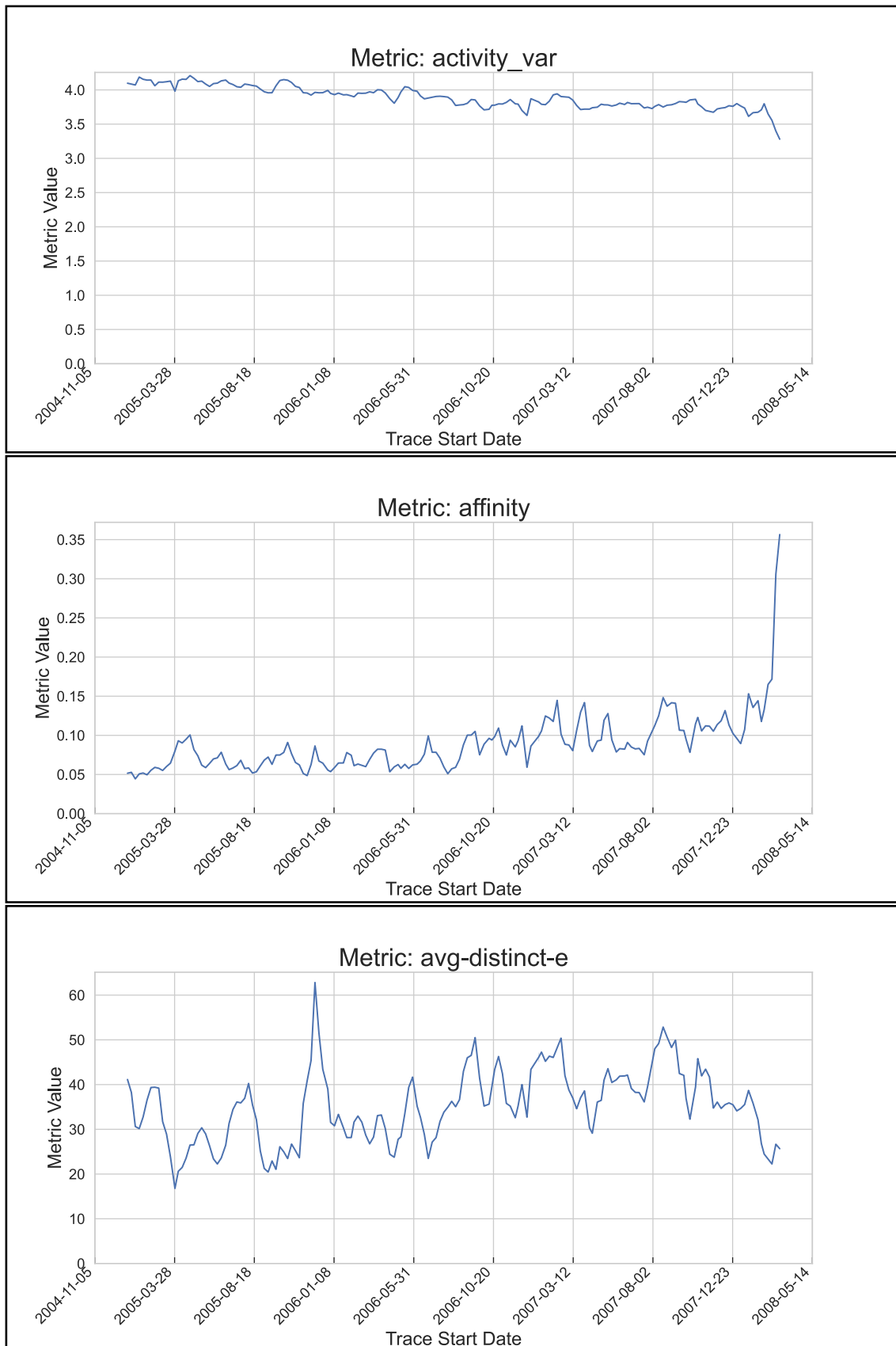


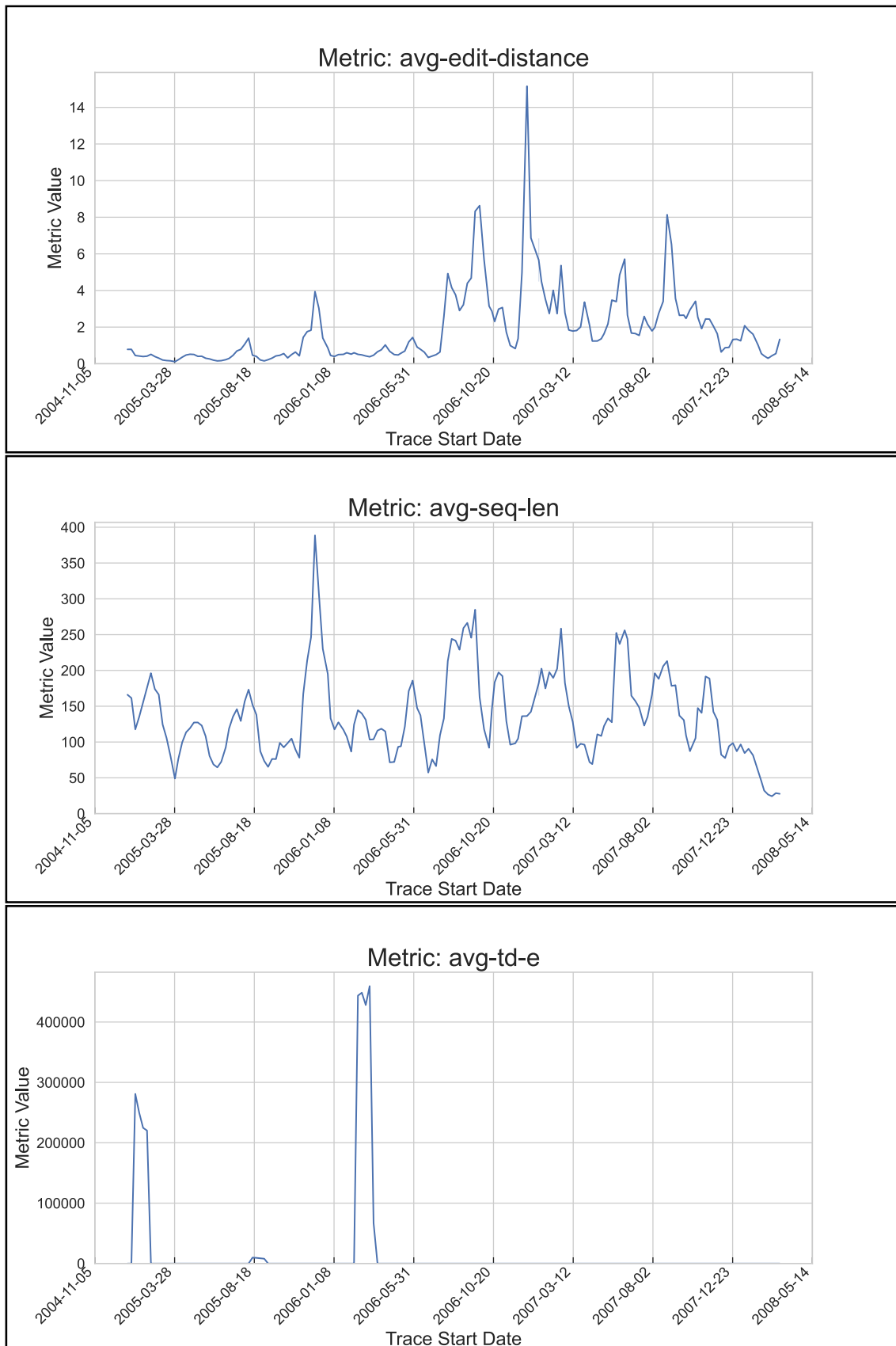


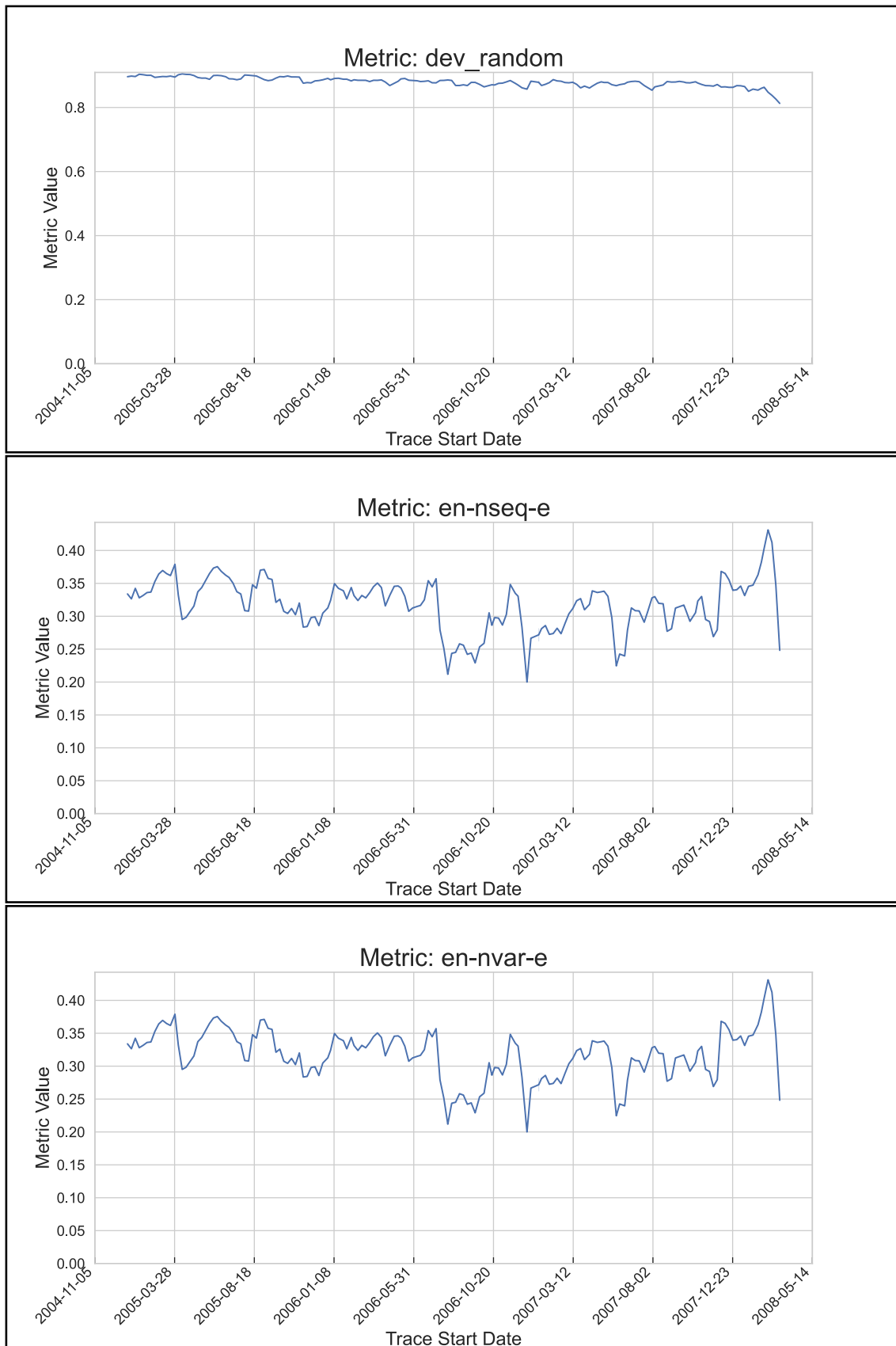


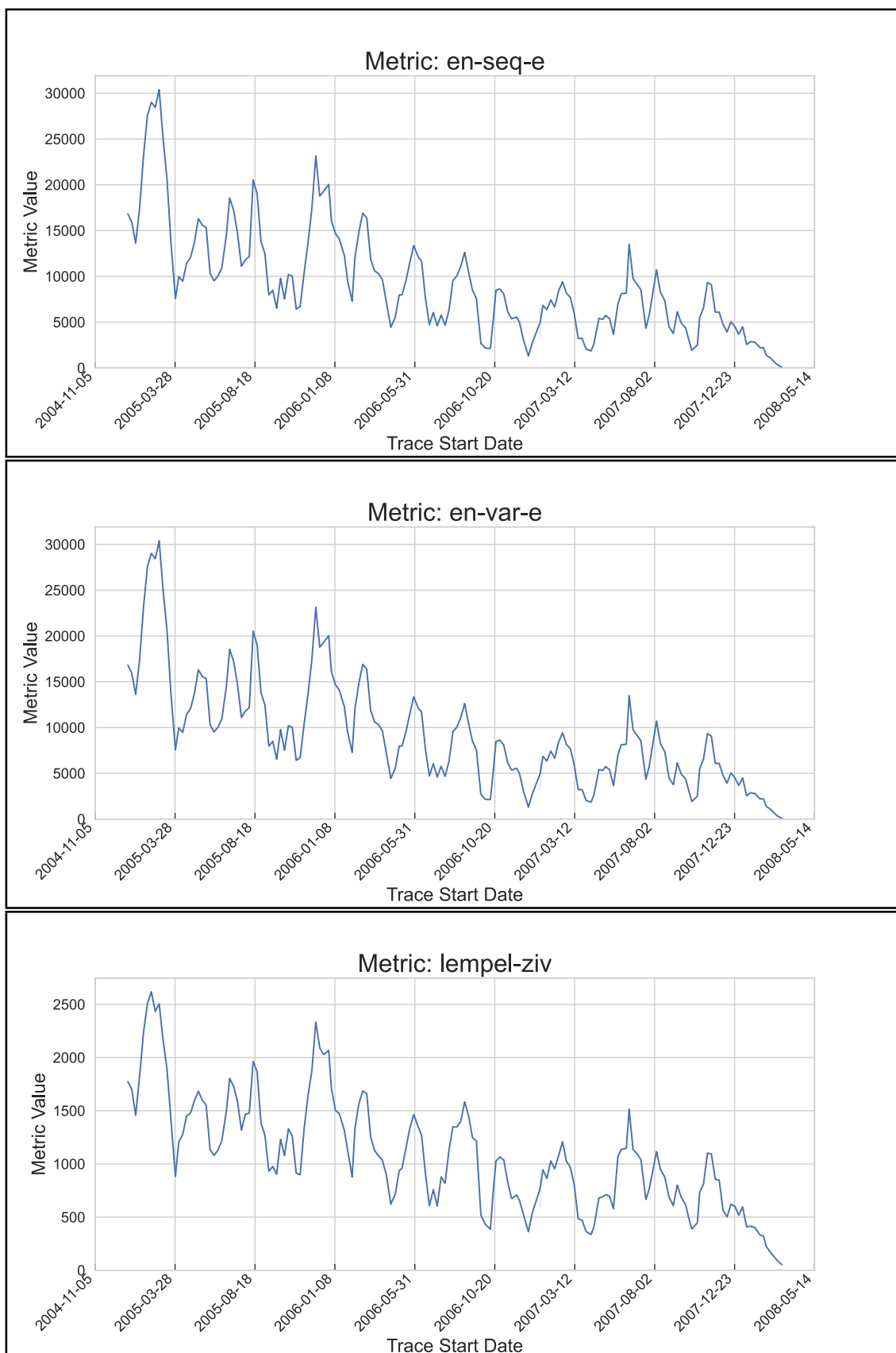
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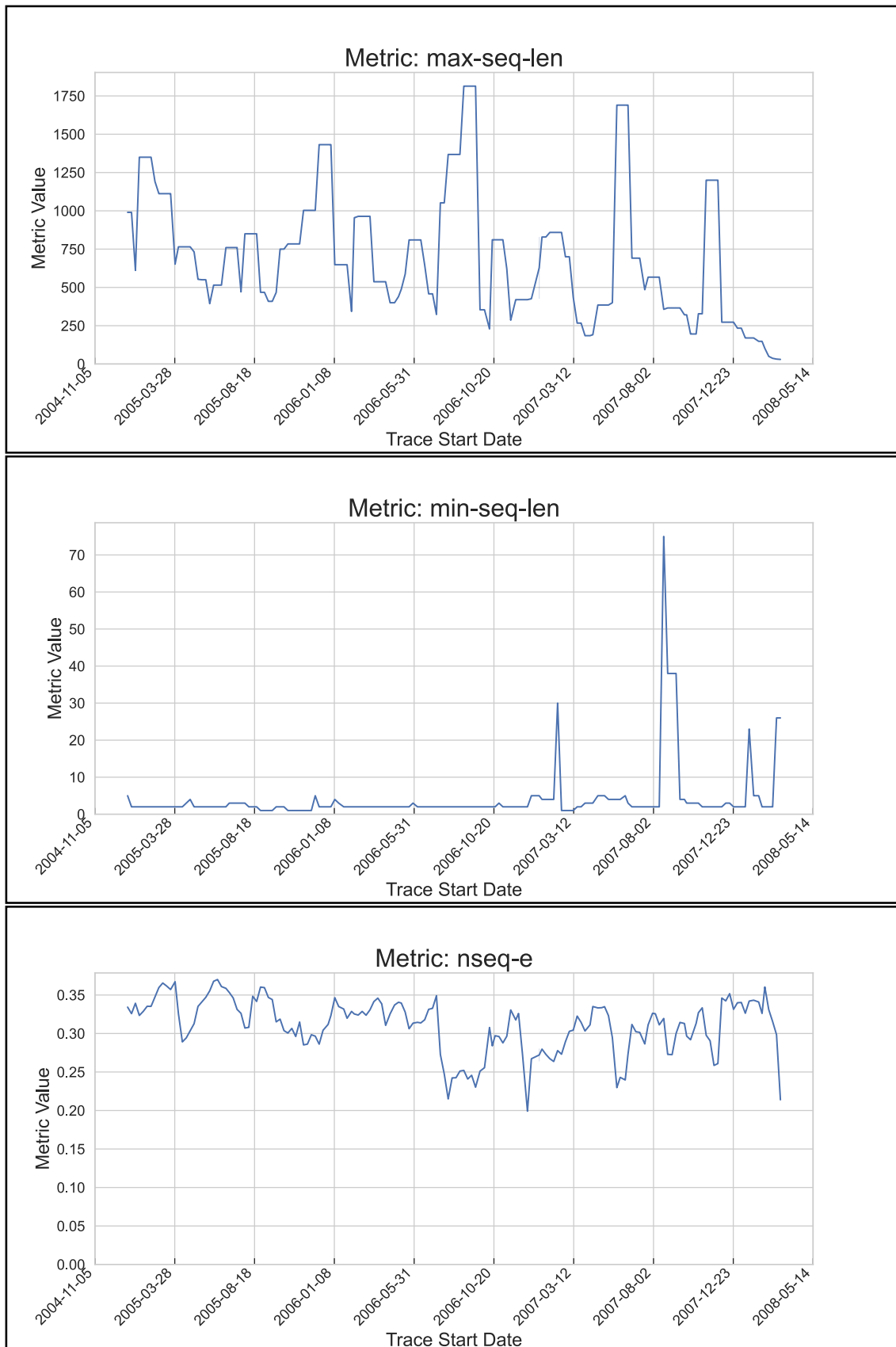


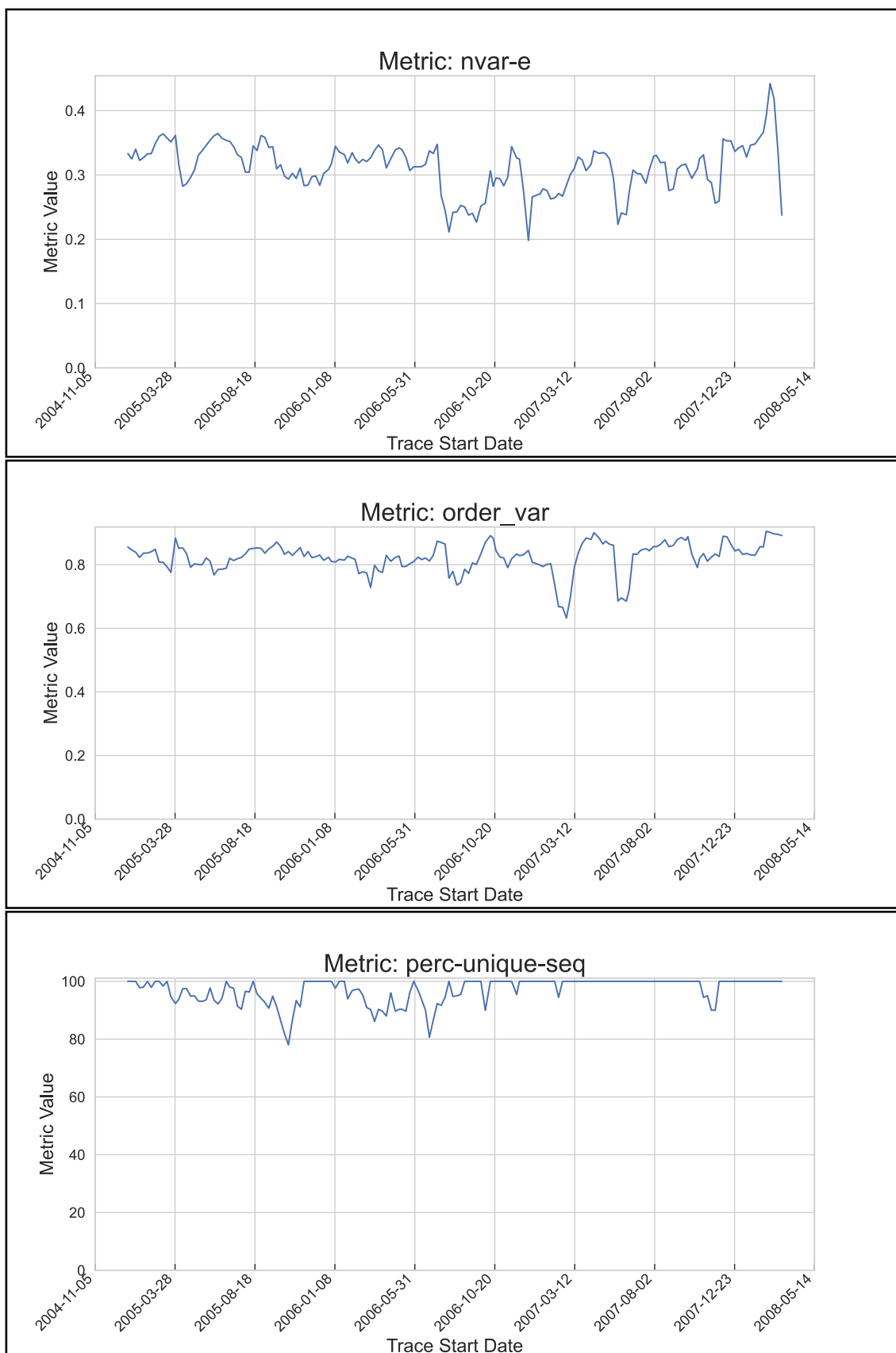


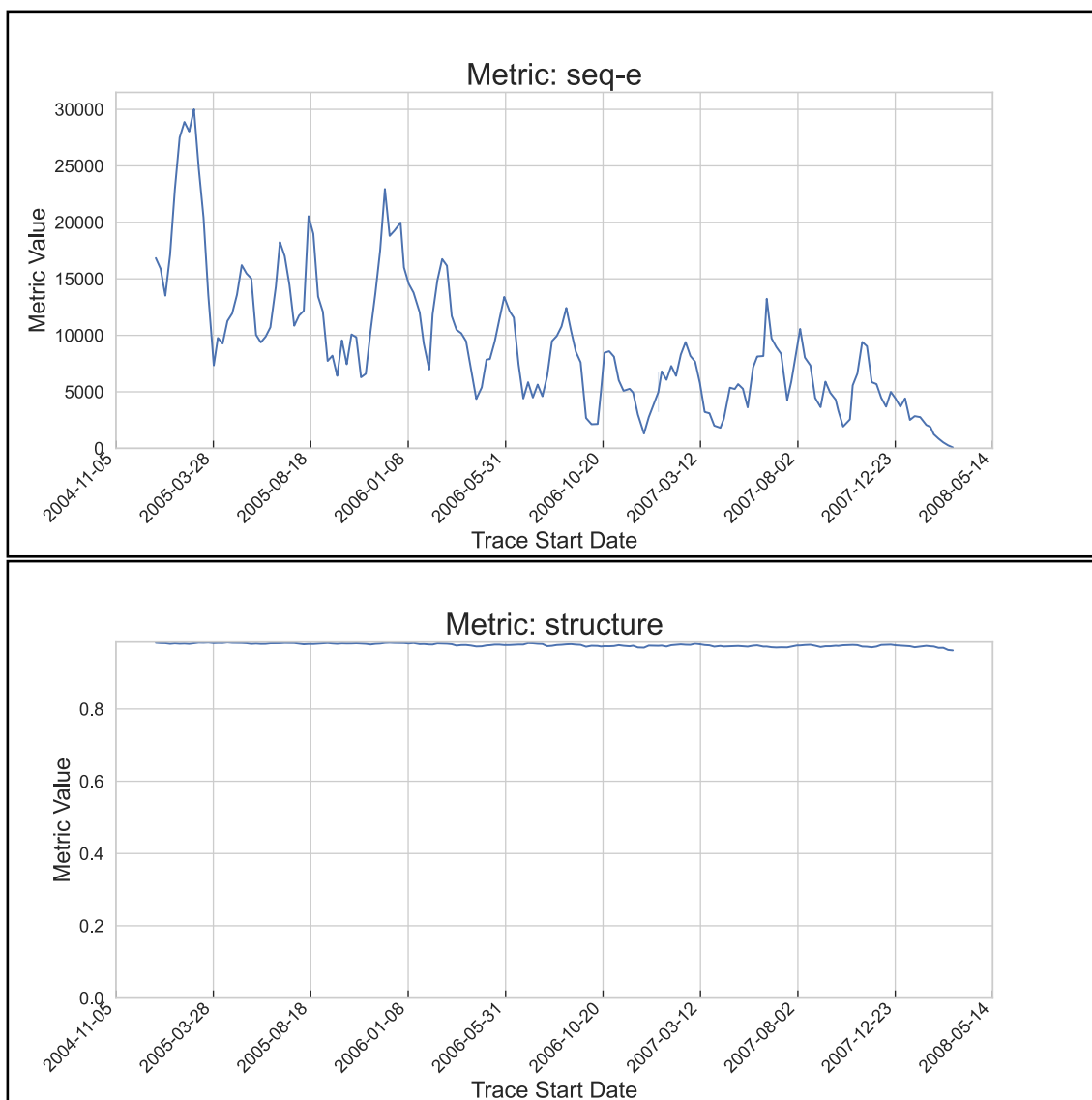


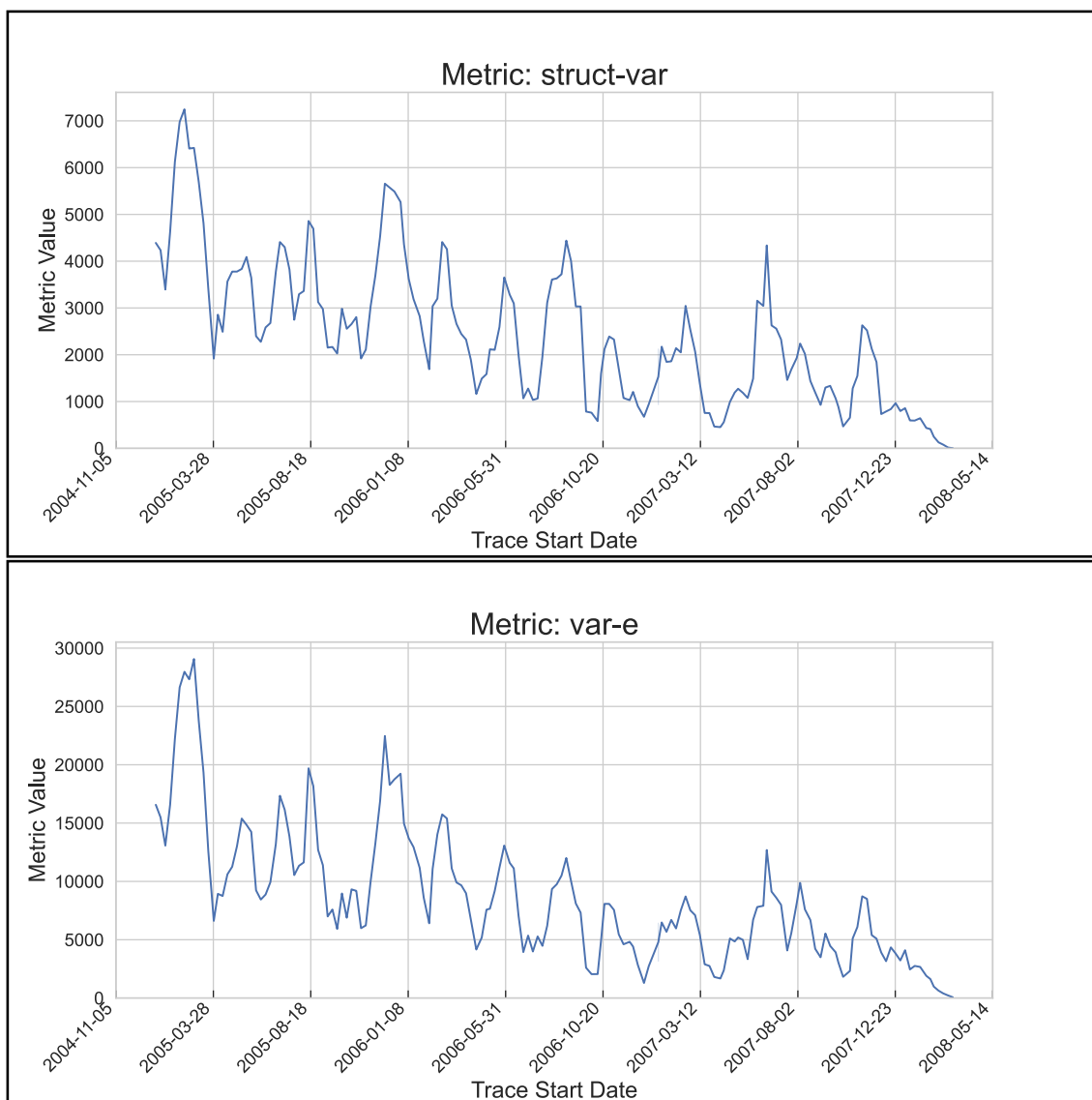










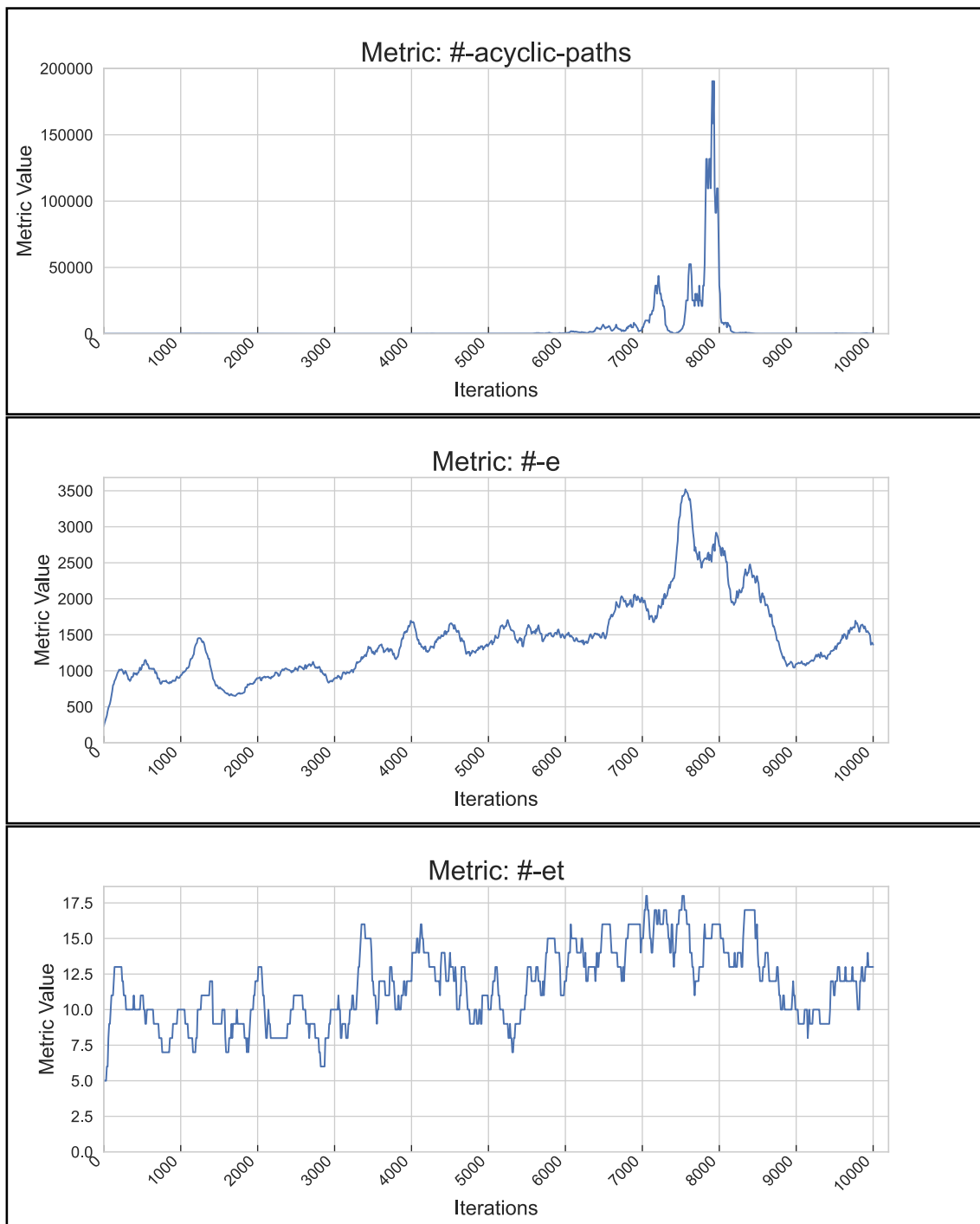


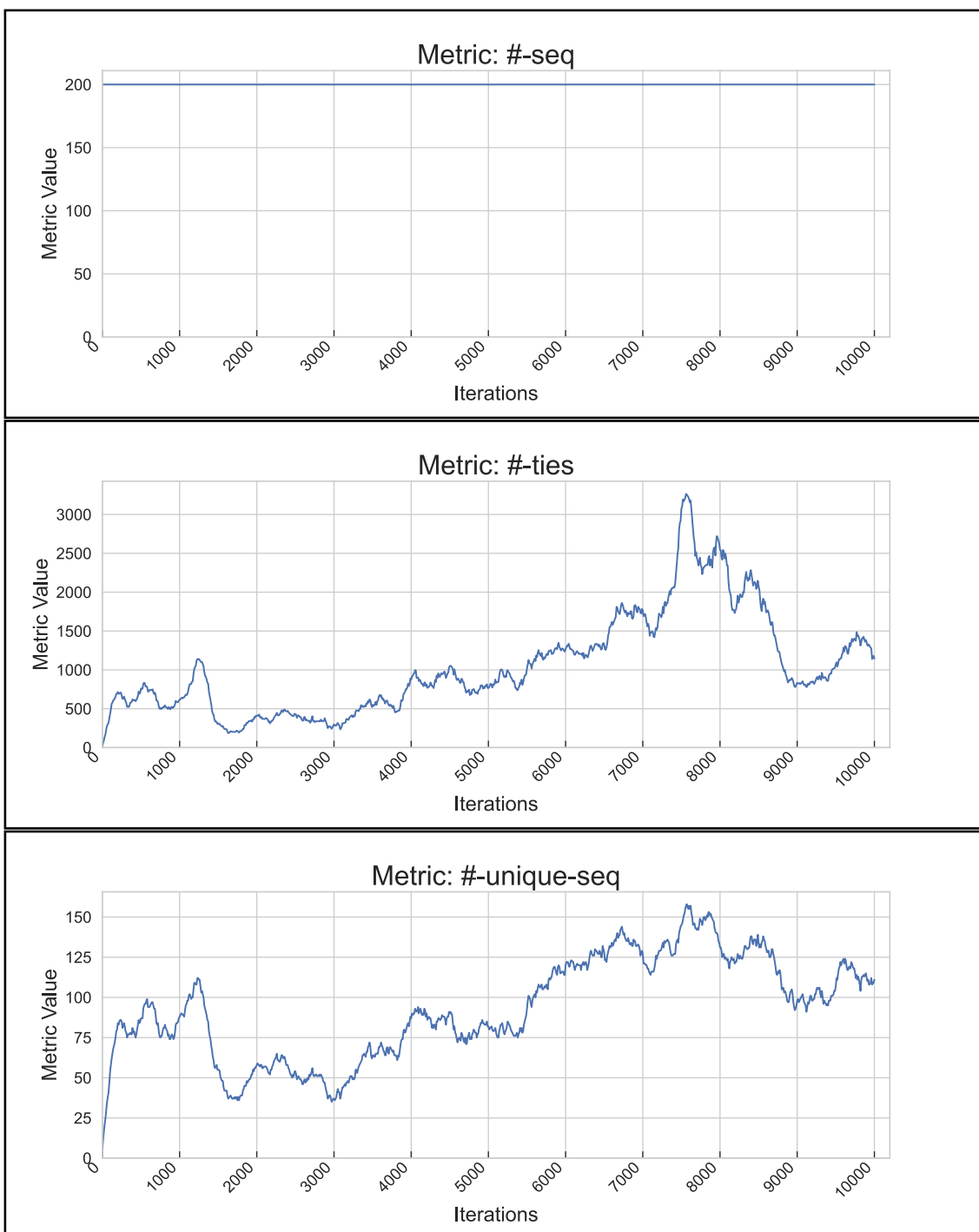
C Process Drift Simulation Additional Figures or Tables

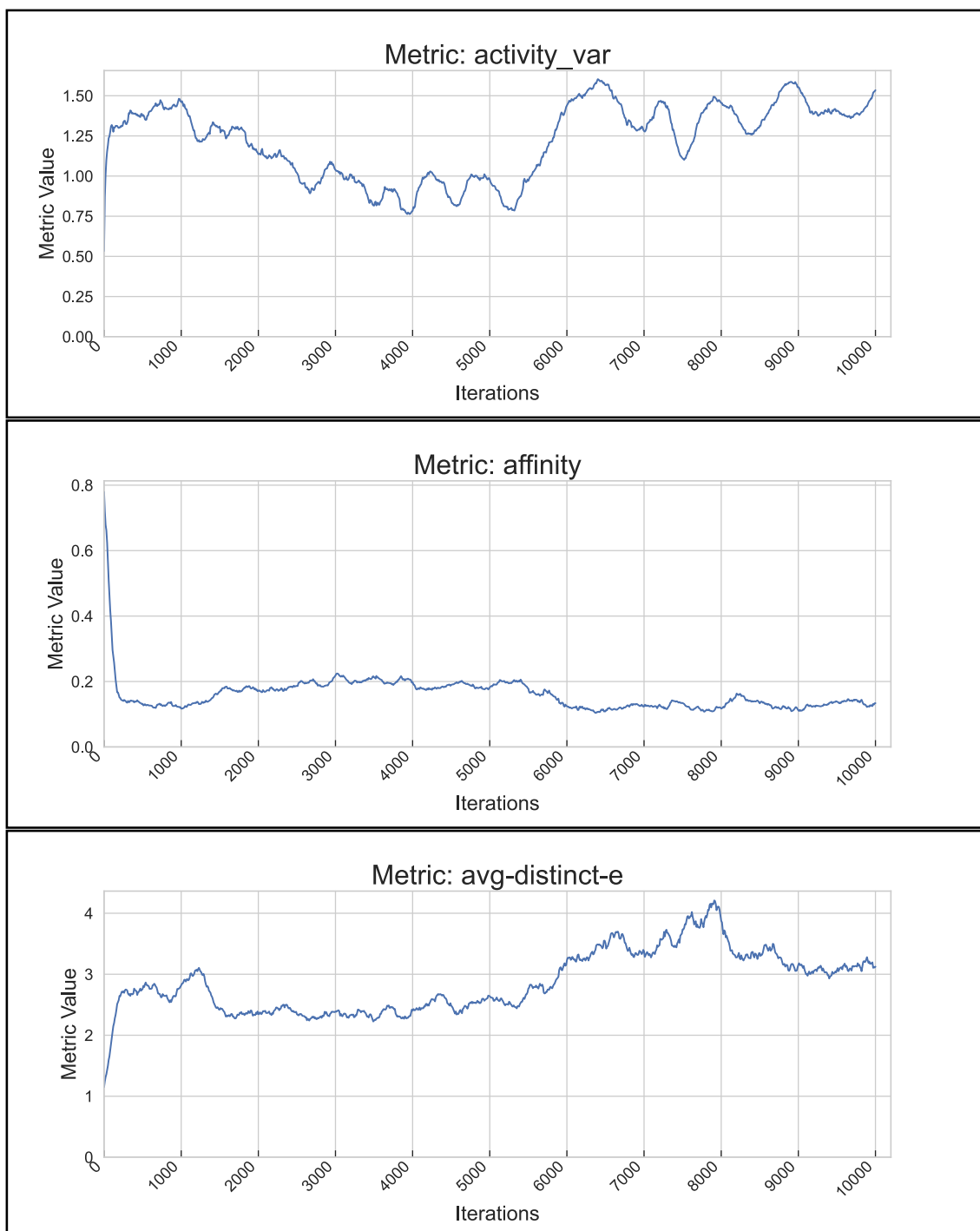
In the following section, all figures showing the plots of the process complexity metrics for the respective datasets can be found. Each metric is shown in one respective figure. All other figures can be found in the GitHub Repo:

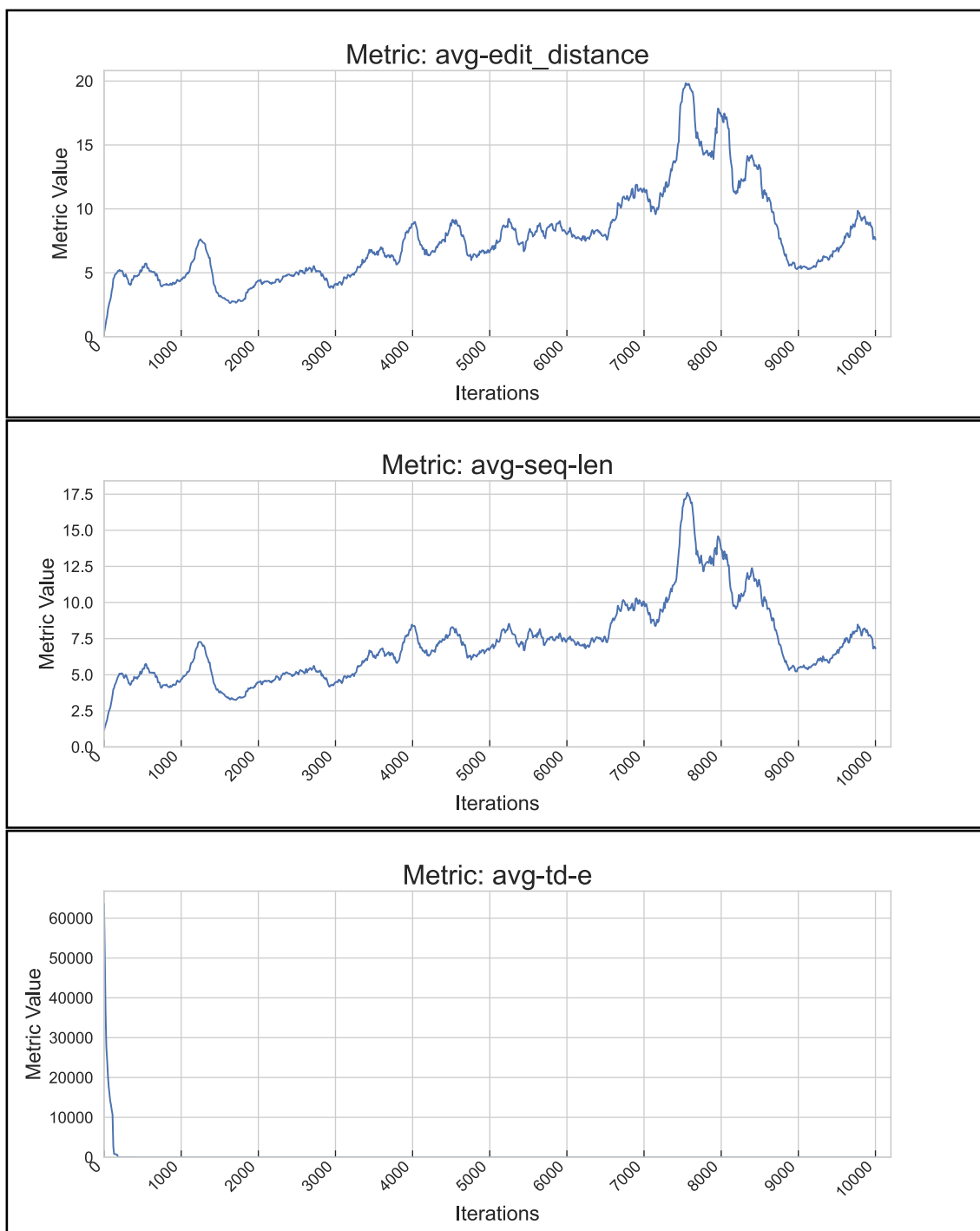
https://github.com/MrWaitomo/WWU_processcomplexity

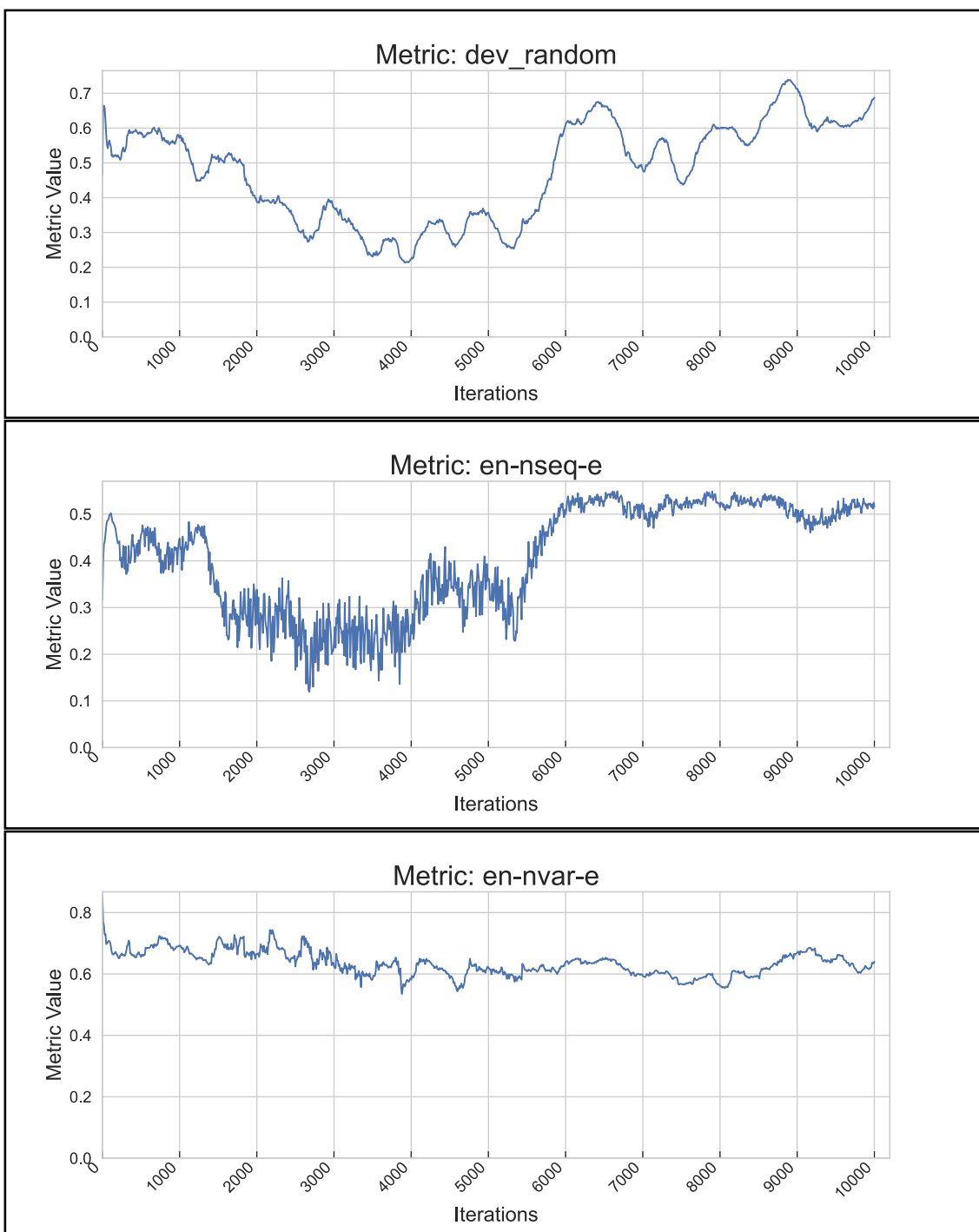
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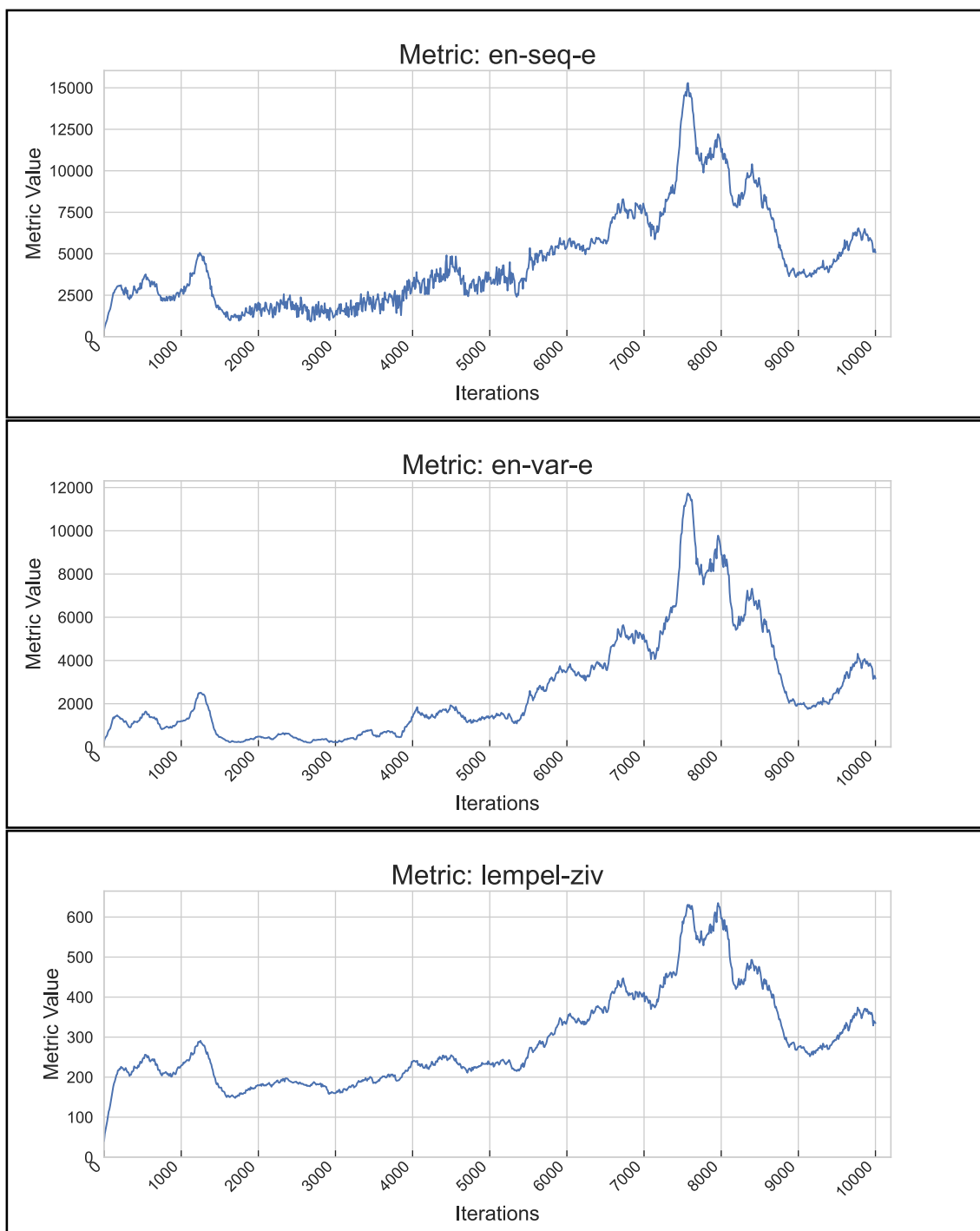


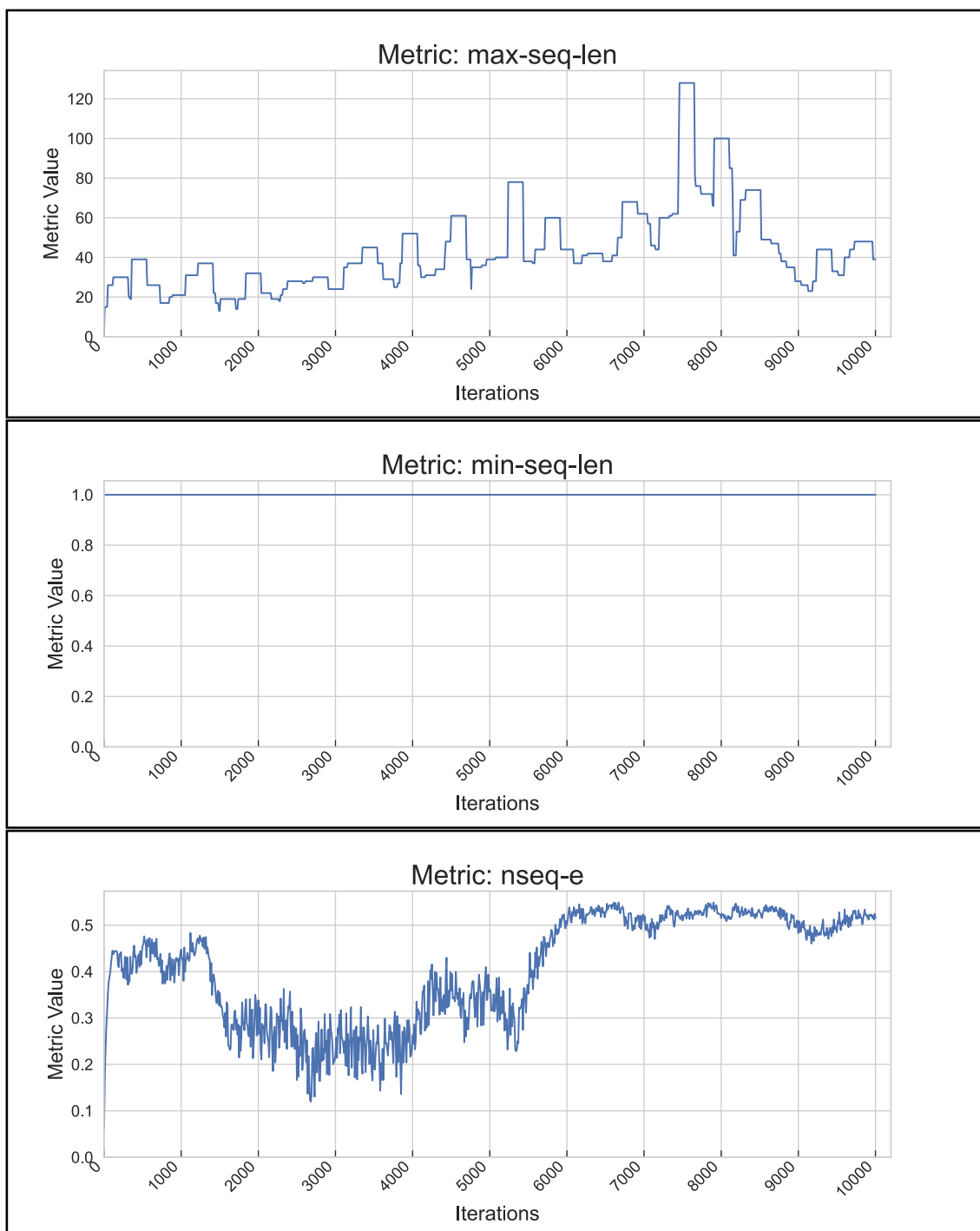


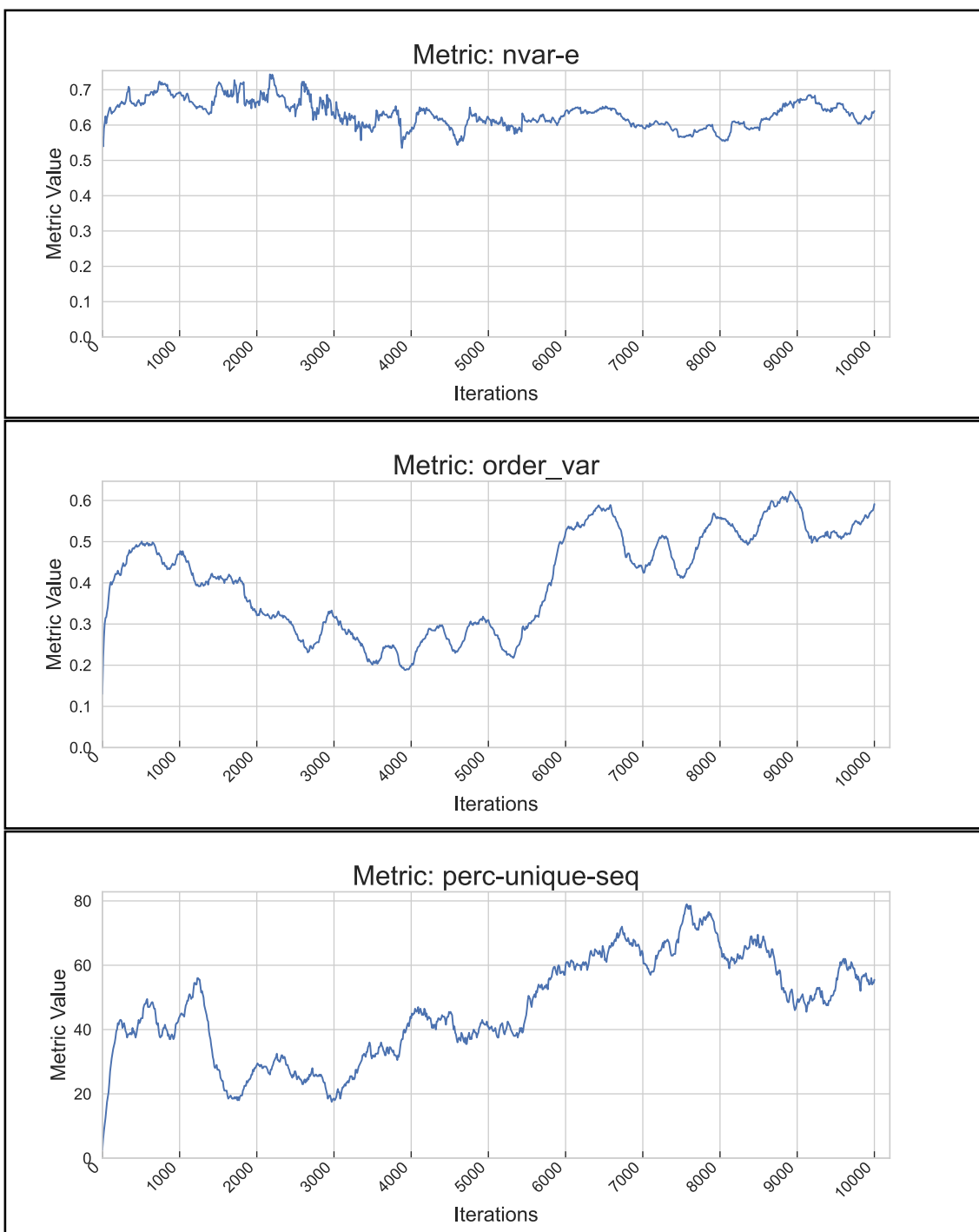


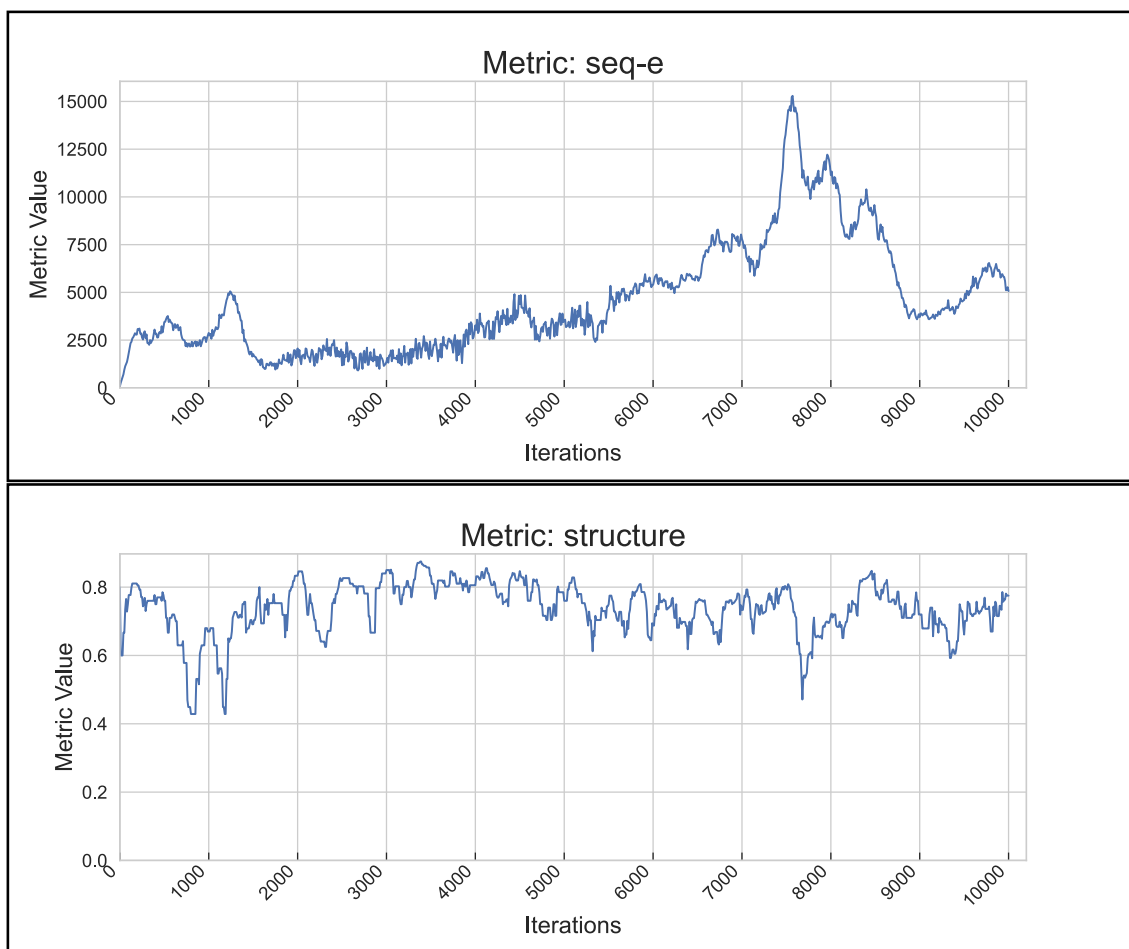


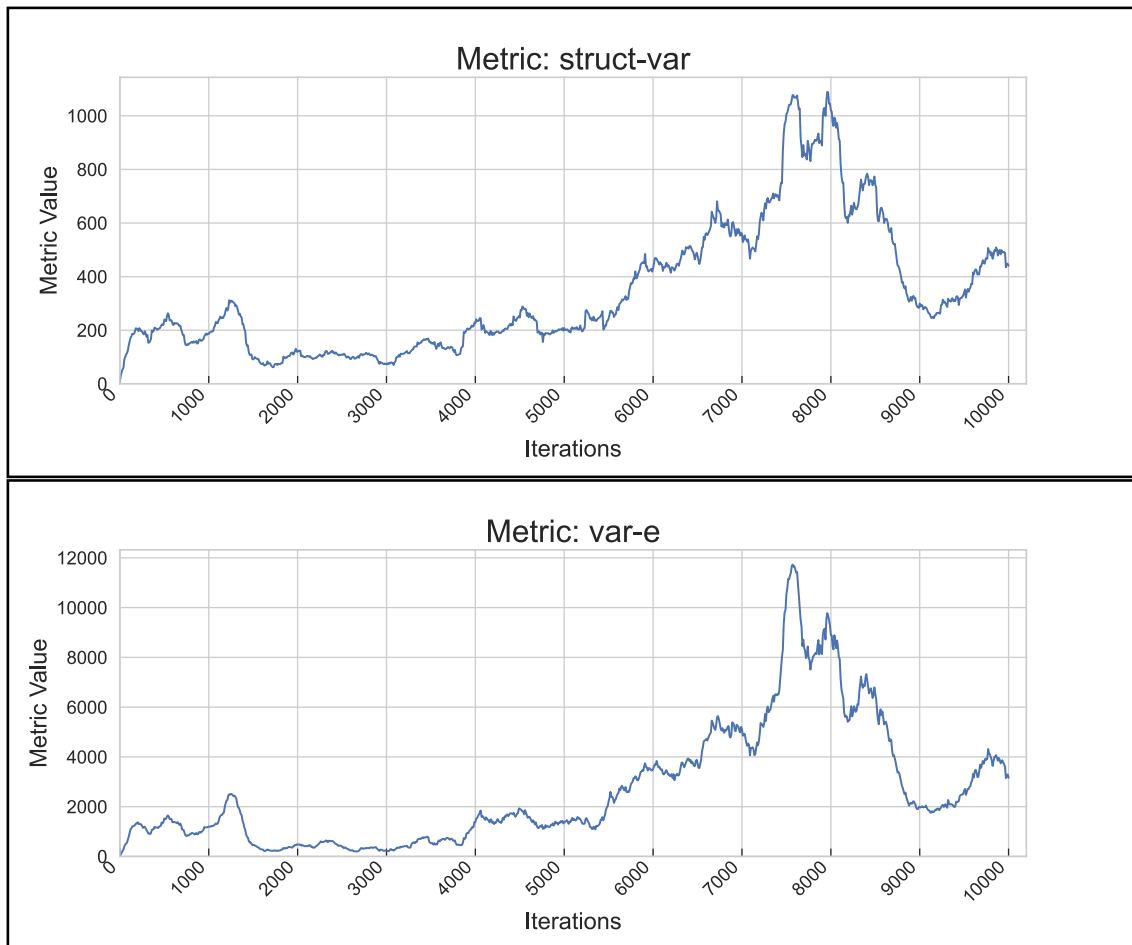


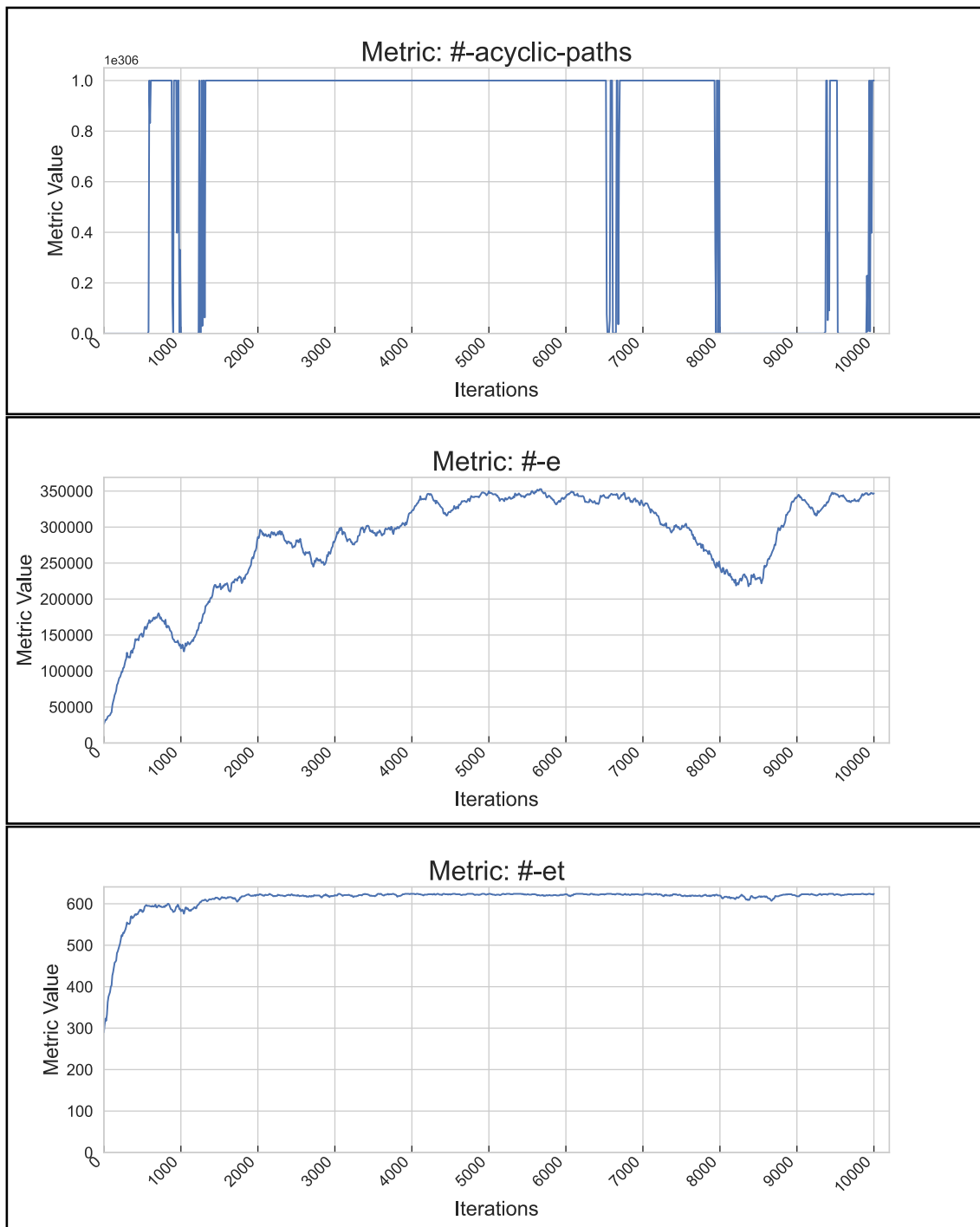


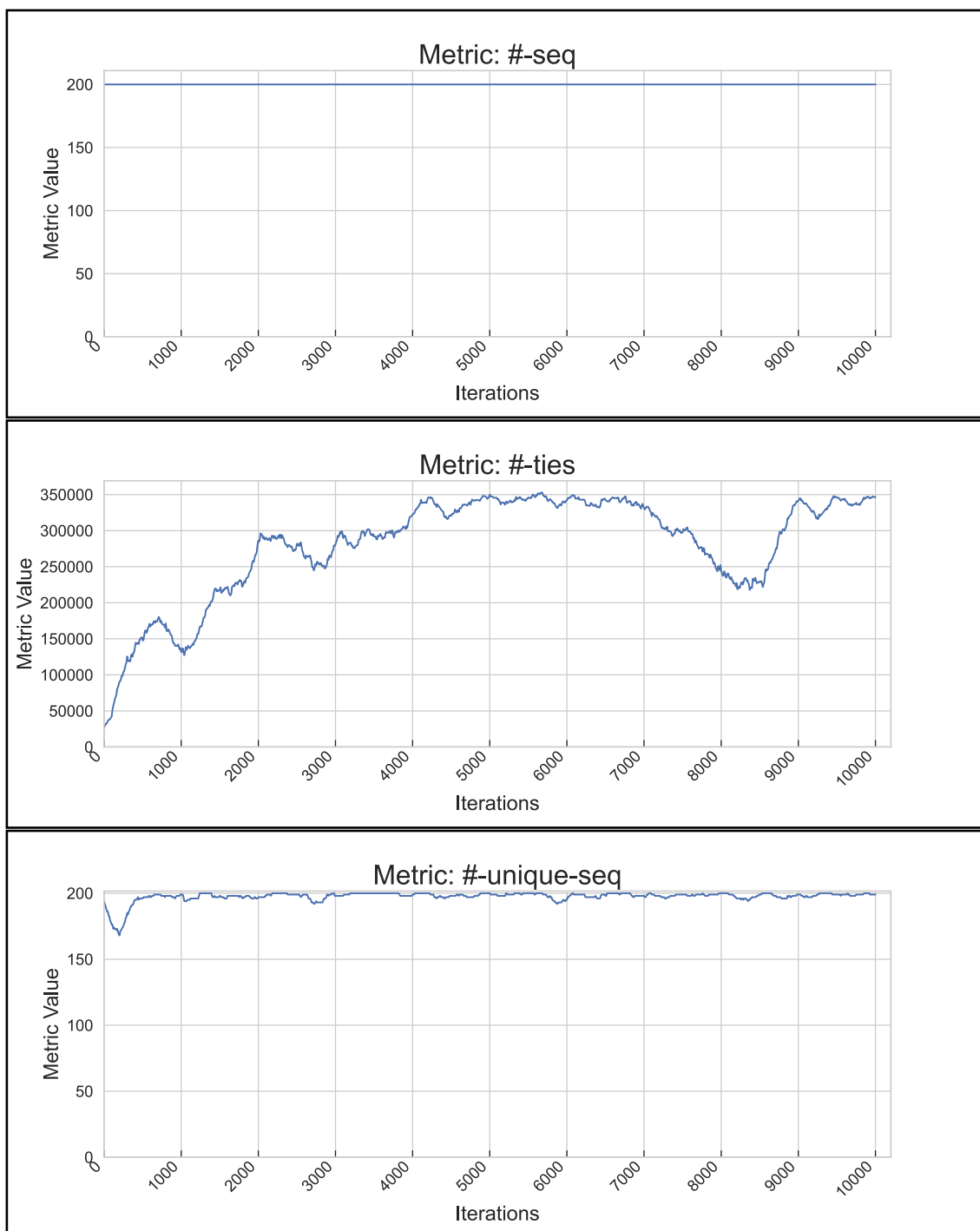


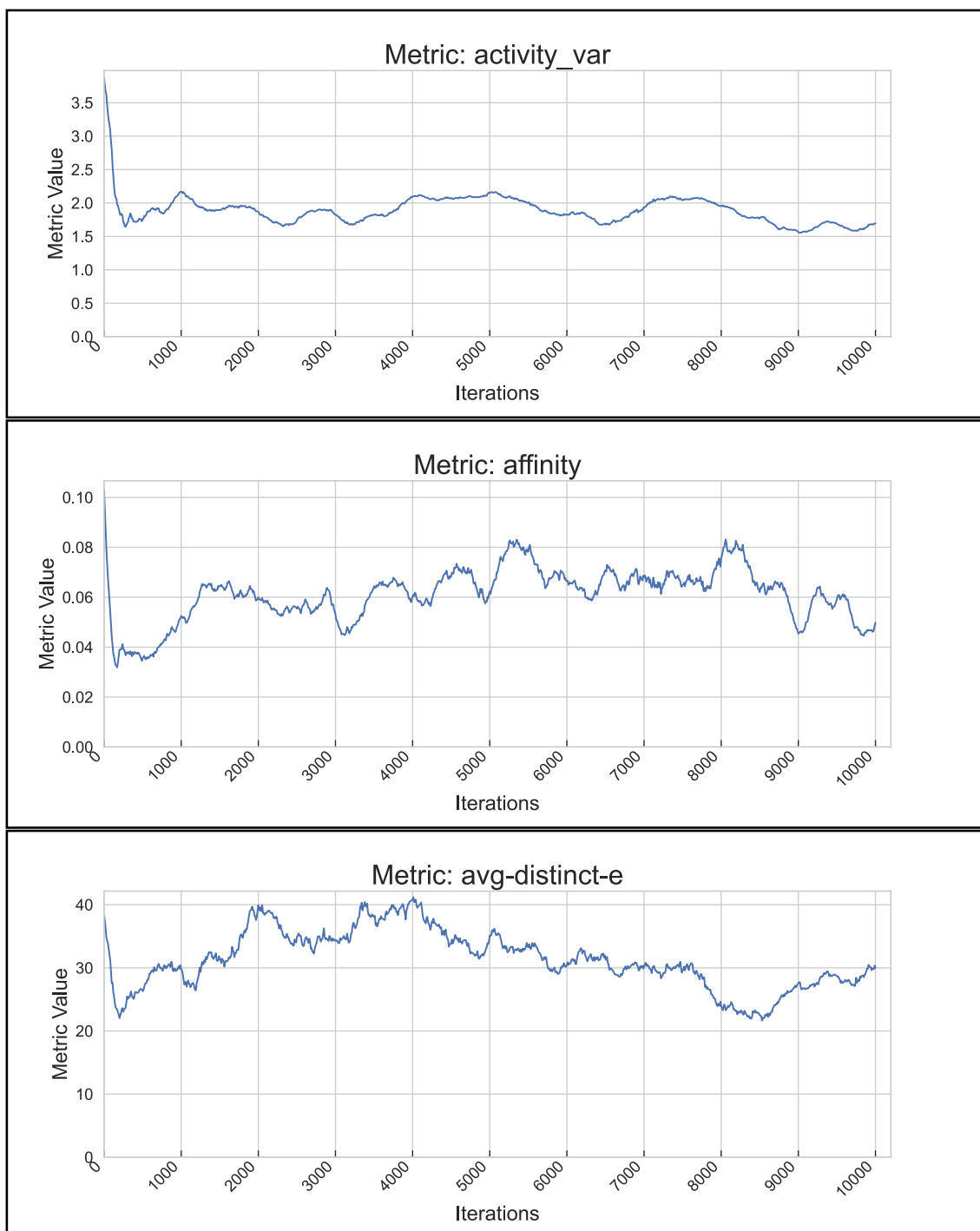


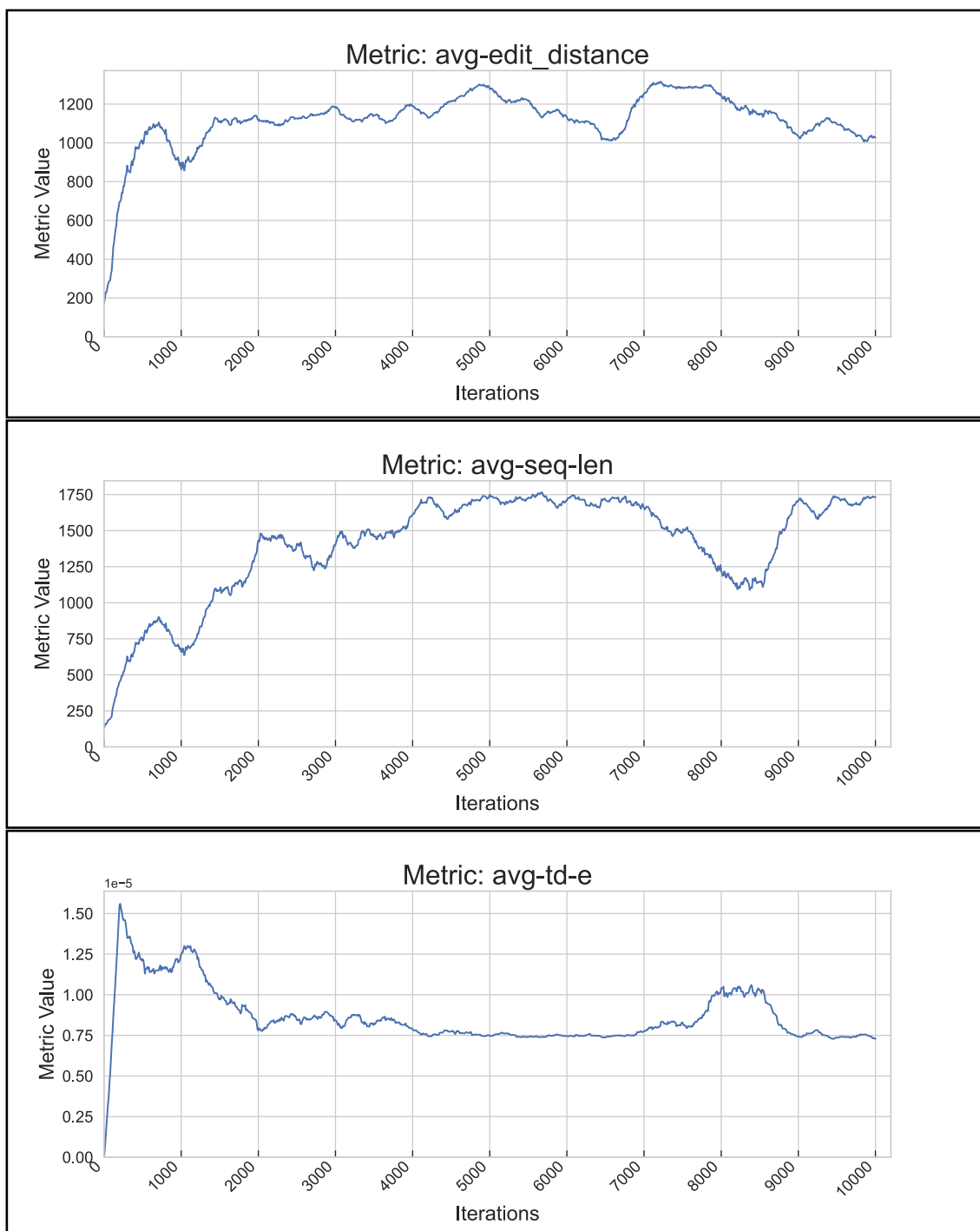


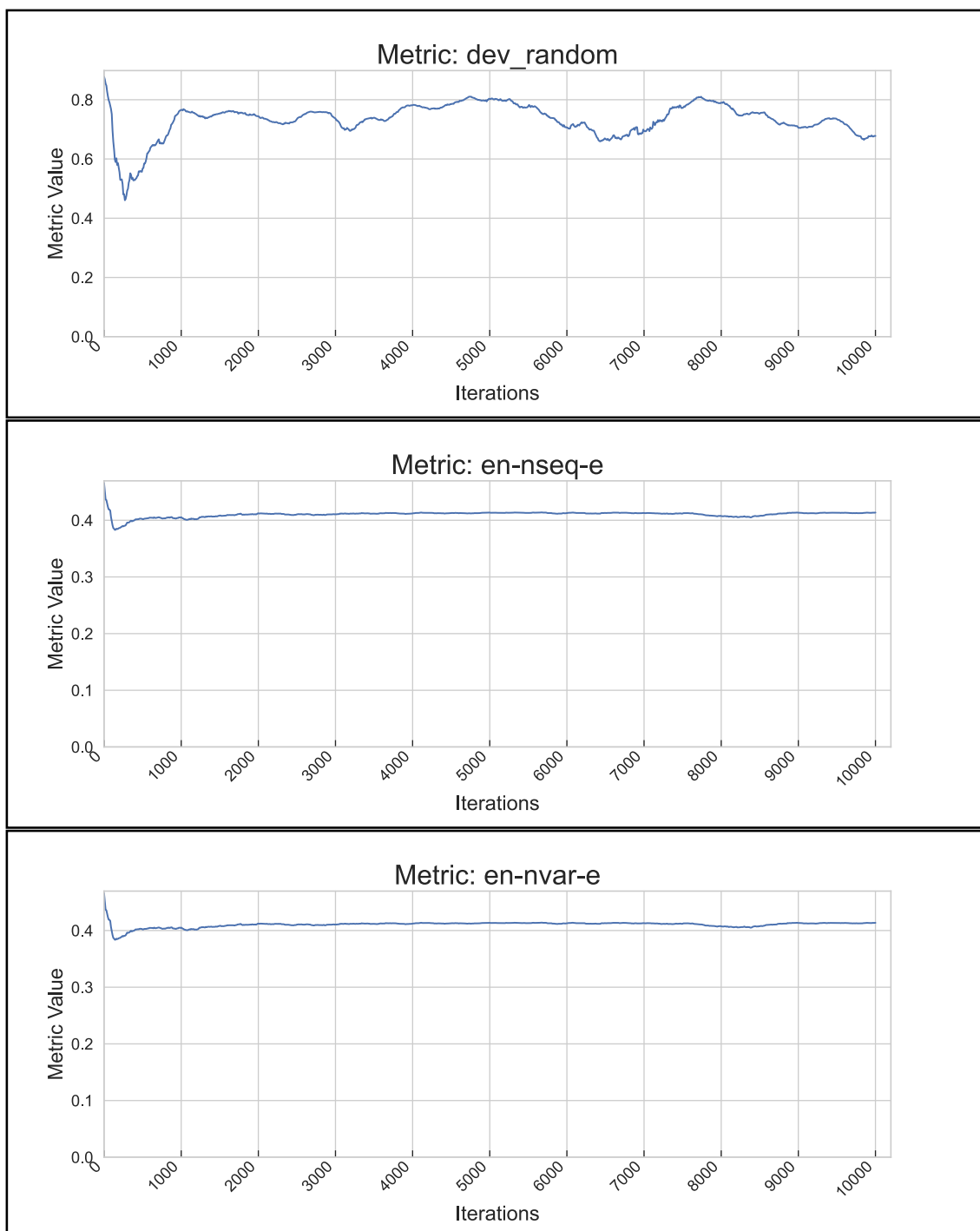


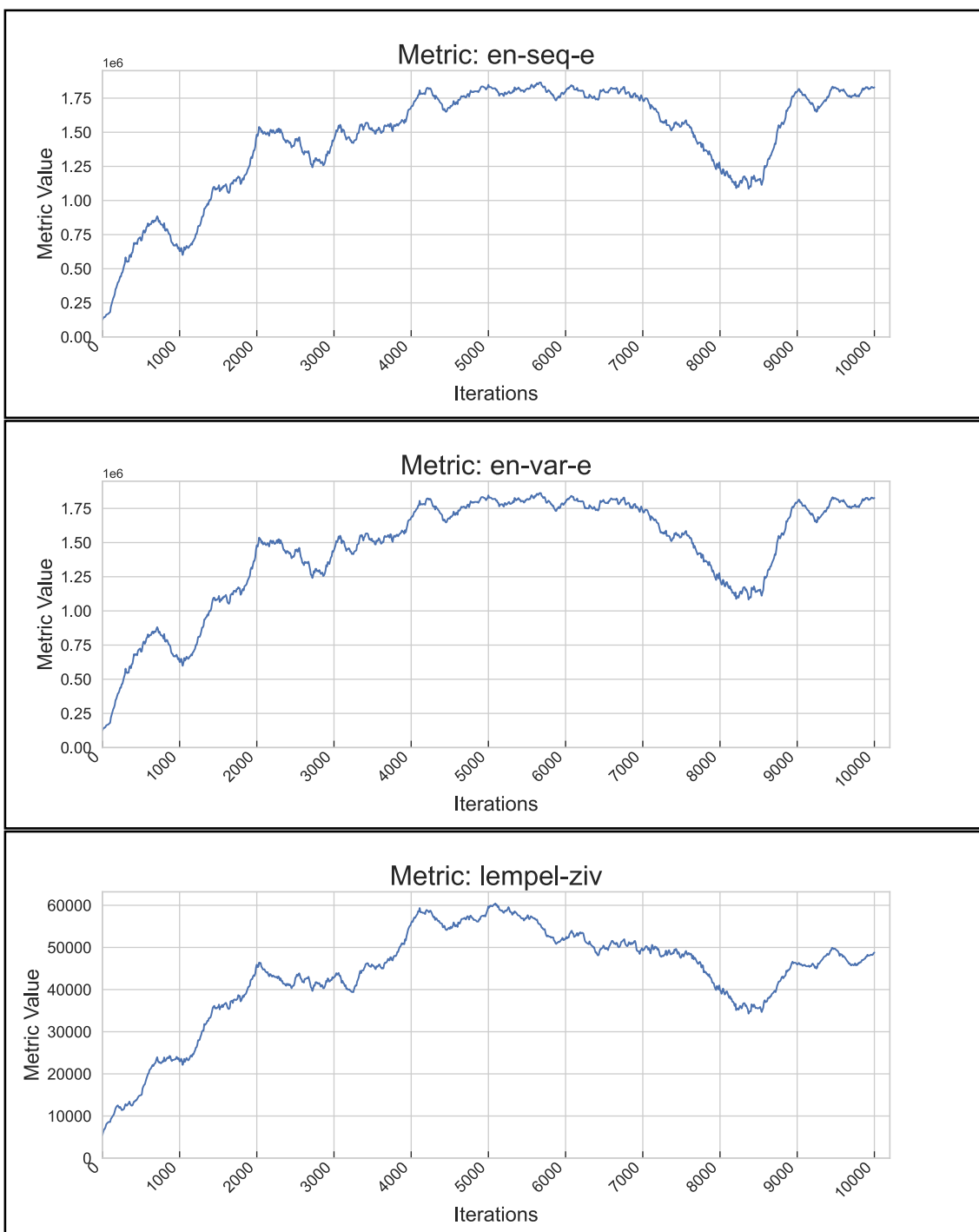
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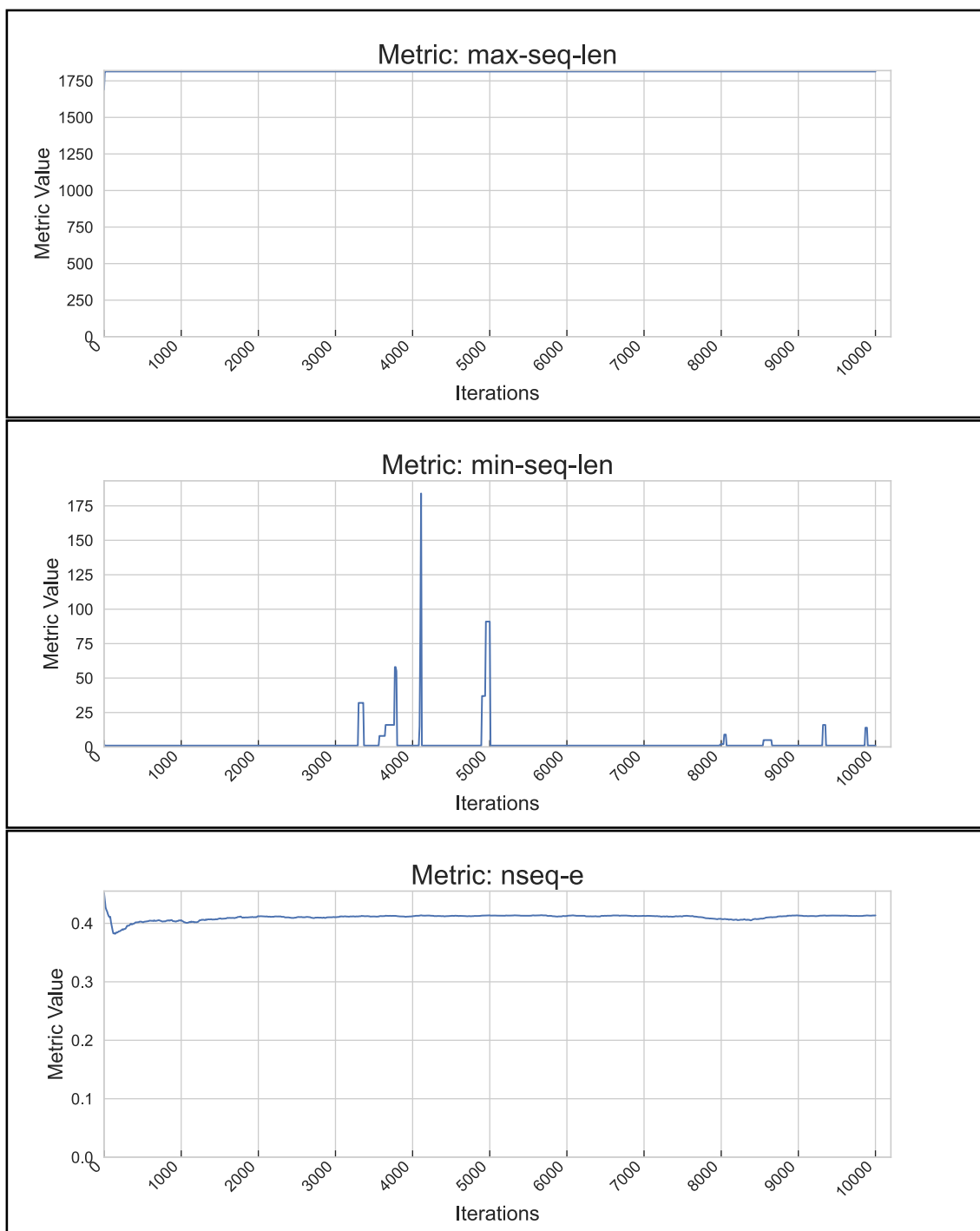


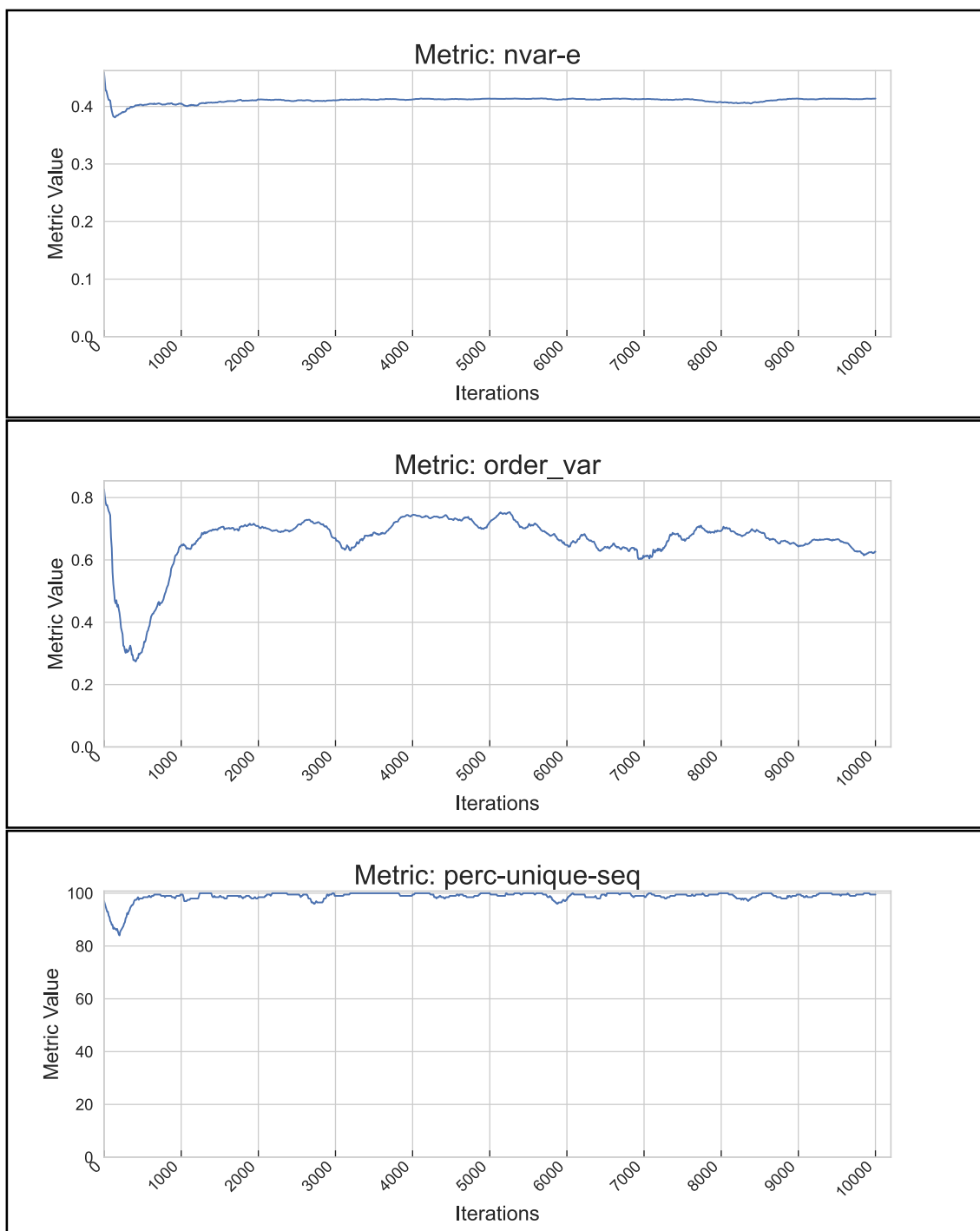


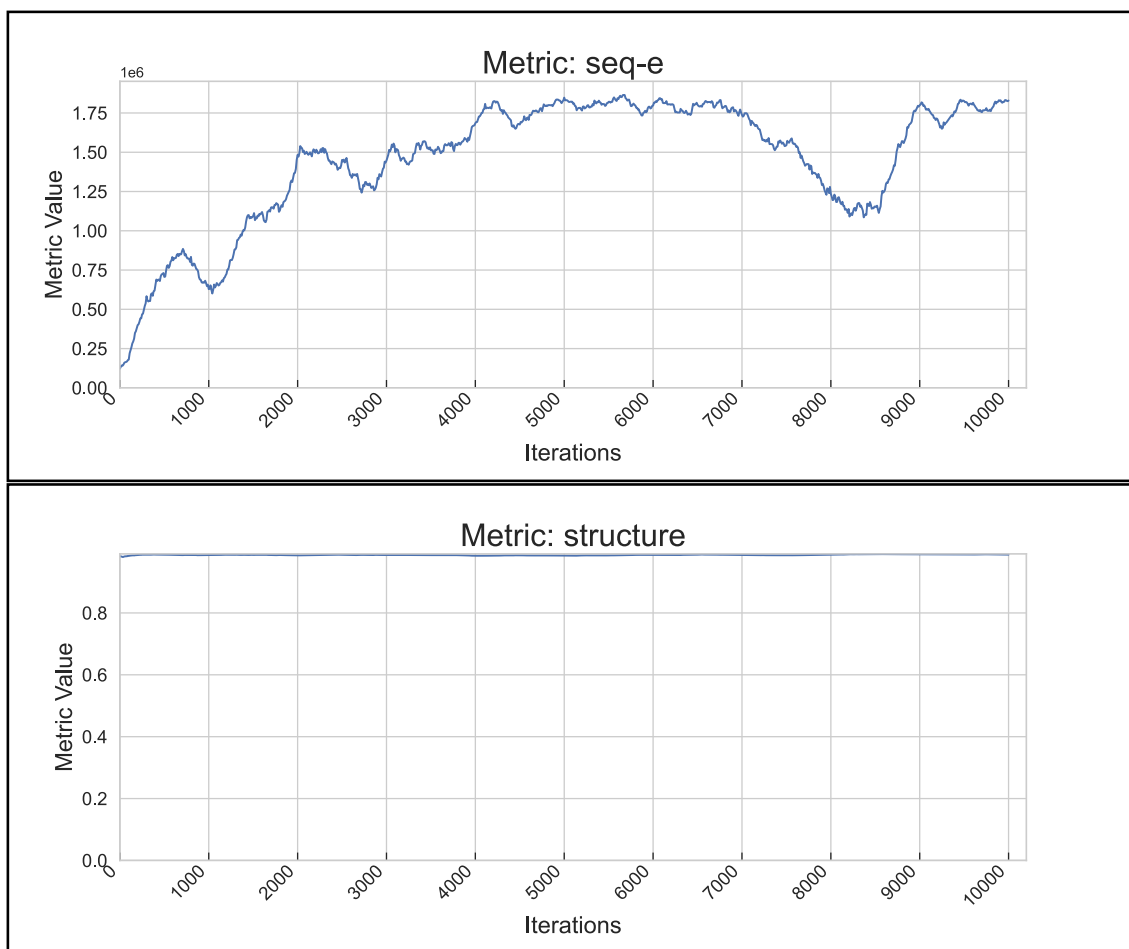


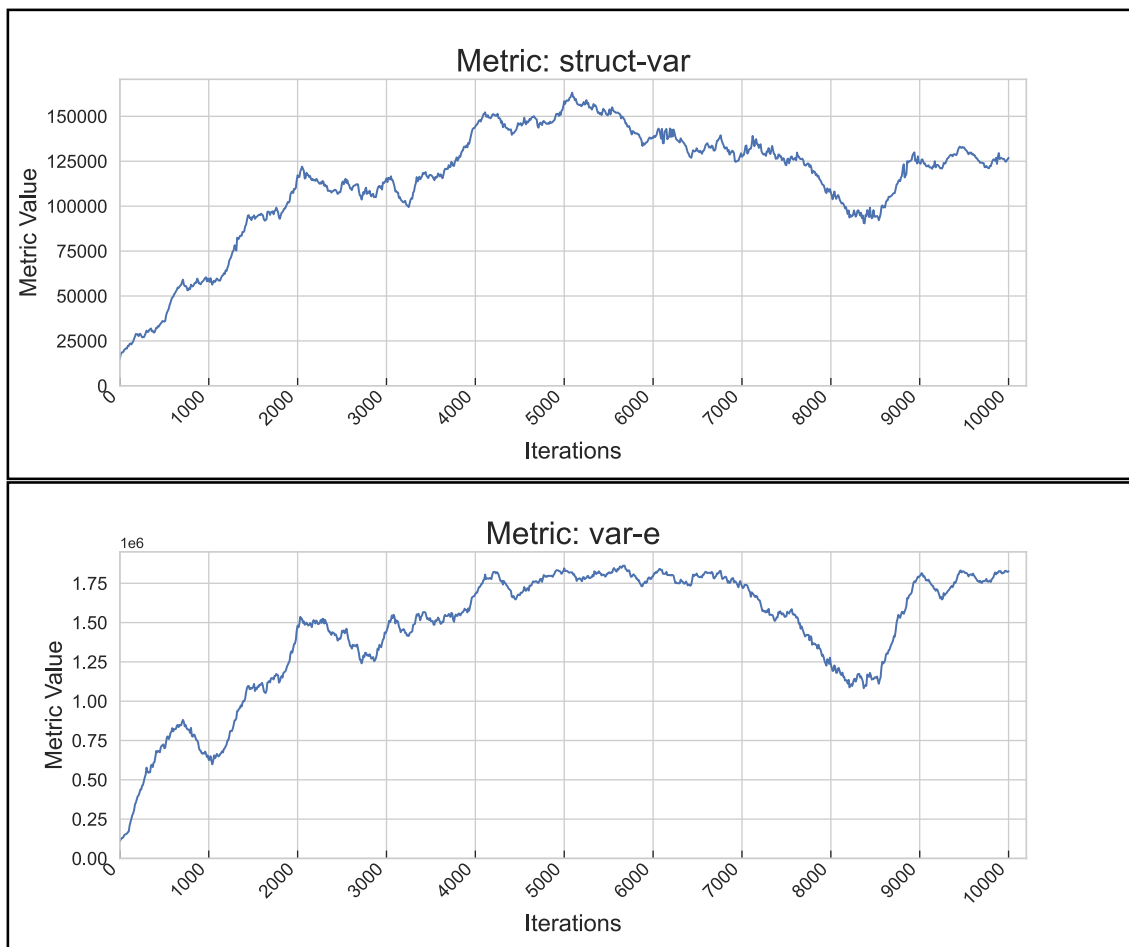












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Fixed window size, first window in the event log.

